# Supplementary material

Who is who: what matters in injuries in professional team sports

Gustavo Pedro Ricou

## S1 Guide

This guide holds the terms the paper uses (Table S1) and the study flow (Figure S1). It also holds the assumed causal structure (Figure S2), and the split of the workload gradient with the within-player slopes (Figure S3). Section S2 gives the data sources, their quality checks and the two substitution eras. Section S3 gives the models. Sections S4 to S7 give the pre-set criteria, the results by league, the sensitivity analyses and the intervals. Sections S8 to S10 give the worldwide survey, the other leagues and sports, and three checks planned before they were run. Section S11 sets who the player was against recent load. Section S12 gives injury history, and Section S13 shows how every number can be reproduced.

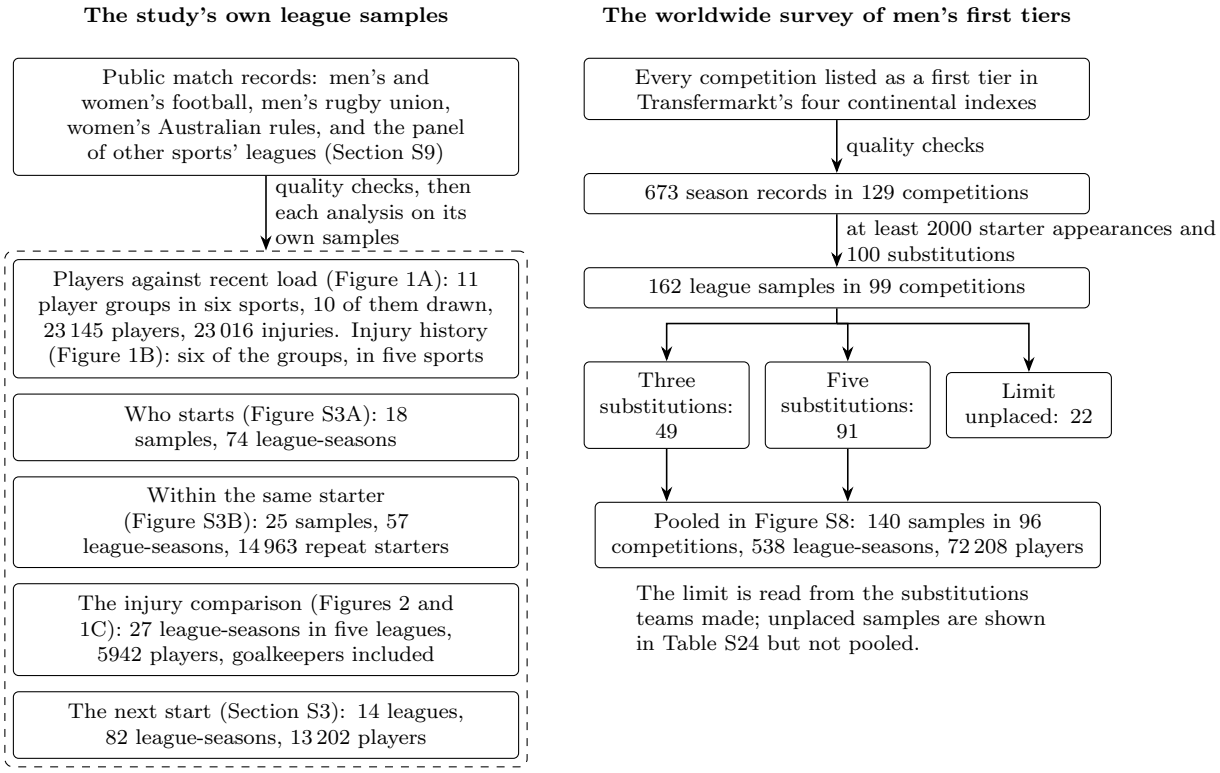



**Figure S1.** Study flow. Left, the league samples and the analysis that uses each, the injury groups first. Right, the worldwide survey, from the continental indexes to the pools of Figure S8. Every count is read from the archive.

The selection drawn in Figure S2 appears well beyond sport. Professional American footballers looked long-lived beside the general population, but not beside replacement players who had played briefly (Venkataramani and colleagues, JAMA 2018). In vaccine studies it is called the healthy vaccinee effect (Lyth and colleagues, Nature Communications 2026). In professional tennis, cumulative load is meant to predict the time lost to injury. But the time lost also shapes that load (Kovalchik, PLOS One 2020). Mothers of twins had more births. The reason was not higher fertility: mothers who gave birth more had more chances to have twins (Rickard and colleagues, Nature Communications 2022).

Our earlier preprint offered two remedies for per-hour rates. The first, reporting per appearance, removes minutes from the denominator. The second, counting only starters' hours, narrows the path through who starts. But arrow (3) of Figure S2 makes starting depend on earlier load. So counting only starters selects players by a result of that load. Such a selection can create a bias of its own (Hernán and colleagues, Epidemiology 2004; Holmberg and Andersen, JAMA 2022).

**Table S1.** The terms the paper uses.

| Term | Meaning |
|---|---|
| League sample | One league's seasons under one substitution limit. |
| Player group | One league, several leagues pooled, or one role in a league (Section S9). |
| Recent match load | Minutes played in league matches in the window, divided by the length of a full match. The window is the previous seven days, the seventh day included. Rugby union uses two weeks. Women's Australian rules uses three weeks, and the share of each match played in place of minutes. Training is not recorded. |
| Recorded minutes | The source's minutes for an appearance. They stop at the full match, 90 minutes in football: added time is not recorded, so a starter who plays the whole match records 90. |
| Workload gradient | How much a player's recorded minutes rise with recent match load: the slope of log recorded minutes on that load. A slope $s$ means $100(e^s - 1)$ per cent more minutes per extra full match. |
| Part due to who starts | The part of the gradient among all players that comes from players with more recent match load being starters rather than substitutes. |
| Same player picked again | The part due to who starts that lies within a player: the same player starting more often after playing more. The rest of it lies between players. |
| Team sheet | The list of players a coach names for a match: the starters and the substitutes. |
| Team-match | One club in one match. |
| Within the same team sheet | A comparison among the players of one club in one match, who share the schedule, the opponent, the score and the weather. |
| Within-player slope | The slope of a starter's log minutes on the same starter's load, with each player's average removed. |
| Between-player part | The gradient among starters minus the within-player slope. |
| Starts per repeat starter | The mean number of starts of the players with at least two starts: how many of a player's own weeks the within-player slope can compare. |
| Odds ratio of substitution | Odds are the chance of an event over the chance of none. This ratio compares a starter's odds of being substituted in a part of the match, per extra full match of recent match load. Below one, loaded starters are kept on. Its logarithm is the substitution coefficient. |
| Baseline rate | Each player's own usual injury rate, set by a Poisson model (Section S11). |
| Median rate ratio | How far apart two players picked at random from one season typically are in injury rate. It is the median of the higher rate over the lower (Section S11). |
| Conditional logistic regression | A logistic regression within strata, such as one player or one team sheet in one period of the match. It takes each stratum's number of events as given, so it removes each stratum's own level. |
| Team selection | The coach's choice of who starts and who stays on. It is not selection bias in the epidemiological sense (Lu and colleagues, Epidemiology 2022). But it acts on a per-hour rate as one. |
| The provider | Transfermarkt, the source of the men's football records (Section S2). |
| Worldwide survey | Every first-tier men's competition in the provider's four continental indexes whose records passed the quality checks. |
| Injury-coded substitution | A substitution whose reason in the provider's match report is injury. Concussion counts as injury. The reason is a label, not a diagnosis (Section S3). |
| Narrower outcome | An injury-coded substitution followed within a week by a recorded absence (Section S3). |
| Loaded, part-loaded, rested | A full match or more in the window; some play but less than a full match; none. |
| Heavy week | A loaded week: a full match or more of recent match load. For a regular starter, a heavy week is usually one full match (Section S10). A light week has less: part of a match, or none. |
| Longer-gap week | A week the calendar rested, 8 to 13 days after the club's previous league match. It is the reference of the comparison the design favours, and it leaves out breaks and missed matches (Section S6). The most common gap is seven days. After a full match, that is a heavy week for a regular starter (Figure S4). |
| Comparison the design favours | Each starter's heavy weeks against the same starter's longer-gap weeks in the same league-season. Section S6 calls it the advised comparison. |
| First match after a pause | A club's first league match after a pause: 9 to 60 days in which none of the five leagues of the injury comparison played. The pauses are mostly international windows and tournaments, in which international players played for their countries (Section S6). |
| Starter hours | The time starters played. A rate per 1000 starter hours counts only starters' time. |
| Back from injury | Absent from the club's previous match, and back from a recorded injury within 28 days. |
| Rested by the calendar, absent | Two causes of rest. Rested by the calendar: the club had no league match in the window. Absent: the club played but the player did not, being dropped, rotated, injured, suspended or unused. The calendar's weeks split by the gap since the club's previous league match. A gap of 8 to 13 days makes a longer-gap week, and one of 14 days or more a break. |
| Hybrid model | A Poisson model with two terms: each starter's share of loaded weeks, and each week's departure from that share. One fit gives the rate ratios within and across starters. A share taken from earlier starts only uses nothing that followed the start it describes. |
| Self-controlled comparison | A comparison of each player's own weeks, as in a self-controlled case series. Only players with an event contribute to it. The within-starter injury comparison is one. |
| Rate ratio | The rate of an event in one group divided by the rate in another. |
| Risk ratio | The chance of an event in one group divided by the chance in another. |
| Time-loss injury | An injury that keeps a player from full training or match play, the consensus definition. |
| Prediction interval | The range in which a new league's own value would fall, given the pooled level and the spread between leagues. |
| $\tau$ | The standard deviation between leagues, after sampling error is removed. |

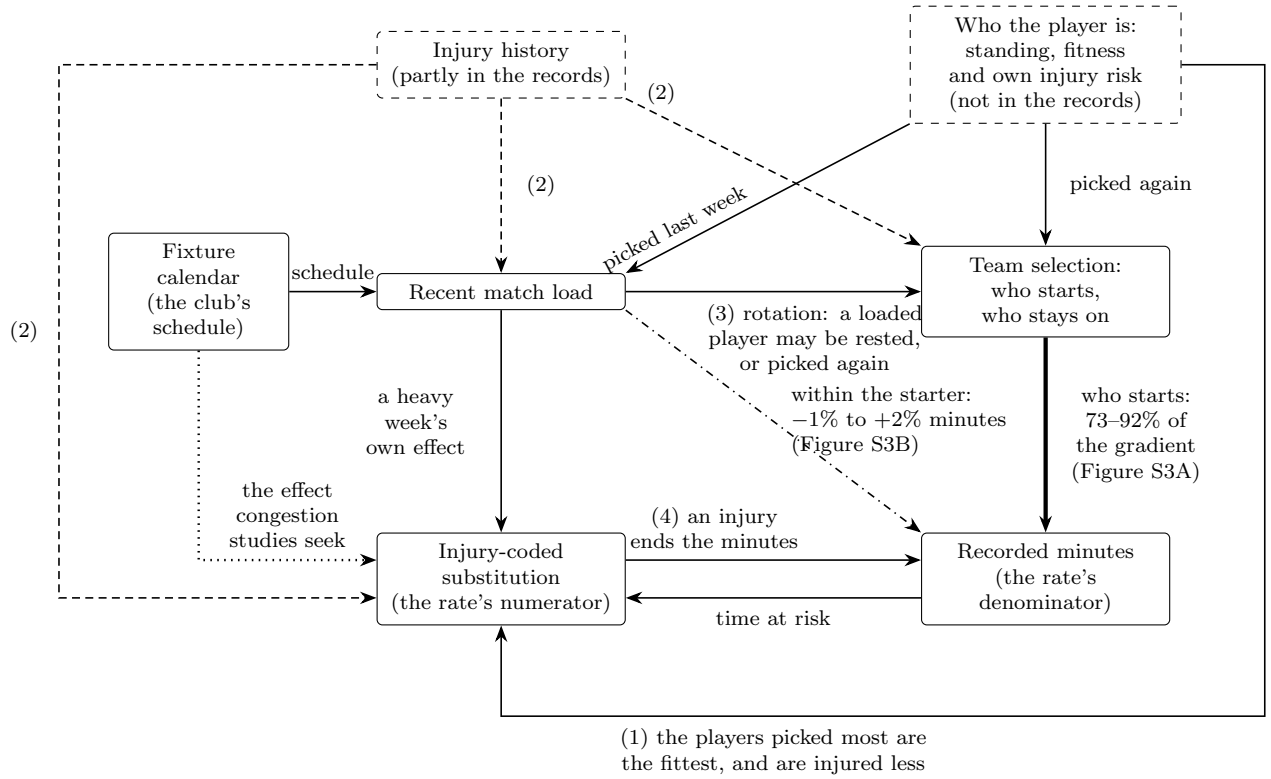



**Figure S2.** Where the measurements sit, drawn as an assumed causal structure. The arrows are assumptions, not estimates. Dashed boxes are missing from the records, or only partly in them. The dotted arrow is the effect congestion studies seek: the effect of matches crowded into short gaps. The dash-dotted arrow is what load does to a starter's own minutes, and the thick arrow is who starts. The top right box, who the player is, holds the player's standing with the coach, fitness and own injury risk. It is missing from the records. It puts the player on the team sheet week after week, so it raises both last week's load and this week's minutes. That shared cause is the part of the gradient due to who starts that lies between players. The fixture calendar sets whether there was a league match in the window at all. When there was none, no player of the club carries any load. The time since the last match acts on injury through recovery, along the dotted arrow. A heavy week's own effect on a player runs from the load. Figure S3B finds the dash-dotted arrow small. Four further arrows complete the structure. (1) Who the player is also acts on injury. The players picked most are the fittest, so a comparison across players attributes their lower risk to load. The same box holds each player's own level of risk, which Figure 1A of the main text measures. (2) Injury history lowers a player's load, and so lowers the share of loaded starts over a player's whole record. It also changes selection and raises the risk of the next injury. So a summary of a player's load that includes the weeks after the player's injuries already holds part of the outcome. That is why the hybrid model's share is also taken from earlier starts only (Section S3). (3) Recent load can itself change selection: a coach rests a loaded player, or picks the same player again because the player played. This arrow is the same player picked again, the part due to who starts that lies within a player. Through it, a player's load changes the player's place in the team, and that place sets the player's later load. Steenland and colleagues describe this as the survivor effect (American Journal of Epidemiology 1996). A model that leaves out a survivor bias of this kind can find a weaker effect, or one in the wrong direction (Buckley and colleagues, Epidemiology 2015). (4) An injury-coded substitution ends the recorded minutes, so the numerator shortens its own denominator. A per-hour rate divides injuries by recorded minutes, and team selection sets those minutes. So team selection enters every comparison of loads made per hour. Minutes and injury act on each other, so the drawing has a loop. It is therefore not a directed acyclic graph, a causal diagram without loops. It maps the paths, and it is not used to choose what to adjust for. Medicine has met the same structure elsewhere. Comparisons across patients had found only a weak link between radiographic osteoarthritis and pain. Comparing each knee with the same person's other knee showed a strong link (Neogi and colleagues, BMJ 2009).

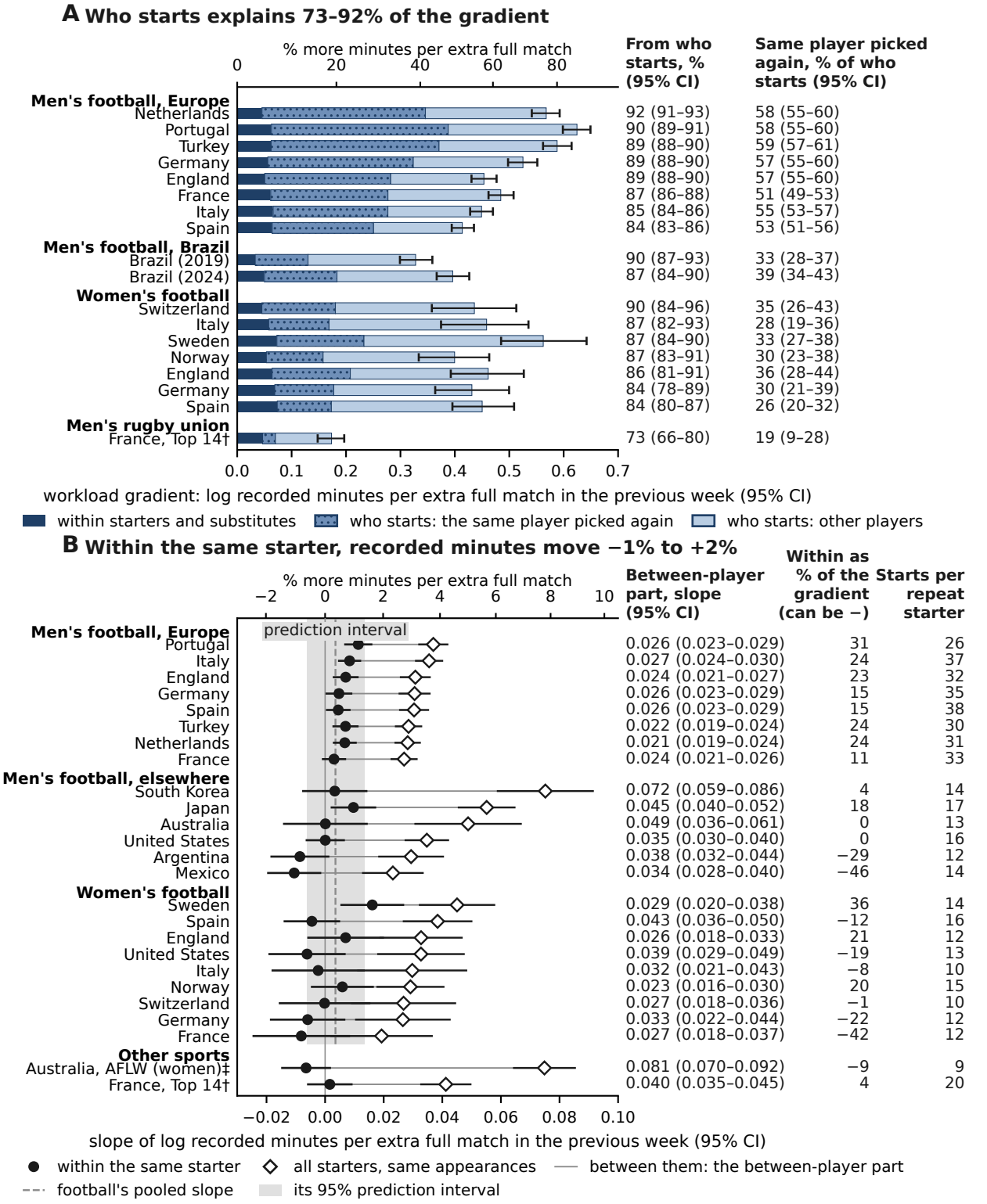



**Figure S3.** Who starts explains most of the workload gradient (A), and within the same starting player recorded minutes change little after a heavy week (B). (A) The gradient among all players who appeared, in 18 league samples. Its darkest part lies within starting players and within substitutes. Two parts are due to who starts: the same player picked again (dotted) and other players. (B) The within-player slope beside the gradient among all starters on the same appearances, in 25 league samples. Football's pooled within-player slope was +0.004 (95% CI +0.001 to +0.006; 95% prediction interval for a new league −0.006 to +0.013). A negative within-player slope means the same player's minutes fell slightly after a heavy week, as fatigue would, and as the estimator's bias in short samples does (Section S6). The estimator recovers only part of a true effect (Figure S6). Corrected for this, the within-player slopes of the 25 samples with a calibration span −3% to +5% minutes per extra full match. Load from the club's fixture list, its matches in the previous seven days, gave a pooled within-player slope of −0.004 per extra club match (−0.006 to −0.002; Table S23). Each panel keeps the samples its analysis was specified on (Table S2). † The Top 14 counts load over two weeks, in full rugby matches. ‡ Women's Australian rules, with no starters, uses all appearances and load over three weeks as the share of a match.

## S2 Data sources, quality checks and the two substitution eras

Men's football comes from Transfermarkt, here called the provider. The eight European leagues come from the provider's public dataset and every other league from its match pages. Women's football comes from two sources. Soccerdonna's match pages give seven leagues, with minutes rebuilt from each substitution's minute. FBref's match reports give France and the United States. Rugby union, the Top 14, comes from itsrugby.fr. Women's Australian rules, the AFLW, comes from AFL Tables and the AFL's own match service. The other sports' leagues come from the panel described in Section S9.

A season entered only if its match sheets carried line-ups and passed the quality checks. Duplicate entries for a player on one date were excluded. No sample size was calculated: every league whose records passed the checks entered. Only the worldwide survey reaches African leagues: four first tiers.

**Which matches count.** Every sample holds one league's matches and nothing else. The analysis script checks it. Recent match load therefore counts that league's matches alone. For the European men's leagues, it is counted from the provider's league appearances. For the other men's leagues and the worldwide survey, it is counted from each league's match pages. For women's football, it is counted from Soccerdonna's and FBref's league match reports. For the Top 14 it is counted from its match sheets, and for the AFLW from its matches. But Figure S3B counts the load over the eight European men's leagues together. So a player who moved between two of them carries both leagues' matches. For the European men's leagues alone, the provider's dataset also holds cup matches, European matches and the national-team tournaments it records. Section S6 adds the cup and European matches to the load, and one of its injury checks adds the tournaments too. National-team matches in the international windows are left out of every sample.

**The fixture calendar.** The load window runs from the seventh day before a match to the day before it, both included. So when a club's previous league match was more than seven days earlier, none of its players carries any load. All of them are rested by the calendar. A player can also have no load when the club did play in the window. That player missed the club's previous match: dropped, rotated, injured, suspended or unused. A team-match is one club in one match. In the eight European men's leagues, 32% to 41% of team-matches followed a gap longer than the window. The most common of these gaps was exactly eight days, in 14% of team-matches (Figure S4). In these leagues, 73% to 82% of rested appearances and 81% to 90% of rested starts were rested by the calendar. Across every sample, the share of team-matches after a gap longer than the window ran from 9% to 62%. The calendar's share of rested appearances ran from 51% to 92% (Table S2).

The calendar's rest is read from each club's previous league match, so the analysis needs each player's club. The provider records it. The Top 14's and the AFLW's sheets record each player's side of the match but not the club. The women's football sources record neither.

Where the source records no club, the club is read from who played with whom. Within a season, two players with at least six appearances are teammates when they share at least five matches. They must also share at least 60 per cent of the matches of the one who played more. Opponents, by contrast, meet two to four times a season. Groups of at least eleven such players are clubs. Every player belongs to the club present in most of that player's matches. Each match holds two clubs, the two with most of their members there. A player whose club is neither of them, such as one who moved in mid-season, gets no club for that match.

The reading was checked on the eight European men's leagues, where the provider names the club. It gave a club to 95.0% to 98.8% of appearances. It moved the share of gaps longer than the window by at most 0.3 percentage points, and the calendar's share of rested appearances by at most 0.7.

The most common gap between league matches is seven days: 27% of the team-matches in the eight European men's leagues and 30% in the seven women's leagues (Figure S4). After a full match, such a week is a heavy week, the usual week of a regular starter. The weeks the calendar rested come in two kinds. A gap of 8 to 13 days, a longer-gap week, came before 22% and 24% of team-matches. A break of 14 days or more is an international window, a winter break or the start of a season. Section S6 gives the injury rates after each kind of gap (Figure S7).

**Ages.** The provider's public dataset records dates of birth for the players of the leagues it covers, the eight European men's leagues among them. In these eight leagues the median age at appearance was 24.4 to 28.7 years (Table S2). For every other sample, the source records no date of birth, or records one for under 95 per cent of its appearances. That is too few to describe the sample. Their ages are left out.

**Provenance.** A public copy of the provider's dataset (github.com/dcaribou/transfermarkt-datasets) was rewritten twice in 2026. In March it was reported to have lost most of its 2024/25 appearance records. A second loss was fixed on 5 May. The snapshot used here was retrieved on 4 August 2026, after both fixes. The copy's main branch then stood at commit af8c44a of 3 August 2026. The copy's last update of the raw data

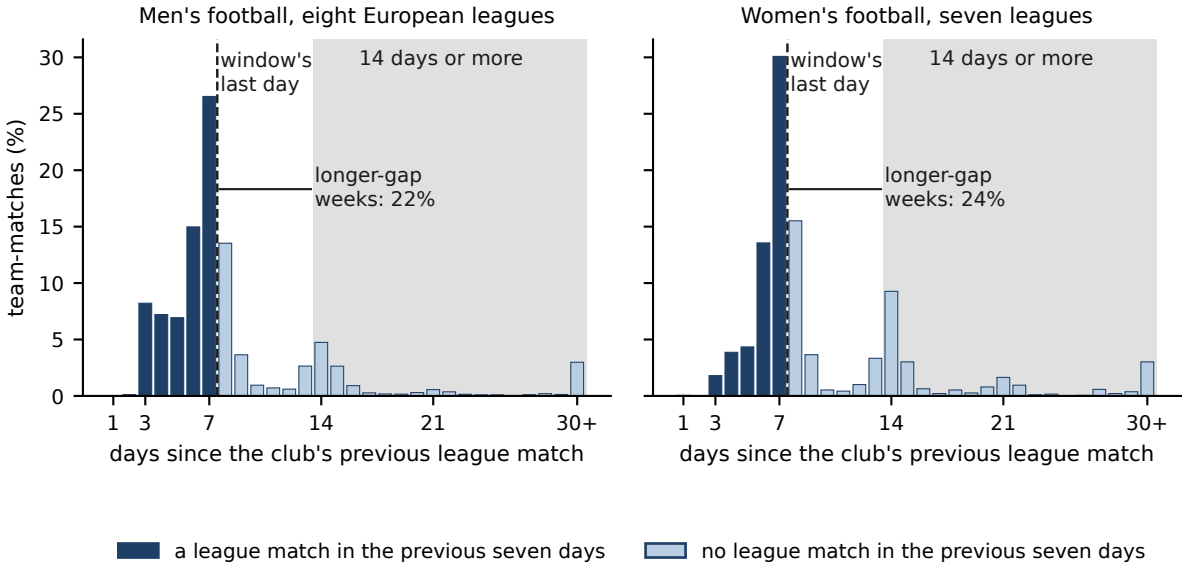



**Figure S4.** Days since the club's previous league match, over every team-match of the eight European men's leagues (left) and of the seven women's leagues of Figure S3A (right). The most common gap is seven days: 27% and 30% of team-matches. Gaps of 8 to 13 days, the longer-gap weeks, make 22% and 24%. The dashed line marks the seventh day, the far end of the load window. After a gap longer than that, no player of the club carries any load. The bracket marks the longer-gap weeks. The shaded band holds the breaks of 14 days or more. Figure S7 gives the injury rates by gap. The last bar holds every gap of 30 days or more, the start of each season among them.

was commit 59fa295 of 11 July 2026. The snapshot's SHA-256 checksum is recorded in the project's archive. In the eight European men's leagues, the snapshot holds 30.4 to 31.3 appearances per match in 2024/25, against 29.9 to 31.3 in the two seasons before. So neither loss affects it.

The women's records were read on 21 and 30 August 2026. They cover matches played from March 2024 to November 2025, all before FBref's data feed from Opta ended on 20 January 2026. FBref was read after that date, so its minutes were checked against Soccerdonna's where both hold the same league-season. These were Italy and Switzerland in 2024/25, read from FBref on 12 September 2026. Of the 5079 appearances found on both, 99.2 per cent agreed within one minute. All of them gave the same role on both sources: starter or substitute.

**The Top 14's replacements.** From 2018 to 2025/26 the Top 14 allowed up to twelve changes a match. Up to four tactically replaced players could also come back on to replace an injured teammate (AFP, 27 April 2026; Rugbyscope, 4 August 2026). Every Top 14 season here falls under those rules.

The source rebuilds a player's minutes from the match sheet's entry and exit events (itsrugby.fr, 24 June 2026). In 2025/26, the season of Figure S3A, the events of 186 of the 188 matches were readable. There, 13 per cent of appearances had more than one stint on the field (13 per cent of starts and 15 per cent of replacements' appearances). In 90 per cent of team-matches, at least one appearance had more than one stint.

When a starter left the field twice, the source counts the minutes up to the last exit (90 per cent of such starters). When a starter came back on and finished, it records the full match (97 per cent). So the time off the field counts as played. A replacement who came on twice is recorded from the last entry (100 per cent), so the first stint is lost. Without the appearances with more than one stint, the share of the gradient due to who starts was 72 per cent, against 73 with them.

The parsed sheets do not mark two things. One is a temporary replacement, for a head-injury assessment or for blood. The other is a round played during an international Test window, when federation release rules shape team sheets.

**The two substitution eras.** Football allowed three substitutions per match until the pandemic. The limit rose to five, first as a temporary rule and then permanently (the International Football Association Board's announcements of 8 May 2020 and 13 June 2022). Most leagues changed in 2020. In the study samples a season's limit comes from the league's adoption date. The women's Australian rules sample is the 2023 season alone, so no change of the interchange rules falls inside it.

In the worldwide survey, a season's limit is read from the mean number of substitutes a team used in a match. A concussion substitute counts as a substitute used, but concussion substitutes do not count against

the limit. Apart from them, a team allowed three cannot average more than three. A season with a mean up to 3.05 is read as three substitutions, and one from 3.60 as five. Between the two, the season is unplaced. The weeks from 8 May to 31 July 2020, when leagues restarted under the temporary rule, are excluded from the survey. The survey's reading classifies a season by its own substitutions, so it was checked where both readings exist. Applied to the eight European men's leagues, it matched the adoption date in all 64 league-seasons.

Teams used more substitutes than the limit in 0.3% of European team-matches under five substitutions and 0.5% under three. In the survey, the shares were 0.2% under five and 2.7% under three. Under three substitutions, six survey samples exceeded their limit in more than one team-match in twenty, so their league may have allowed more. Leaving them out gave a pooled odds ratio of 0.69 under three substitutions, against 0.70 with them (Table S23).

The survey's odds ratios barely move between the eras: 0.70 under three substitutions and 0.68 under five. So the extra substitutions changed how many were made, not whom coaches keep on. The landing page of UEFA's September 2026 report on the competition landscape says that clubs now spread match time across more players and use their full substitution allowance. The first holds here: more players share the minutes, which is the part due to who starts in Figure S3A. Loaded starters are still kept on as well.

**Minimum sizes.** The survey needs at least 2000 starter appearances and 100 substitutions per sample. These minimums were set when the survey was first fitted, not before. Doubled, they left 42 and 80 samples under three and five substitutions, with odds ratios of 0.70 and 0.68 (Table S23).

**Rules tied to age.** In some survey leagues, a rule was tied to players' age. It governed the substitutions allowed (kind A), a starting place (B), a place on the pitch (C) or a quota of minutes (D). The K League's under-22 rule, for example, allowed five substitutions only when two under-22 players took part. From mid-2019 to 2022, the Chinese Super League required an under-23 player on the pitch at all times. Under a rule of kind A, the limit varies from team to team. A young starter picked by the rule may be taken off early whatever the starter's load.

Rules of these kinds in force since 2010 were sought in every survey competition's rulebooks, federation notices and press, and found in 36 of them. A sample is flagged when the dates of any of its seasons overlap a rule's: 41 samples in 30 competitions (Table S24). Without them, 37 and 64 samples remained under three and five substitutions, with odds ratios of 0.70 and 0.69 (Table S23), against 0.70 and 0.68 with them. Payments for fielding young players and rules on the bench or on club-trained players were not counted. For some competitions, most of them in Asia and Africa, no rulebook could be read, so a rule there may have been missed.

Only one of the study's own men's samples outside Europe falls under such a rule: South Korea's, the 2024 season. The rule is the K League's under-22 rule above. That sample's starters' gradient is the steepest of the men's samples (Figure S3B). The Mexican sample, 2023–24, falls between Liga MX's quotas of 2018–20 and those from its 2024 opening tournament. So no quota applied to it. The rules, their dates and sources are in the archive.

**The European men's leagues and the survey.** The survey reads the provider's match pages, league by league. The study's European men's leagues come from the provider's public dataset instead. So no match pages were read for them, and the survey leaves them out. Its open circles in Figure S8 are the study's leagues outside Europe.

**The samples.** Table S2 describes every league sample: its seasons, players, starts and appearances from the bench, and the mean minutes of each. It gives how many starts last to the final whistle and the starts per repeat starter. It shows how recent match load is spread, what share of rested appearances the calendar rested, and the players' ages where the source records them.

**Table S2.** Every league sample. Bench apps. are appearances by a substitute who took the field. Load is recent match load in full matches. Starts to the end last to the final whistle. Starts per repeat starter is the mean number of starts of players with at least two. Rested and loaded are the shares of appearances with no load in the window and with at least one full match. Long gaps are the share of team-matches whose club had no league match in the load window. Rest from the calendar is the share of rested appearances whose club had none. Age is at appearance, in years, where the source records dates of birth for at least 95 per cent of appearances. The IQR, the interquartile range, holds the middle half of the ages. Women's Australian rules records the share of the match played, not minutes, and has no starters or substitutes, so its columns for starts and the bench carry dashes.

| League sample | Seasons | Players | Starts | Bench apps. | Minutes per start | Minutes per bench app. | Starts to the end (%) | Starts per repeat starter | Load, mean | Rested (%) | Loaded (%) | Long gaps (%) | Rest from the calendar (%) | Age, median (IQR) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Figure S3A: the split of the gradient* | | | | | | | | | | | | | | |
| France, men, rugby union (Top 14) | 2025/26 | 681 | 5640 | 2846 | 68 | 27 | 51 | 10 | 0.49 | 43 | 26 | 25 | 59 | – |
| Brazil, men (2019) | 2019 | 693 | 8272 | 2163 | 84 | 23 | 73 | 15 | 0.62 | 39 | 42 | 21 | 51 | – |
| Brazil, men (2024) | 2024 | 703 | 8316 | 3468 | 81 | 22 | 58 | 15 | 0.54 | 41 | 31 | 30 | 66 | – |
| Germany, women | 2024/25 | 304 | 2904 | 1095 | 82 | 21 | 62 | 12 | 0.25 | 67 | 16 | 62 | 92 | – |
| England, women | 2024/25 | 297 | 2904 | 1020 | 82 | 21 | 65 | 12 | 0.33 | 58 | 22 | 51 | 89 | – |
| Spain, women | 2024/25 | 412 | 5258 | 2011 | 82 | 21 | 62 | 16 | 0.37 | 51 | 24 | 43 | 84 | – |
| Italy, women | 2024/25 | 245 | 1980 | 757 | 82 | 21 | 61 | 10 | 0.31 | 59 | 20 | 52 | 88 | – |
| Norway, women | 2025 | 247 | 2970 | 967 | 83 | 20 | 67 | 15 | 0.45 | 50 | 29 | 44 | 90 | – |
| Switzerland, women | 2024/25 | 254 | 1957 | 698 | 82 | 22 | 64 | 10 | 0.31 | 59 | 21 | 52 | 88 | – |
| Sweden, women | 2025 | 345 | 3982 | 1296 | 84 | 19 | 67 | 14 | 0.39 | 51 | 27 | 43 | 86 | – |
| England, men | 2017/18 to 2024/25 | 1580 | 65 186 | 19 131 | 84 | 19 | 72 | 51 | 0.51 | 44 | 36 | 34 | 77 | 26.7 (24.0–29.5) |
| Spain, men | 2017/18 to 2024/25 | 1727 | 63 480 | 22 862 | 82 | 20 | 65 | 45 | 0.44 | 47 | 29 | 35 | 75 | 27.4 (24.2–30.4) |
| Italy, men | 2017/18 to 2024/25 | 1769 | 65 215 | 23 440 | 82 | 21 | 65 | 46 | 0.47 | 44 | 31 | 32 | 73 | 26.5 (23.9–29.9) |
| Germany, men | 2017/18 to 2024/25 | 1481 | 52 549 | 18 487 | 83 | 19 | 66 | 44 | 0.44 | 47 | 29 | 36 | 77 | 26.2 (23.4–29.2) |
| France, men | 2017/18 to 2024/25 | 1881 | 59 927 | 20 078 | 83 | 19 | 67 | 39 | 0.45 | 46 | 31 | 35 | 77 | 25.5 (22.7–29.0) |
| Netherlands, men | 2017/18 to 2024/25 | 1762 | 50 434 | 16 513 | 83 | 20 | 68 | 36 | 0.43 | 48 | 30 | 39 | 82 | 24.5 (22.0–27.7) |
| Portugal, men | 2017/18 to 2024/25 | 1994 | 50 538 | 18 282 | 83 | 19 | 65 | 31 | 0.38 | 51 | 26 | 41 | 80 | 25.7 (23.4–28.6) |
| Turkey, men | 2017/18 to 2024/25 | 2044 | 58 169 | 20 353 | 83 | 19 | 67 | 35 | 0.40 | 50 | 28 | 40 | 79 | 28.7 (25.6–31.4) |
| *Figure S3B: within the same starter* | | | | | | | | | | | | | | |
| Spain, men | 2020/21 to 2024/25 | 1365 | 40 753 | 16 669 | 81 | 20 | 60 | 38 | 0.43 | 47 | 26 | 35 | 76 | 27.5 (24.2–30.6) |
| France, men | 2020/21 to 2024/25 | 1401 | 37 129 | 14 155 | 83 | 19 | 63 | 33 | 0.44 | 46 | 28 | 36 | 78 | 25.5 (22.7–29.0) |
| England, men | 2022/23 to 2024/25 | 925 | 23 518 | 8595 | 83 | 18 | 66 | 32 | 0.47 | 44 | 31 | 35 | 79 | 26.3 (23.6–29.4) |
| Italy, men | 2020/21 to 2024/25 | 1448 | 42 294 | 17 291 | 81 | 21 | 60 | 37 | 0.46 | 44 | 28 | 33 | 74 | 26.5 (23.9–29.7) |
| Germany, men | 2020/21 to 2024/25 | 1135 | 32 353 | 12 930 | 82 | 19 | 62 | 35 | 0.42 | 47 | 26 | 37 | 78 | 26.3 (23.6–29.3) |
| Netherlands, men | 2020/21 to 2024/25 | 1300 | 31 954 | 11 954 | 82 | 20 | 64 | 31 | 0.42 | 48 | 27 | 39 | 82 | 24.4 (22.0–27.7) |
| Portugal, men | 2020/21 to 2024/25 | 1540 | 33 122 | 13 540 | 82 | 19 | 61 | 26 | 0.37 | 51 | 23 | 41 | 81 | 25.5 (23.3–28.5) |
| Turkey, men | 2020/21 to 2024/25 | 1653 | 38 985 | 15 192 | 82 | 19 | 64 | 30 | 0.42 | 48 | 27 | 38 | 78 | 28.5 (25.3–31.2) |
| Germany, women | 2024/25 | 304 | 2904 | 1095 | 82 | 21 | 62 | 12 | 0.25 | 67 | 16 | 62 | 92 | – |
| England, women | 2024/25 | 297 | 2904 | 1020 | 82 | 21 | 65 | 12 | 0.33 | 58 | 22 | 51 | 89 | – |
| Spain, women | 2024/25 | 412 | 5258 | 2011 | 82 | 21 | 62 | 16 | 0.37 | 51 | 24 | 43 | 84 | – |
| Italy, women | 2024/25 | 245 | 1980 | 757 | 82 | 21 | 61 | 10 | 0.31 | 59 | 20 | 52 | 88 | – |
| Norway, women | 2025 | 247 | 2970 | 967 | 83 | 20 | 67 | 15 | 0.45 | 50 | 29 | 44 | 90 | – |
| Switzerland, women | 2024/25 | 254 | 1957 | 698 | 82 | 22 | 64 | 10 | 0.31 | 59 | 21 | 52 | 88 | – |
| Sweden, women | 2025 | 345 | 3982 | 1296 | 84 | 19 | 67 | 14 | 0.39 | 51 | 27 | 43 | 86 | – |
| France, women | 2024/25 | 299 | 2772 | 1125 | 82 | 20 | 59 | 12 | 0.25 | 67 | 15 | 62 | 92 | – |
| United States, women | 2024 | 356 | 3872 | 1503 | 82 | 20 | 61 | 13 | 0.41 | 47 | 26 | 40 | 84 | – |
| Japan, men | 2024 | 588 | 8206 | 3145 | 82 | 20 | 61 | 17 | 0.42 | 47 | 27 | 39 | 81 | – |
| Argentina, men | 2024 | 926 | 8291 | 3257 | 81 | 22 | 60 | 12 | 0.35 | 56 | 22 | 44 | 78 | – |
| Mexico, men | 2023/24 | 562 | 6512 | 2569 | 81 | 22 | 60 | 14 | 0.48 | 44 | 29 | 34 | 76 | – |
| Australia, men | 2023/24 | 316 | 3388 | 1252 | 82 | 22 | 63 | 13 | 0.40 | 49 | 26 | 41 | 81 | – |
| United States, men | 2024 | 790 | 10 714 | 3869 | 82 | 21 | 64 | 16 | 0.61 | 32 | 38 | 21 | 62 | – |
| South Korea, men | 2024 | 421 | 4994 | 1985 | 80 | 25 | 60 | 14 | 0.44 | 48 | 27 | 37 | 76 | – |
| Australia, women, Australian rules (2023) | 2023 | 506 | – | – | – | – | – | – | 1.51 | 14 | 70 | 9 | 66 | – |

| League sample | Seasons | Players | Starts | Bench apps. | Minutes per start | Minutes per bench app. | Starts to the end (%) | Starts per repeat starter | Load, mean | Rested (%) | Loaded (%) | Long gaps (%) | Rest from the calendar (%) | Age, median (IQR) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| France, men, rugby union (Top 14) | 2023/24 to 2025/26 | 1038 | 16 387 | 8395 | 67 | 27 | 49 | 20 | 0.48 | 43 | 25 | 26 | 61 | – |

# S3 Models

## Recent match load

Recent match load is the minutes a player recorded in the window before a match, divided by the length of a full match. The window runs from the seventh day before the match to the day before it, both included (Section S2). In football the window is seven days and the match 90 minutes. In the European men's samples only league matches count. Section S6 counts every club match instead. In rugby union the window is two weeks and the match 80 minutes, both in the split of the gradient and in the within-player slope. In women's Australian rules the window is three weeks, and the load is the share of each match played, added over the window. Its gradient uses the log of the share played in place of log minutes.

## The workload gradient

The gradient is the slope of log recorded minutes on recent match load. Each appearance counts at least one minute, the floor, so that its log exists. It is fitted by least squares with standard errors clustered by player. Intervals are Wald intervals, built from each estimate and its standard error. The gradient among all players uses every appearance, and the starters' gradient uses starters alone. In football the model adds four day-of-week terms, the sine and cosine of one and two cycles per week. Rugby union and Australian rules play on three or four days of the week, so the four terms cannot be told apart there. The Top 14's gradient of Table S6 uses one indicator per day played in their place. So does every rugby union and Australian rules refit in Section S6. The Australian rules gradient of Table S6 was first estimated without day-of-week terms.

A coach substitutes several players in one match, so standard errors were also clustered by player and by team-match together. In the eight European men's leagues, these standard errors were 1.00 to 1.15 times the player-clustered ones for the gradients among all players and among starters. For the within-player slope they were 0.98 to 1.07 times the player-clustered ones. With them, the within-player slope's interval stayed below the starters' gradient in all eight leagues.

## Splitting the gradient

Let $x$ be recent match load and $y$ log minutes. The two roles $g$, starters and substitutes, have shares $\pi_g$. Then the slope of $y$ on $x$ is exactly

$$\frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)} \;=\; \frac{\sum_g \pi_g \,\mathrm{Cov}_g(x,y)}{\mathrm{Var}(x)} \;+\; \frac{\sum_g \pi_g (\bar{x}_g - \bar{x})(\bar{y}_g - \bar{y})}{\mathrm{Var}(x)}.$$

The first term is the gradient within starters and within substitutes. The second is the part due to who starts. The identity is Kitagawa's split of a difference between two rates into a composition part and a rate part (Journal of the American Statistical Association 1955). It holds in every sample, up to the computer's rounding. These slopes carry no day-of-week terms, so they differ slightly from the adjusted gradients of Figure S3B.

The share due to who starts is an estimate of each league's share, so it is given a 95% interval from 1000 bootstrap draws over players (Table S5). A substitute who comes on late records a minute or two, and the log of the one-minute floor is zero. So very short appearances carry large weight. The floor was therefore also raised to five and ten minutes. The share ran from 74% to 93% with the floor at five minutes, and from 74% to 93% at ten. On the minutes scale, without logs, it ran from 73% to 91% (Table S13).

Let $m$ be the mean log minutes of each appearance's role, starter or substitute. The part due to who starts is the covariance of load with $m$, over $\mathrm{Var}(x)$. By the law of total covariance it divides exactly in two:

$$\mathrm{Cov}(x,m) \;=\; \mathrm{E}_i\left[\,\mathrm{Cov}_i(x,m)\right] \;+\; \mathrm{Cov}(\bar{x}_i, \bar{m}_i),$$

where $i$ is the player. The first part is the same player picked again, or dropped, as the load changes. The second is players who start more also carrying more load. Both are divided by $\mathrm{Var}(x)$, so the three parts add up to the gradient. One bootstrap over players gives the interval of the same player's share of the part due to who starts.

The same player's share of the part due to who starts was 19% to 59% over the 18 samples. In the eight European men's leagues, whose samples run over eight seasons, it was 51% to 59% (Table S5). The gradient within the same player, over all appearances there (starts and bench appearances together), was 0.23 to 0.37: 52% to 60% of the gradient. Per extra full match, the same player was 12 to 19 percentage points more likely to start. So the finding that minutes barely change within the same player holds once the player starts, not before.

The same identity was applied within each team sheet, with each team-match's own means removed. Every comparison is then between players of one club in one match, who share the schedule, the opponent, the score

and the weather. The gradient within team sheets was 0.28 to 1.39. Who starts explained 78% to 93% of it (Tables S5 and S13). In the eight European men's leagues, the gradient rose within team sheets. Among all players it was 0.41 to 0.62, and 0.83 to 1.29 within team sheets. The starters' gradient was 0.019 to 0.029, and 0.069 to 0.097 within team sheets. So the calendar dilutes the gradient: it rests whole teams at once. It does not create the gradient.

Team selection, the coach's choice of who starts and who stays on, is not the selection Darwin made famous. It has a selector, and the selector writes the team sheet.

## The within-player slope

The within-player slope is the least-squares slope of a starter's log minutes on recent match load, after each player's own averages are removed. Only starters with at least two starts enter, and a sample needs at least ten such players. Its standard error comes from 1000 bootstrap draws over players. The model carries no day-of-week terms. When they were added, they changed a within-player slope by at most 0.003 (Table S11). The between-player part is the starters' gradient refitted on the same appearances minus the within-player slope. One bootstrap over players gives both slopes in each draw, and so the part's own 95% interval.

For the eight European men's leagues, the slope uses the seasons under five substitutions. The starters' gradient first estimated (Table S6) spans the whole period from July 2017 to April 2025. For the Top 14, the starters' gradient was first estimated on one season and a three-week window. It is refitted here on the slope's own three seasons and two-week window, so the split and the within-player comparison share one rugby window. Figure S3B draws every starters' gradient refitted on exactly the within-player appearances (Table S11). In all 25 samples, the within-player slope's 95% interval lay wholly below the starters' gradient (Table S14).

Sports that grew from a common ancestor can support one another's results, but they are not repeats of one test. So the slopes are pooled over football alone. Over football, the pooled slope was +0.004 (95% CI +0.001 to +0.006; 23 samples). It was +0.005 over men's football and +0.001 over women's. For comparison, the pool over every sport gave +0.003 (Table S23).

## Keeping loaded starters on

A starter's chance of being substituted is modelled in each of three periods of the match: to the 60th minute, to the 75th and to the end. The substitution model is a discrete-time hazard: the chance of coming off in a period, for a starter still on the field at its start. It is fitted by logistic regression. The model adjusts for the period and the day of the week. Its coefficient on recent match load is the log odds ratio per extra full match. Standard errors are clustered by player. In the eight European men's leagues, standard errors clustered by player and by team-match together were 0.76 to 0.97 times these (Table S17). There are no player, league or season effects, so it compares different starters.

Goalkeepers were 9% of starts in the eight European men's leagues. Of their starts, 0.6% to 2.1% ended in a substitution, against 26% to 44% of outfield starts. Their mean recent load was 0.50 to 0.71 full matches, against 0.44 to 0.59 for outfield starters. In the eight European men's leagues the model was therefore refitted without goalkeepers, with outfield position (defender, midfielder, forward), and within the same player (Table S17). The worldwide survey has no position term (Table S3), although one starter in eleven is a goalkeeper.

The within-player model is the same hazard with one intercept per player, fitted by maximum likelihood with the intercepts profiled out. It compares each player's weeks only with the same player's other weeks. One intercept per player carries a bias of order one over the player's number of person-periods. A person-period is one period of one start. So the within-player model was also fitted by conditional logistic regression, stratified by player. That fit removes the intercepts exactly. The conditional likelihood is computed exactly, by a recursion over the ways all of a stratum's events could fall among its rows (Gail, Lubin and Rubinstein, Biometrika 1981). It uses no approximation for tied events. Ties are the rule in the team-sheet fit below, whose strata are one team-match and period: several substitutions often fall in one stratum. An approximation would bias the estimate towards no association.

Within the same team sheet, the model is fitted by conditional logistic regression stratified by team-match and period. It compares the starters of one club in one match who are still on the field at the period's start. They share the schedule, the opponent, the weather and the score at the period's start. So the fit also answers whether the score explains the association (Table S18). The model was also fitted with one load coefficient per period, and with the score at each period's start (level, leading or trailing). The score is read from the match events, whose goals add up to the final score in 100.0% of team-matches (Table S18).

**Table S3.** How the worldwide survey differs from the study samples.

| | Study samples | Worldwide survey |
|---|---|---|
| Source | The provider's dataset for the European leagues, match pages elsewhere | Match pages, every league in the continental indexes |
| Substitution limit | The league's adoption date | Substitutes used per team-match |
| Seasons | Each sample's own period | Every season that passed the quality checks, pooled per league and limit |
| Minimum sizes | None | 2000 starter appearances and 100 substitutions |
| Model | As in Section S3 | The same. Appearances above 90 minutes were to be dropped, but no survey appearance exceeded 90 minutes |
| Position | In the eight European men's leagues, refitted without goalkeepers and with outfield position | None |
| Mexico | The opening and closing tournaments merged | The closing tournament alone |

## The observed and predicted fall in substitutions

The observed fall is how much less often starters are substituted per extra full match of recent match load. It is the unadjusted linear slope of being substituted on that load among starters. A model with one coefficient $\theta$ on load in every period of the match predicts it. If the chance of lasting the match is $1 - s = e^{-H}$, with the cumulative hazard $H = H_0 e^{\theta x}$ proportional in load, then $ds/dx = \theta H(1 - s) = \theta(1 - s)\ln[1/(1 - s)]$, the fall first reported, at the sample's substitution rate $s$. That formula is exact for a coefficient on the complementary log-log scale, while $\theta$ was fitted with a logit link. The fall was therefore also computed from the fitted logistic model itself, as the average over starters of each starter's own change in the chance of being substituted. It was computed from a complementary log-log refit as well.

The 15 samples of Table S4 include no European men's league. Across them, the observed fall averaged 0.88 of the logistic model's prediction (95% CI 0.86 to 0.91, a $t$-interval; coefficient of variation 5.8 per cent). It averaged 0.88 of the log-log refit's prediction, against 0.81 of the formula as first estimated. Part of the difference from the formula was the link.

**Table S4.** The observed and predicted fall in substitutions with recent match load. The log-odds is the sample's own substitution coefficient. The last three columns divide the observed fall by each prediction.

| League sample | Log-odds | Observed fall | Predicted, hazard formula | Observed over formula | Observed over logistic model | Observed over log-log refit |
|---|---|---|---|---|---|---|
| Brazil, men (2019) | −0.389 | −0.074 | −0.089 | 0.83 | 0.89 | 0.88 |
| Brazil, men (2024) | −0.286 | −0.076 | −0.091 | 0.84 | 0.93 | 0.93 |
| Germany, women | −0.167 | −0.036 | −0.049 | 0.72 | 0.79 | 0.80 |
| England, women | −0.348 | −0.087 | −0.098 | 0.89 | 0.97 | 0.97 |
| Spain, women | −0.399 | −0.094 | −0.119 | 0.79 | 0.87 | 0.87 |
| Italy, women | −0.301 | −0.072 | −0.090 | 0.80 | 0.88 | 0.89 |
| Norway, women | −0.370 | −0.079 | −0.098 | 0.80 | 0.87 | 0.86 |
| Switzerland, women | −0.258 | −0.060 | −0.073 | 0.81 | 0.88 | 0.89 |
| Sweden, women | −0.443 | −0.088 | −0.118 | 0.75 | 0.81 | 0.81 |
| France, women | −0.216 | −0.050 | −0.067 | 0.75 | 0.83 | 0.83 |
| United States, women | −0.458 | −0.113 | −0.138 | 0.82 | 0.91 | 0.90 |
| Japan, men | −0.536 | −0.132 | −0.160 | 0.82 | 0.92 | 0.91 |
| Argentina, men | −0.271 | −0.072 | −0.083 | 0.87 | 0.96 | 0.96 |
| Mexico, men | −0.264 | −0.068 | −0.081 | 0.83 | 0.91 | 0.91 |
| Australia, men | −0.545 | −0.125 | −0.160 | 0.78 | 0.86 | 0.85 |

## Japan and market value

In Japan under five substitutions, loaded starters also came off later. The substitution minute rose by 0.043 in log minutes per extra full match (95% CI 0.023 to 0.063). As the first author's nine-sport preprint noted, adding market value left that unchanged. That rests on one season, 2024. The market-value test uses the substituted starters of 2024 who had a recorded value, 93 per cent of them. Starters without one were excluded. Adding the standardised log market value and the day-of-week terms gave a slope of 0.043 (0.023 to 0.063). Value itself had $P = 0.19$. Within a match, the minute a loaded starter comes off is still the coach's choice of who stays

on. The analysis cannot say why loaded starters come off later.

## Injury-coded substitutions

A substitution is injury-coded when the provider's match report gives the injury code. The provider uses one code for injury and concussion alike, so a concussion substitution is injury-coded. The reasons are read from the provider's match events in the snapshot (Section S2). A league-season enters only if at least 90 per cent of its substitutions carry some reason, and only if it allowed five substitutions throughout. So the 27 league-seasons that entered, from 2020/21 to 2025/26, have near-complete reasons, not complete ones (Table S8). A plan written before any of them was compared chose them. The archive holds the plan. An earlier version of this comparison used the seven league-seasons of a separate collection of match reports (Section S6). The reason code agreed with the match events' own description: 100.0% of injury-coded substitutions were described as injuries, and 100.0% of those described as injuries carried the code. Both fields come from the provider's own match page, so this is internal consistency, not a validation. The video benchmark and the recorded absences below are the outside checks on the outcome.

The rate per starter appearance divides the injury-coded substitutions of listed starters by the starter appearances. Each match whose line-ups list twenty-two starters gives twenty-two starter appearances. The seasons are pooled by restricted maximum likelihood: 0.0323 per starter appearance (95% CI 0.0309 to 0.0336), about one start in 31. Across league-seasons it ran from 0.024 to 0.037.

For the comparison of denominators, every appearance with recorded minutes enters. Loaded, part-loaded and rested players are compared per appearance, per recorded hour and per starter hour. Per appearance, the comparison estimates the risk per match played. Per recorded hour, it estimates the rate per minute on the field. That rate assumes every minute carries the same risk. Like any hazard, it also compares only those still on the field (Hernán, Epidemiology 2010). Per starter hour, it estimates the same rate among starters. Counting only starters' hours can also bias that rate, because earlier load shapes who is picked to start (Hernán and colleagues, Epidemiology 2004).

The assumption that every minute carries the same risk is doubtful. In elite men's football, more anterior cruciate ligament injuries occur in the first 45 minutes of play than later (Buckthorpe and colleagues, Knee Surgery, Sports Traumatology, Arthroscopy 2025). In rugby union, starters and replacements had the same injury incidence per hour overall, but replacements' was lower in the third and fourth quarters (Roberts and colleagues, Sports Medicine 2024). Counting events over accumulated time assumes a constant risk, a practice medicine has long questioned (Windeler and Lange, BMJ 1995).

The ratios are Mantel–Haenszel ratios stratified by league-season, with Greenland–Robins intervals. Poisson regressions with league-season terms and standard errors clustered by player check them (a modified Poisson model for the risk per appearance). Rested players are also split by the cause of their rest (Section S2): 77% of rested appearances, and 85% of rested starts, were rested by the calendar. The calendar's weeks are further split by the gap since the club's previous league match. The groups are gaps of 8 to 13 days, gaps of 14 to 60 days, and gaps beyond that or no previous league match on file. Every contrast is made with each of these groups as the reference. Absence from the club's previous match is split by the provider's injury histories. Either a recorded injury covered the date of that match, or one ended in the 28 days before this one, or none is recorded.

Loaded, part-loaded and absent players are each compared with players rested by the calendar. Loaded and part-loaded players are also compared with those absent from the club's previous match. The ratios against all rested players are also stratified by league-season and whether the club played in the window. They are stratified by team-match as well. Within a team-match, every rested player missed the club's previous match (Table S20).

Among starters, a discrete-time hazard of the injury-coded substitution was fitted over the three periods of the match. Other substitutions end the time at risk. The model has outfield position and league-season terms, with standard errors clustered by player. Its odds ratio was 0.84 for loaded against rested starters (95% CI 0.79 to 0.89) and 1.11 for part-loaded starters (1.05 to 1.18). With a complementary log-log link, the hazard ratio was 0.84 for loaded starters.

Within the same starters, a conditional Poisson model compares a starter's own loaded, part-loaded and rested weeks. It is stratified by player, with the log of the starter hours as its offset. It uses the 6859 injury-coded substitutions of the 2695 starters with at least one. Only starters with an event contribute, each against the same starter's other weeks. So the comparison works as a self-controlled case series (Petersen, Douglas and Whitaker, BMJ 2016). Its key assumption is that an event does not change later exposure, here the later load (Whitaker and colleagues, Statistics in Medicine 2018; Farrington and colleagues, Biostatistics 2009). That assumption is at risk here, because an injury is followed by absence and part-loaded returns.

So the model was refitted under the checks that the self-controlled case series design calls for (Section S6). The first counts only each starter's first injury-coded substitution and keeps every start. Cutting a starter's history at the first event would place that event at the end of a run of starts, where load accumulates, and bias the

comparison upwards. The second uses strata by player and league-season, which remove trends across seasons. The injury-coded rate differed between league-seasons, so this fit is the main text's within-starter estimate. The risk of a further injury stays raised for about twelve weeks after a return from an acute hamstring injury (Zhang and colleagues, Orthopaedic Journal of Sports Medicine 2026). So the third leaves out the 28 and the 84 days after an injury-coded substitution.

The hybrid model puts within-starter and across-starter comparisons into one Poisson fit (Curran and Bauer, Annual Review of Psychology 2011). Each load group enters as the starter's share of weeks in that group (across starters) and as the week's departure from that share (within the starter). The fit has league-season terms, the log of the starter hours as the offset and standard errors clustered by player. The difference of the two coefficients, with its interval from their joint covariance, is the selection among starters.

A share taken over all of a starter's starts includes the starts after the starter's injuries. An injury interrupts a run of loaded starts: the player misses matches, and the next starts are rested, absent or part-loaded. So the more often a starter is injured, the lower that share, whatever the starter's standing. An across-starter term built on that share therefore holds part of the outcome it is compared with. Wang, Kaufman and Shrier name this error for training load: an exposure defined with information from after the event (Journal of Clinical Epidemiology 2024, cited in the main text). Target-trial thinking names it in general (Hernán and colleagues, Journal of Clinical Epidemiology 2016).

The model was therefore refitted with the share taken from each starter's earlier starts only. Such a share is a pre-sample mean in the sense of count models (Blundell, Griffith and Windmeijer, Journal of Econometrics 2002). The refits use the starts with at least five and at least 10 earlier ones. Each is set beside the share over all starts, fitted on the same starts.

**What the injury code counts.** The code is broader than injuries that force a substitution. In the provider's injury histories, 39% of injury-coded substitutions were followed within a week by the start of a recorded absence, against 4% of tactical ones. The histories record longer injuries better than shorter ones, so this share is a lower bound. The injury-coded rate is close in size to UEFA's figure for every time-loss match injury (Ekstrand and colleagues, British Journal of Sports Medicine 2021). Injuries that force a substitution should be fewer than all time-loss match injuries. So the closeness is no validation: the reason also counts withdrawals that are not time-loss injuries.

The Premier League's on-field injuries that required a substitution numbered 135 on video in 2018/19 (Argibay-González and colleagues, International Journal of Environmental Research and Public Health 2022). The video count, checked against club records, was 0.36 a match. The provider coded 192 substitutions in that league and season as injuries, 1.4 times the video count. Yet 69% of that season's substitutions carry no reason at all. With near-complete reasons, the 27 league-seasons have 0.54 to 0.86 injury-coded substitutions a match (Table S8), 1.5 to 2.4 times the video's count a match. The code therefore also covers precautionary and minor withdrawals. A coach's decisions may shape them.

The outcome also misses injuries. Few contact head and neck injuries end in an immediate substitution, so the outcome under-represents head injury. Public records capture prominent players' injuries better. So how fully an injury is recorded may rise with the player's standing.

No video audit of the study seasons was feasible; the video benchmark above is of one season with most reasons missing. An injury-coded substitution is therefore an index test with unknown capture. So is footage for concussion, where most visible signs are specific but insensitive (Townsend and colleagues, British Journal of Sports Medicine 2025). The outcome's capture may also differ between the groups compared. A team that has made its last permitted substitution cannot replace an injured player. Loaded starters are the ones still on the field late in matches, so their injuries could go unrecorded more often. One check in Section S6 tests this: it keeps only the team-matches with a substitution left. The comparison was repeated with a narrower outcome: an injury-coded substitution followed within a week by a recorded absence. It was first a check. After review, the main text gives every football injury result on both outcomes. Among outfield players, a recorded absence followed 41% of loaded players' injury-coded substitutions, against 34% to 39% in the other load groups. It followed 38% of outfield starters' injury-coded substitutions, against 58% of outfield substitutes'. So the injury reason is confirmed at least as often after a heavy week. An absence can also be a staff decision, and staff might record one more readily after a heavy week.

Rules on treatment also change which knocks become substitutions. From 2026/27 the Premier League keeps a treated player off the field for one minute. Major League Soccer made treated players leave the field. Stoppages for players down then fell, and far more of them ended in a substitution (the league's data, reported in 2024). The rates here come from seasons before such rules. A reader who applies them to later seasons should allow for that.

**Added time and pauses.** The provider's minutes and FBref's stop each half at 45 minutes. So first-half added time is missing for everyone on the field at half-time, and second-half added time for everyone at the

final whistle. A substitution made in added time is recorded at the 90th minute. That gives the substitute one minute, the floor. A public analysis of these providers' minutes measured the time they leave out (McKinley, American Soccer Analysis, December 2024). It put the unrecorded share at about a third of substitutes' time, against about a tenth of starters'. The per-hour ratios were therefore recomputed with added time given to every appearance on the field at the final whistle. A fuller bound also gave it to every appearance on the field at half-time (71% of appearances).

International players are recorded as rested after an international window, when they played for their countries. A pause here is 9 to 60 days in which none of the five leagues played. The 27 league-seasons held 24 pauses: the international windows, the 2022 World Cup and the short summer break of 2020. The first league match of each club after each pause was also left out (11% of appearances).

Section S6 gives these checks and those for detection and return from injury.

### The next start

A starter substituted by the 60th minute started the club's next league match with an adjusted probability of 42%, against 85% after finishing it. The risk ratio was 0.496 (95% CI 0.490 to 0.502). The probabilities were 46% and 85% without the substitutions the match events describe as injuries. So most of the drop remains without injuries. The model gives the chance of a start after an early substitution. It does not give the chance of an injury given an absence. The records cannot estimate that.

The model is a logistic regression on the seasons under five substitutions of 14 European leagues. The outcome is whether a starter started the club's next league match of the same season. The comparison is a substitution at or before the 60th minute against finishing the match. A second term marks a later substitution. The model adjusts for recent match load, familiarity, league and season, with standard errors clustered by player. Familiarity is the mean number of matches started together with the club's other starters over the previous year. It enters the model standardised.

The odds ratio of a common outcome overstates the risk ratio, so the fitted model is read as probabilities by marginal standardisation. Every analysed start is predicted twice, once as an early substitution and once as a finished match. The predictions are averaged, with delta-method intervals from the clustered covariance.

Before adjustment, 39% of early-substituted starters and 86% of finishers started the next match. After it, the chance of the next start was 42.9 percentage points lower after an early substitution (95% CI 42.3 to 43.4). The odds ratio from the one fit was 0.121 (0.118 to 0.124). Each league is also fitted alone. The pooled odds ratio of the 14 leagues was 0.12, and their risk ratios ran from 0.44 to 0.57 (Table S10).

Goalkeepers almost never come off and almost always start again, so the model was refitted without them. The probabilities were then 42% and 84% (risk ratio 0.497, 0.491 to 0.503). Of the 44 633 goalkeepers' starts, those that ended early were followed by a start 28% of the time, and those finished 91%.

Rotation is routine in a cup tie a few days later. So the model added the club's next match in any competition (league, 85% of starts; continental, 8%; cup or other, 7%). It also added the days until that match and until the next league match, in bands (up to three, four to five, six to seven, eight or more). The risk ratio was 0.497 (0.491 to 0.504). Without goalkeepers as well, the probabilities were 42% and 84% (risk ratio 0.499, 0.493 to 0.505), close to those of the main fit.

A conditional logistic regression stratified by player compares each player's own starts. Its odds ratio was 0.127 (0.123 to 0.131). With one intercept per player, it was 0.118 (0.115 to 0.122). The fits within the same player use the starts of the players whose next start varies. On those starts, the probabilities were 42% and 85% (risk ratio 0.499). The model without injuries removes every substitution whose match-event description names an injury. That field agrees with the reason code wherever both exist (above). The model gave a risk ratio of 0.544 (0.537 to 0.551). Its odds ratio was 0.143: the odds were 7.0-fold lower.

### Missing data

Missing records were never imputed. Seasons without line-ups and duplicate player-date entries were dropped before any analysis.

## S4 Pre-set criteria

Two criteria were written into the analysis script that tests them. Both were saved with it to the project's repository on 31 August 2026. There was no public protocol, so the repository cannot show that the criteria came before the results. Every other analysis is descriptive or exploratory. That includes the later checks that were written down before they ran (Sections S6 and S10).

The first criterion was that the range of within-player slopes be at most half the range of the starters' gradients. The ratio of ranges was 0.48, so it held narrowly. A second reading was made after the fact. It

compared standard deviations instead of ranges, and it removed sampling error from the within-player slopes first. Sampling error made up 0.54 of their variance. The ratio of standard deviations was 0.33, also under half.

Both ratios use the starters' gradients as first estimated. The Top 14's gradient was first estimated on one season and a three-week window. It was later refitted on three seasons and the two-week window (Section S3). The Top 14 lies at neither end of the range of gradients. So the refit leaves the ratio of ranges and the verdict unchanged. It moves the ratio of standard deviations only in its fourth decimal.

The second criterion was that, across the men's football leagues, the between-player part would rise with how firmly a league keeps loaded starters on. That firmness was measured by the league's substitution coefficient. A more negative coefficient means loaded starters are kept on more firmly, so its sign was reversed. The test was a Spearman correlation, which compares the leagues' ranks on the two measures. It needed a two-sided p below 0.05. Over 14 leagues, the correlation was positive, at 0.43. But p was 0.12, so the criterion failed. The 95% CI, by Fisher's $z$, ran from −0.13 to 0.78. It holds both no link and a strong positive one, so the data cannot tell either way.

# S5 Results by league

**Table S5.** The split of the gradient among all players. Gradients are unadjusted slopes in log recorded minutes per extra full match: among all players, among starters alone, and among all players within team sheets. The first has 95% intervals from 1000 bootstrap draws over players. The share of the gradient due to who starts is in per cent with its interval. So is the same player's share of that part. That share comes from the same player being picked again after playing. The rest of that part comes from differences between players.

| League sample | Window (days) | Appearances | All players (95% CI) | Among starters | Within team sheets | Share from who starts, % (95% CI) | Same player, % of it (95% CI) |
|---|---|---|---|---|---|---|---|
| France, men, rugby union (Top 14) | 14 | 8486 | 0.173 (0.148 to 0.197) | 0.036 | 0.284 | 73 (66 to 80) | 19 (9 to 28) |
| Brazil, men (2019) | 7 | 10 435 | 0.328 (0.299 to 0.359) | 0.024 | 0.561 | 90 (87 to 93) | 33 (28 to 37) |
| Brazil, men (2024) | 7 | 11 784 | 0.396 (0.367 to 0.426) | 0.033 | 0.729 | 87 (84 to 90) | 39 (34 to 43) |
| Germany, women | 7 | 3999 | 0.431 (0.364 to 0.500) | 0.022 | 1.259 | 84 (78 to 89) | 30 (21 to 39) |
| England, women | 7 | 3924 | 0.461 (0.393 to 0.527) | 0.034 | 1.136 | 86 (81 to 91) | 36 (28 to 44) |
| Spain, women | 7 | 7269 | 0.450 (0.396 to 0.509) | 0.033 | 0.988 | 84 (80 to 87) | 26 (20 to 32) |
| Italy, women | 7 | 2737 | 0.458 (0.375 to 0.535) | 0.028 | 1.077 | 87 (82 to 93) | 28 (19 to 36) |
| Norway, women | 7 | 3937 | 0.399 (0.334 to 0.463) | 0.027 | 1.086 | 87 (83 to 91) | 30 (23 to 38) |
| Switzerland, women | 7 | 2655 | 0.436 (0.358 to 0.513) | 0.023 | 1.095 | 90 (84 to 96) | 35 (26 to 43) |
| Sweden, women | 7 | 5278 | 0.562 (0.485 to 0.642) | 0.041 | 1.395 | 87 (84 to 90) | 33 (27 to 38) |
| England, men | 7 | 84 317 | 0.453 (0.431 to 0.477) | 0.019 | 0.974 | 89 (88 to 90) | 57 (55 to 60) |
| Spain, men | 7 | 86 342 | 0.414 (0.394 to 0.435) | 0.024 | 0.827 | 84 (83 to 86) | 53 (51 to 56) |
| Italy, men | 7 | 88 655 | 0.449 (0.428 to 0.470) | 0.027 | 0.883 | 85 (84 to 86) | 55 (53 to 57) |
| Germany, men | 7 | 71 036 | 0.525 (0.498 to 0.552) | 0.026 | 1.073 | 89 (88 to 90) | 57 (55 to 60) |
| France, men | 7 | 80 005 | 0.484 (0.462 to 0.508) | 0.026 | 0.983 | 87 (86 to 88) | 51 (49 to 53) |
| Netherlands, men | 7 | 66 947 | 0.568 (0.541 to 0.593) | 0.022 | 1.268 | 92 (91 to 93) | 58 (55 to 60) |
| Portugal, men | 7 | 68 820 | 0.624 (0.599 to 0.649) | 0.029 | 1.289 | 90 (89 to 91) | 58 (55 to 60) |
| Turkey, men | 7 | 78 522 | 0.588 (0.562 to 0.615) | 0.023 | 1.210 | 89 (88 to 90) | 59 (57 to 61) |

**Table S6.** The within-player slopes. Repeat starters are players with at least two starts. The starters' gradient is each sample's own gradient among starters as first estimated, with its interval. For the Top 14, the table gives the gradient refitted on its within-player appearances with the two-week window. The pre-set criteria use every gradient as first estimated, so for the Top 14 they use the earlier value (Section S4). The standard error of the within-player slope is from a bootstrap over players.

| League sample | Repeat starters | Starters' gradient (95% CI) | Within-player slope | SE |
|---|---|---|---|---|
| Spain, men | 1082 | 0.026 (0.023 to 0.030) | 0.004 | 0.002 |
| France, men | 1113 | 0.025 (0.021 to 0.029) | 0.003 | 0.002 |
| England, men | 732 | 0.020 (0.017 to 0.023) | 0.007 | 0.002 |
| Italy, men | 1127 | 0.030 (0.026 to 0.034) | 0.008 | 0.002 |
| Germany, men | 914 | 0.029 (0.025 to 0.033) | 0.005 | 0.002 |
| Netherlands, men | 1033 | 0.024 (0.020 to 0.027) | 0.007 | 0.002 |
| Portugal, men | 1254 | 0.031 (0.027 to 0.035) | 0.011 | 0.002 |
| Turkey, men | 1309 | 0.026 (0.022 to 0.030) | 0.007 | 0.002 |
| Germany, women | 235 | 0.027 (0.010 to 0.043) | −0.006 | 0.007 |
| England, women | 249 | 0.033 (0.019 to 0.047) | 0.007 | 0.007 |
| Spain, women | 328 | 0.038 (0.027 to 0.050) | −0.005 | 0.005 |
| Italy, women | 192 | 0.030 (0.011 to 0.049) | −0.002 | 0.008 |
| Norway, women | 204 | 0.029 (0.018 to 0.041) | 0.006 | 0.005 |
| Switzerland, women | 199 | 0.027 (0.009 to 0.045) | 0.000 | 0.008 |
| Sweden, women | 275 | 0.045 (0.032 to 0.058) | 0.016 | 0.006 |
| France, women | 235 | 0.019 (0.002 to 0.037) | −0.008 | 0.008 |
| United States, women | 293 | 0.033 (0.018 to 0.048) | −0.006 | 0.007 |
| Japan, men | 478 | 0.055 (0.045 to 0.065) | 0.010 | 0.004 |
| Argentina, men | 713 | 0.029 (0.018 to 0.041) | −0.009 | 0.005 |
| Mexico, men | 456 | 0.023 (0.013 to 0.034) | −0.011 | 0.005 |
| Australia, men | 250 | 0.049 (0.030 to 0.067) | 0.000 | 0.007 |
| United States, men | 663 | 0.035 (0.027 to 0.042) | 0.000 | 0.003 |
| South Korea, men | 347 | 0.075 (0.059 to 0.092) | 0.003 | 0.006 |
| Australia, women, Australian rules (2023) | 476 | 0.072 (0.062 to 0.082) | −0.006 | 0.004 |
| France, men, rugby union (Top 14) | 806 | 0.041 (0.032 to 0.050) | 0.002 | 0.004 |

**Table S7.** The substitution coefficient in each of the 15 study samples under three substitutions, with standard errors clustered by player. Pooled by restricted maximum likelihood, the coefficient is −0.353 (95% CI −0.376 to −0.331; $\tau = 0.024$). That is an odds ratio of 0.70 (0.69 to 0.72). The worldwide survey found the same level.

| League | Substitution coefficient | SE |
|---|---|---|
| Spain | −0.320 | 0.032 |
| France | −0.414 | 0.032 |
| England | −0.310 | 0.023 |
| Italy | −0.335 | 0.032 |
| Germany | −0.438 | 0.037 |
| Netherlands | −0.323 | 0.034 |
| Portugal | −0.338 | 0.036 |
| Turkey | −0.360 | 0.034 |
| Greece | −0.314 | 0.035 |
| Denmark | −0.368 | 0.042 |
| Brazil | −0.389 | 0.044 |
| Argentina | −0.326 | 0.053 |
| Australia | −0.396 | 0.079 |
| Japan | −0.387 | 0.060 |
| Mexico | −0.376 | 0.054 |

**Table S8.** The league-seasons of the injury comparison, and the Premier League's 2017/18 and 2018/19 seasons. Reasons given is the share of substitutions carrying any reason. Injury-coded per match counts the substitutions the provider codes as injuries. The injury-coded rate is per starter appearance (Section S3). The two earlier seasons miss too many reasons to enter the comparison. For 2018/19, a video count found 135 injuries requiring a substitution. Section S3 uses it as a benchmark. The team's fifth counts the injury-coded substitutions made as the team's last permitted one. After it, the team could not replace a further injured player. These were 10% of injury-coded substitutions. Of team-matches, 60% used every substitution and 0.6% used more, as a concussion substitute allows.

| League | Season | Matches | Reasons given | Injury-coded per match | Injury-coded per start | Injury-coded substitutions | Of them, the team's fifth |
|---|---|---|---|---|---|---|---|
| Spain | 2020–21 | 380 | 0.981 | 0.60 | 0.0261 | 220 | 20 |
| Spain | 2021–22 | 380 | 1.000 | 0.79 | 0.0345 | 298 | 41 |
| Spain | 2022–23 | 380 | 0.973 | 0.81 | 0.0354 | 307 | 32 |
| Spain | 2023–24 | 380 | 0.981 | 0.80 | 0.0352 | 305 | 46 |
| Spain | 2024–25 | 380 | 0.997 | 0.79 | 0.0349 | 300 | 41 |
| Spain | 2025–26 | 380 | 0.997 | 0.71 | 0.0306 | 271 | 35 |
| France | 2020–21 | 380 | 0.985 | 0.54 | 0.0240 | 207 | 17 |
| France | 2021–22 | 380 | 0.998 | 0.62 | 0.0269 | 232 | 20 |
| France | 2022–23 | 380 | 1.000 | 0.75 | 0.0337 | 285 | 24 |
| France | 2023–24 | 306 | 1.000 | 0.66 | 0.0293 | 203 | 19 |
| France | 2024–25 | 306 | 1.000 | 0.72 | 0.0307 | 219 | 23 |
| France | 2025–26 | 306 | 1.000 | 0.69 | 0.0300 | 209 | 24 |
| England | 2022–23 | 380 | 0.902 | 0.72 | 0.0317 | 273 | 17 |
| England | 2023–24 | 380 | 0.997 | 0.86 | 0.0374 | 328 | 24 |
| England | 2024–25 | 380 | 1.000 | 0.68 | 0.0294 | 259 | 27 |
| England | 2025–26 | 380 | 1.000 | 0.72 | 0.0310 | 272 | 19 |
| Italy | 2020–21 | 380 | 0.998 | 0.76 | 0.0331 | 285 | 18 |
| Italy | 2021–22 | 380 | 0.979 | 0.85 | 0.0374 | 319 | 24 |
| Italy | 2022–23 | 380 | 0.993 | 0.83 | 0.0360 | 317 | 46 |
| Italy | 2023–24 | 380 | 0.965 | 0.81 | 0.0350 | 307 | 42 |
| Italy | 2024–25 | 380 | 0.999 | 0.81 | 0.0343 | 308 | 40 |
| Italy | 2025–26 | 380 | 1.000 | 0.78 | 0.0344 | 297 | 43 |
| Germany | 2020–21 | 306 | 1.000 | 0.76 | 0.0327 | 232 | 19 |
| Germany | 2021–22 | 306 | 0.994 | 0.76 | 0.0330 | 232 | 18 |
| Germany | 2022–23 | 306 | 0.997 | 0.71 | 0.0306 | 212 | 20 |
| Germany | 2023–24 | 306 | 0.950 | 0.76 | 0.0331 | 232 | 27 |
| Germany | 2025–26 | 306 | 0.974 | 0.75 | 0.0327 | 229 | 25 |
| England | 2017–18 | 380 | 0.427 | 0.60 | – | – | – |
| England | 2018–19 | 380 | 0.310 | 0.51 | – | – | – |

**Table S9.** Injury-coded substitutions by recent match load, with the load counted from league matches and from every club match. Ratios are against rested players (Mantel–Haenszel, stratified by league-season). In the last rows, rest is split by its cause and each ratio is against players rested by the calendar. Loaded players, mostly starters, played 81 minutes per appearance. Rested players, many of them substitutes, played 60. With every club match counted, the loaded-to-rested ratio was 1.20 per appearance (95% CI 1.14 to 1.27) and 0.90 per 1000 hours (0.85 to 0.95). It was 0.85 per 1000 starter hours (0.81 to 0.90).

| Load from | Players | Previous week | Appearances | Minutes | Injuries | Per app. | Per 1000 h | Ratio per app. (95% CI) | Ratio per hour (95% CI) |
|---|---|---|---|---|---|---|---|---|---|
| league | all players | none | 136 122 | 60 | 3085 | 0.0227 | 22.8 | 1 | 1 |
| league | all players | part | 79 521 | 55 | 1934 | 0.0243 | 26.3 | 1.07 (1.02 to 1.14) | 1.15 (1.09 to 1.22) |
| league | all players | full or more | 80 386 | 81 | 2139 | 0.0266 | 19.7 | 1.18 (1.12 to 1.25) | 0.87 (0.82 to 0.92) |
| league | starters | none | 88 080 | 82 | 2923 | 0.0332 | 24.3 | 1 | 1 |
| league | starters | part | 48 877 | 77 | 1819 | 0.0372 | 28.9 | 1.12 (1.06 to 1.19) | 1.18 (1.12 to 1.25) |
| league | starters | full or more | 74 710 | 85 | 2117 | 0.0283 | 19.9 | 0.86 (0.81 to 0.91) | 0.82 (0.78 to 0.87) |
| all club | all players | none | 116 162 | 59 | 2579 | 0.0222 | 22.6 | – | – |
| all club | all players | part | 83 265 | 55 | 1998 | 0.0240 | 26.2 | – | – |
| all club | all players | full or more | 96 602 | 79 | 2581 | 0.0267 | 20.4 | – | – |
| all club | starters | none | 74 087 | 82 | 2449 | 0.0331 | 24.2 | – | – |
| all club | starters | part | 50 769 | 77 | 1871 | 0.0369 | 28.6 | – | – |
| all club | starters | full or more | 86 811 | 85 | 2539 | 0.0292 | 20.7 | – | – |
| by cause | all players | none: calendar | 104 590 | 64 | 2477 | 0.0237 | 22.1 | 1 | 1 |
| by cause | all players | none: absent | 31 532 | 44 | 608 | 0.0193 | 26.4 | 0.81 (0.75 to 0.89) | 1.20 (1.09 to 1.31) |
| by cause | all players | part | 79 521 | 55 | 1934 | 0.0243 | 26.3 | 1.03 (0.97 to 1.09) | 1.19 (1.12 to 1.26) |
| by cause | all players | full or more | 80 386 | 81 | 2139 | 0.0266 | 19.7 | 1.13 (1.07 to 1.20) | 0.90 (0.85 to 0.95) |
| by cause | starters | none: calendar | 74 861 | 82 | 2369 | 0.0316 | 23.1 | 1 | 1 |
| by cause | starters | none: absent | 13 219 | 80 | 554 | 0.0419 | 31.3 | 1.33 (1.21 to 1.45) | 1.36 (1.24 to 1.49) |
| by cause | starters | part | 48 877 | 77 | 1819 | 0.0372 | 28.9 | 1.18 (1.11 to 1.25) | 1.25 (1.17 to 1.32) |
| by cause | starters | full or more | 74 710 | 85 | 2117 | 0.0283 | 19.9 | 0.90 (0.85 to 0.95) | 0.87 (0.82 to 0.92) |

**Table S10.** The next start after a substitution by the 60th minute, league by league. The log-odds compares starting the club's next match after an early substitution with starting it after finishing the match. The table also gives the adjusted chance of that start after each, with their risk ratio. Pooled by restricted maximum likelihood, the log-odds is −2.13 (95% CI −2.22 to −2.04; $\tau = 0.15$), an odds ratio of 0.12. One fit over all 490 125 starts gives a risk ratio of 0.496.

| League | Starts followed by another match | Log-odds | SE | Next start after an early substitution | Next start after finishing | Risk ratio (95% CI) |
|---|---|---|---|---|---|---|
| Belgium | 34 676 | −2.330 | 0.052 | 43% | 88% | 0.48 (0.46 to 0.51) |
| Denmark | 16 254 | −2.190 | 0.072 | 47% | 88% | 0.53 (0.50 to 0.57) |
| Spain | 48 338 | −1.871 | 0.042 | 41% | 81% | 0.50 (0.49 to 0.52) |
| France | 43 776 | −2.240 | 0.052 | 37% | 84% | 0.44 (0.42 to 0.47) |
| England | 32 560 | −2.196 | 0.055 | 45% | 87% | 0.51 (0.49 to 0.54) |
| Greece | 22 959 | −1.920 | 0.058 | 38% | 80% | 0.47 (0.44 to 0.50) |
| Italy | 48 699 | −1.934 | 0.040 | 42% | 83% | 0.51 (0.49 to 0.53) |
| Germany | 39 085 | −2.201 | 0.051 | 42% | 86% | 0.48 (0.46 to 0.51) |
| Netherlands | 38 699 | −2.132 | 0.048 | 51% | 89% | 0.57 (0.55 to 0.59) |
| Portugal | 38 761 | −2.097 | 0.046 | 44% | 86% | 0.52 (0.49 to 0.54) |
| Russia | 30 217 | −2.158 | 0.048 | 41% | 85% | 0.48 (0.46 to 0.51) |
| Scotland | 24 823 | −2.384 | 0.059 | 40% | 87% | 0.46 (0.43 to 0.48) |
| Turkey | 45 009 | −2.261 | 0.042 | 37% | 85% | 0.44 (0.42 to 0.46) |
| Ukraine | 26 269 | −1.970 | 0.052 | 41% | 83% | 0.49 (0.47 to 0.52) |

# S6 Sensitivity analyses

**The same appearances.** Table S6 prints each sample's starters' gradient as first estimated, except the Top 14's. Table S11 refits each gradient on exactly the appearances, seasons and window of its within-player slope. The refit is adjusted for the day of the week, and Figure S3B draws it. Every within-player interval stayed below its refitted gradient, in all 25 samples. The between-player part's own interval lay above zero in all 25 samples. Its lowest bound was 0.016.

**Table S11.** The starters' gradient refitted on the within-player slope's own appearances, the within-player slope with and without day-of-week terms, and the between-player part with its 95% interval from one bootstrap over players that gives both slopes in each draw. Below means the within-player slope's 95% interval lies wholly below the starters' gradient.

| League sample | Starts | Starters' gradient (95% CI) | Within-player slope (SE) | With day-of-week terms | Between-player part (95% CI) | Below |
|---|---|---|---|---|---|---|
| Spain, men | 40 753 | 0.030 (0.025 to 0.035) | 0.004 (0.002) | 0.005 | 0.026 (0.023 to 0.029) | yes |
| France, men | 37 129 | 0.027 (0.022 to 0.032) | 0.003 (0.002) | 0.003 | 0.024 (0.021 to 0.026) | yes |
| England, men | 23 518 | 0.031 (0.026 to 0.036) | 0.007 (0.002) | 0.008 | 0.024 (0.021 to 0.027) | yes |
| Italy, men | 42 294 | 0.035 (0.031 to 0.040) | 0.008 (0.002) | 0.010 | 0.027 (0.024 to 0.030) | yes |
| Germany, men | 32 353 | 0.031 (0.025 to 0.036) | 0.005 (0.002) | 0.004 | 0.026 (0.023 to 0.029) | yes |
| Netherlands, men | 31 954 | 0.028 (0.024 to 0.033) | 0.007 (0.002) | 0.007 | 0.021 (0.019 to 0.024) | yes |
| Portugal, men | 33 122 | 0.037 (0.032 to 0.042) | 0.011 (0.002) | 0.012 | 0.026 (0.023 to 0.029) | yes |
| Turkey, men | 38 985 | 0.028 (0.024 to 0.033) | 0.007 (0.002) | 0.008 | 0.022 (0.019 to 0.024) | yes |
| Germany, women | 2904 | 0.027 (0.010 to 0.043) | −0.006 (0.007) | −0.004 | 0.033 (0.022 to 0.044) | yes |
| England, women | 2904 | 0.033 (0.019 to 0.047) | 0.007 (0.007) | 0.005 | 0.026 (0.018 to 0.033) | yes |
| Spain, women | 5258 | 0.038 (0.027 to 0.050) | −0.005 (0.005) | −0.002 | 0.043 (0.036 to 0.050) | yes |
| Italy, women | 1980 | 0.030 (0.011 to 0.049) | −0.002 (0.008) | 0.001 | 0.032 (0.021 to 0.043) | yes |
| Norway, women | 2970 | 0.029 (0.018 to 0.041) | 0.006 (0.005) | 0.007 | 0.023 (0.016 to 0.030) | yes |
| Switzerland, women | 1957 | 0.027 (0.009 to 0.045) | 0.000 (0.008) | 0.001 | 0.027 (0.018 to 0.036) | yes |
| Sweden, women | 3982 | 0.045 (0.032 to 0.058) | 0.016 (0.006) | 0.015 | 0.029 (0.020 to 0.038) | yes |
| France, women | 2772 | 0.019 (0.002 to 0.037) | −0.008 (0.008) | −0.005 | 0.027 (0.018 to 0.037) | yes |
| United States, women | 3872 | 0.033 (0.018 to 0.048) | −0.006 (0.007) | −0.009 | 0.039 (0.029 to 0.049) | yes |
| Japan, men | 8206 | 0.055 (0.045 to 0.065) | 0.010 (0.004) | 0.009 | 0.045 (0.040 to 0.052) | yes |
| Argentina, men | 8291 | 0.029 (0.018 to 0.041) | −0.009 (0.005) | −0.009 | 0.038 (0.032 to 0.044) | yes |
| Mexico, men | 6512 | 0.023 (0.013 to 0.034) | −0.011 (0.005) | −0.011 | 0.034 (0.028 to 0.040) | yes |
| Australia, men | 3388 | 0.049 (0.030 to 0.067) | 0.000 (0.007) | 0.001 | 0.049 (0.036 to 0.061) | yes |
| United States, men | 10 714 | 0.035 (0.027 to 0.042) | 0.000 (0.003) | 0.000 | 0.035 (0.030 to 0.040) | yes |
| South Korea, men | 4994 | 0.075 (0.059 to 0.092) | 0.003 (0.006) | 0.003 | 0.072 (0.059 to 0.086) | yes |
| Australia, women, Australian rules (2023) | 4158 | 0.075 (0.064 to 0.086) | −0.006 (0.004) | −0.007 | 0.081 (0.070 to 0.092) | yes |
| France, men, rugby union (Top 14) | 16 387 | 0.041 (0.032 to 0.050) | 0.002 (0.004) | 0.001 | 0.040 (0.035 to 0.045) | yes |

**Every club match.** In the eight European men's leagues the provider's dataset holds every club competition, so recent match load was recomputed with cup and European matches included. Both the starters' gradient and the within-player slope were refitted on the same seasons under five substitutions (Table S12). The within-player slope fell in all eight leagues, from between +0.003 and +0.012 to between 0.000 and +0.009. Every interval stayed below its starters' gradient. Matches in the international windows are not in the dataset and stay uncounted.

**Table S12.** League matches only against every club match, on the same seasons.

| League | Load counted from | Starters' gradient | Within-player slope | SE | Mean load (minutes) |
|---|---|---|---|---|---|
| Spain | league matches | 0.030 | 0.0044 | 0.0021 | 46 |
| Spain | all club matches | 0.025 | 0.0027 | 0.0020 | 56 |
| France | league matches | 0.027 | 0.0030 | 0.0021 | 47 |
| France | all club matches | 0.025 | 0.0006 | 0.0020 | 52 |
| England | league matches | 0.031 | 0.0070 | 0.0022 | 52 |
| England | all club matches | 0.028 | 0.0056 | 0.0021 | 63 |
| Italy | league matches | 0.035 | 0.0083 | 0.0020 | 50 |
| Italy | all club matches | 0.032 | 0.0057 | 0.0019 | 59 |
| Germany | league matches | 0.031 | 0.0047 | 0.0023 | 46 |
| Germany | all club matches | 0.023 | 0.0003 | 0.0021 | 58 |
| Netherlands | league matches | 0.028 | 0.0067 | 0.0021 | 47 |
| Netherlands | all club matches | 0.021 | 0.0032 | 0.0019 | 57 |
| Portugal | league matches | 0.037 | 0.0117 | 0.0025 | 42 |
| Portugal | all club matches | 0.035 | 0.0092 | 0.0025 | 48 |
| Turkey | league matches | 0.029 | 0.0070 | 0.0023 | 46 |
| Turkey | all club matches | 0.026 | 0.0055 | 0.0022 | 49 |

**Other windows.** Recent match load was also measured over four and fourteen days in football, one and three weeks in rugby union, and one and two weeks in women's Australian rules. Over the longer windows, who starts explained 71% to 92% of the gradient (Table S13). Every within-player interval stayed below its starters' gradient, in all 24 samples with a longer window (Table S14). Over the shorter windows, who starts explained 73% to 91% of the gradient in the men's samples. Every within-player interval stayed below its starters' gradient, in all 15 men's samples.

But women's football mostly plays once a week, so a four-day window rarely holds an earlier match. Over four days, the within-player interval stayed below its starters' gradient in one of the nine women's leagues. In seven others, only 1 to 7 per cent of starts carried any load. Their within-player standard errors were three to seven times their seven-day size. In the remaining one league, no start carried any load. These four-day estimates are too imprecise to read either way.

**Load in categories.** Recent match load was also grouped into none, part of a match, and a full match or more. Between loaded and rested players, who starts explained 76% to 93% of the difference in log minutes (Table S13). After a full match or more, a player's own minutes differed from that player's rested starts by $-0.015$ to $+0.018$ in log minutes. Across all starters, the difference was $+0.035$ to $+0.103$ (Table S15). Every within-player interval stayed below the starters' contrast, in all 25 samples.

**The pandemic seasons.** Leaving out 2019–20 and 2020–21 changes only the European men's leagues, the samples that contain them. Who starts explained 85% to 92% of the gradient (Table S13). England's within-player sample, under five substitutions, holds neither season. So the within-player slope was refitted in the other seven leagues. Every within-player interval stayed below its starters' gradient (Table S14).

**One rugby window, and one set of seasons.** With the two-week window used throughout, the Top 14's share due to who starts is 0.73. With the three-week window of an earlier analysis, it was 0.71. Over the three seasons of Figure S3B, it is 0.71. So the window and the seasons barely move the share.

**Table S13.** The share of the gradient due to who starts, under each variation. The shorter window is four days in football and one week in rugby union; the longer, fourteen days and three weeks. Loaded against rested splits the difference between a full match or more and no load. The floor counts every appearance as ten minutes at least before the log is taken. The minutes scale uses minutes without logs. Within team sheets, each team-match's own means are removed. A dash marks a variation that does not apply, or a window that held too few loaded appearances to fit. A share outside zero to one arises when the two parts have opposite signs. That can happen over four days in weekly women's leagues, where the gradient itself is near zero.

| League sample | Own window | Shorter window | Longer window | Loaded against rested | Without the pandemic seasons | Floor of 10 minutes | Minutes scale | Within team sheets |
|---|---|---|---|---|---|---|---|---|
| France, men, rugby union (Top 14) | 0.731 | 0.740 | 0.709 | 0.755 | – | 0.737 | 0.726 | 0.776 |
| Brazil, men (2019) | 0.899 | 0.887 | 0.890 | 0.894 | – | 0.905 | 0.878 | 0.912 |
| Brazil, men (2024) | 0.873 | 0.885 | 0.869 | 0.865 | – | 0.879 | 0.863 | 0.901 |
| Germany, women | 0.840 | 0.759 | 0.836 | 0.823 | – | 0.862 | 0.869 | 0.876 |
| England, women | 0.861 | 1.110 | 0.861 | 0.861 | – | 0.873 | 0.848 | 0.898 |
| Spain, women | 0.837 | 0.590 | 0.830 | 0.835 | – | 0.860 | 0.843 | 0.852 |
| Italy, women | 0.874 | – | 0.846 | 0.877 | – | 0.891 | 0.878 | 0.913 |
| Norway, women | 0.866 | 0.844 | 0.854 | 0.895 | – | 0.893 | 0.865 | 0.908 |
| Switzerland, women | 0.896 | 0.855 | 0.870 | 0.910 | – | 0.896 | 0.891 | 0.893 |
| Sweden, women | 0.870 | 0.820 | 0.862 | 0.890 | – | 0.883 | 0.866 | 0.883 |
| England, men | 0.889 | 0.844 | 0.886 | 0.906 | 0.890 | 0.913 | 0.900 | 0.906 |
| Spain, men | 0.844 | 0.728 | 0.849 | 0.869 | 0.848 | 0.870 | 0.860 | 0.875 |
| Italy, men | 0.853 | 0.745 | 0.853 | 0.877 | 0.856 | 0.874 | 0.862 | 0.873 |
| Germany, men | 0.893 | 0.844 | 0.881 | 0.909 | 0.893 | 0.912 | 0.894 | 0.910 |
| France, men | 0.873 | 0.795 | 0.865 | 0.892 | 0.873 | 0.894 | 0.878 | 0.889 |
| Netherlands, men | 0.920 | 0.913 | 0.916 | 0.927 | 0.921 | 0.931 | 0.915 | 0.931 |
| Portugal, men | 0.899 | 0.845 | 0.896 | 0.911 | 0.907 | 0.914 | 0.901 | 0.912 |
| Turkey, men | 0.894 | 0.896 | 0.884 | 0.914 | 0.892 | 0.914 | 0.899 | 0.908 |

**Table S14.** The within-player comparison over other windows and without the pandemic seasons. Starts with load is the share of starts with any load in the window. Below means the within-player slope's 95% interval lies wholly below the starters' gradient, both refitted on the same appearances.

| League sample | Variation | Starts with load (%) | Within-player slope (SE) | Starters' gradient (95% CI) | Below |
|---|---|---|---|---|---|
| Spain, men | 4 days | 17 | 0.000 (0.004) | 0.023 (0.015 to 0.032) | yes |
| Spain, men | 14 days | 84 | 0.007 (0.002) | 0.031 (0.028 to 0.035) | yes |
| Spain, men | no pandemic seasons | 56 | 0.005 (0.003) | 0.034 (0.028 to 0.039) | yes |
| France, men | 4 days | 12 | −0.006 (0.004) | 0.013 (0.004 to 0.022) | yes |
| France, men | 14 days | 85 | 0.006 (0.002) | 0.031 (0.027 to 0.034) | yes |
| France, men | no pandemic seasons | 57 | 0.001 (0.002) | 0.026 (0.020 to 0.031) | yes |
| England, men | 4 days | 20 | 0.008 (0.003) | 0.026 (0.019 to 0.033) | yes |
| England, men | 14 days | 86 | 0.007 (0.002) | 0.029 (0.025 to 0.033) | yes |
| Italy, men | 4 days | 17 | 0.003 (0.004) | 0.030 (0.021 to 0.039) | yes |
| Italy, men | 14 days | 85 | 0.008 (0.002) | 0.031 (0.028 to 0.035) | yes |
| Italy, men | no pandemic seasons | 58 | 0.009 (0.002) | 0.039 (0.033 to 0.045) | yes |
| Germany, men | 4 days | 8 | 0.002 (0.004) | 0.030 (0.018 to 0.042) | yes |
| Germany, men | 14 days | 86 | 0.006 (0.002) | 0.033 (0.029 to 0.036) | yes |
| Germany, men | no pandemic seasons | 57 | 0.004 (0.003) | 0.031 (0.024 to 0.037) | yes |
| Netherlands, men | 4 days | 11 | −0.002 (0.004) | 0.021 (0.012 to 0.030) | yes |
| Netherlands, men | 14 days | 85 | 0.009 (0.002) | 0.030 (0.027 to 0.034) | yes |
| Netherlands, men | no pandemic seasons | 55 | 0.006 (0.002) | 0.028 (0.023 to 0.033) | yes |
| Portugal, men | 4 days | 5 | 0.008 (0.006) | 0.026 (0.015 to 0.038) | yes |
| Portugal, men | 14 days | 82 | 0.007 (0.002) | 0.032 (0.028 to 0.035) | yes |
| Portugal, men | no pandemic seasons | 50 | 0.008 (0.003) | 0.035 (0.029 to 0.041) | yes |
| Turkey, men | 4 days | 17 | 0.006 (0.003) | 0.025 (0.017 to 0.033) | yes |
| Turkey, men | 14 days | 83 | 0.005 (0.002) | 0.025 (0.022 to 0.028) | yes |
| Turkey, men | no pandemic seasons | 52 | 0.005 (0.003) | 0.029 (0.024 to 0.035) | yes |
| Germany, women | 4 days | 4 | −0.022 (0.021) | 0.003 (−0.040 to 0.046) | no |
| Germany, women | 14 days | 69 | −0.007 (0.005) | 0.028 (0.016 to 0.040) | yes |
| England, women | 4 days | 3 | −0.106 (0.045) | −0.053 (−0.148 to 0.042) | no |
| England, women | 14 days | 75 | 0.011 (0.005) | 0.031 (0.021 to 0.042) | yes |
| Spain, women | 4 days | 3 | −0.022 (0.017) | −0.006 (−0.054 to 0.043) | no |
| Spain, women | 14 days | 83 | −0.003 (0.004) | 0.040 (0.030 to 0.049) | yes |
| Italy, women | 4 days | 0 | – | – | – |
| Italy, women | 14 days | 77 | 0.006 (0.007) | 0.042 (0.027 to 0.056) | yes |
| Norway, women | 4 days | 22 | 0.010 (0.008) | 0.030 (0.013 to 0.047) | yes |
| Norway, women | 14 days | 84 | 0.002 (0.004) | 0.024 (0.016 to 0.032) | yes |
| Switzerland, women | 4 days | 2 | 0.012 (0.024) | 0.041 (−0.009 to 0.091) | no |
| Switzerland, women | 14 days | 70 | 0.004 (0.005) | 0.032 (0.019 to 0.045) | yes |
| Sweden, women | 4 days | 3 | 0.023 (0.020) | 0.037 (0.002 to 0.072) | no |
| Sweden, women | 14 days | 84 | 0.012 (0.004) | 0.040 (0.031 to 0.050) | yes |
| France, women | 4 days | 1 | −0.053 (0.038) | −0.037 (−0.119 to 0.045) | no |

| League sample | Variation | Starts with load (%) | Within-player slope (SE) | Starters' gradient (95% CI) | Below |
|---|---|---|---|---|---|
| France, women | 14 days | 77 | −0.002 (0.006) | 0.039 (0.024 to 0.054) | yes |
| United States, women | 4 days | 7 | −0.056 (0.020) | −0.024 (−0.069 to 0.020) | no |
| United States, women | 14 days | 85 | 0.003 (0.005) | 0.043 (0.033 to 0.054) | yes |
| Japan, men | 4 days | 20 | −0.001 (0.005) | 0.053 (0.039 to 0.067) | yes |
| Japan, men | 14 days | 82 | 0.004 (0.002) | 0.038 (0.032 to 0.044) | yes |
| Argentina, men | 4 days | 14 | −0.006 (0.008) | 0.017 (0.001 to 0.033) | yes |
| Argentina, men | 14 days | 82 | −0.009 (0.004) | 0.033 (0.025 to 0.041) | yes |
| Mexico, men | 4 days | 24 | −0.009 (0.006) | 0.020 (0.004 to 0.037) | yes |
| Mexico, men | 14 days | 82 | −0.012 (0.003) | 0.021 (0.014 to 0.028) | yes |
| Australia, men | 4 days | 8 | −0.003 (0.013) | 0.036 (0.004 to 0.068) | yes |
| Australia, men | 14 days | 86 | −0.001 (0.005) | 0.045 (0.034 to 0.056) | yes |
| United States, men | 4 days | 31 | −0.005 (0.005) | 0.031 (0.019 to 0.044) | yes |
| United States, men | 14 days | 88 | 0.002 (0.002) | 0.031 (0.026 to 0.036) | yes |
| South Korea, men | 4 days | 19 | 0.009 (0.008) | 0.081 (0.055 to 0.106) | yes |
| South Korea, men | 14 days | 84 | 0.005 (0.004) | 0.070 (0.058 to 0.082) | yes |
| Australia, women, Australian rules (2023) | 7 days | 55 | 0.003 (0.009) | 0.112 (0.088 to 0.137) | yes |
| Australia, women, Australian rules (2023) | 14 days | 85 | −0.001 (0.006) | 0.107 (0.092 to 0.122) | yes |
| France, men, rugby union (Top 14) | 7 days | 41 | 0.003 (0.005) | 0.048 (0.035 to 0.060) | yes |
| France, men, rugby union (Top 14) | 21 days | 77 | 0.004 (0.003) | 0.044 (0.037 to 0.051) | yes |

**Table S15.** Recent match load in categories. Each entry is the difference in log minutes from rested starts, for a full match or more and for part of a match, within the same player and among all starters.

| League sample | Within player, full match (SE) | All starters, full match (95% CI) | Within player, part (SE) | All starters, part (95% CI) |
|---|---|---|---|---|
| Spain, men | 0.008 (0.003) | 0.045 (0.038 to 0.051) | −0.010 (0.003) | −0.057 (−0.064 to −0.049) |
| France, men | 0.008 (0.003) | 0.041 (0.035 to 0.046) | −0.019 (0.003) | −0.063 (−0.070 to −0.055) |
| England, men | 0.013 (0.003) | 0.043 (0.037 to 0.050) | −0.012 (0.004) | −0.058 (−0.067 to −0.049) |
| Italy, men | 0.012 (0.002) | 0.047 (0.041 to 0.053) | −0.015 (0.003) | −0.056 (−0.063 to −0.048) |
| Germany, men | 0.010 (0.003) | 0.044 (0.037 to 0.050) | −0.014 (0.003) | −0.059 (−0.067 to −0.051) |
| Netherlands, men | 0.010 (0.002) | 0.043 (0.037 to 0.048) | −0.014 (0.003) | −0.061 (−0.069 to −0.053) |
| Portugal, men | 0.018 (0.003) | 0.055 (0.049 to 0.060) | −0.013 (0.004) | −0.060 (−0.069 to −0.052) |
| Turkey, men | 0.013 (0.003) | 0.044 (0.038 to 0.049) | −0.016 (0.003) | −0.060 (−0.067 to −0.052) |
| Germany, women | 0.001 (0.007) | 0.055 (0.036 to 0.073) | −0.021 (0.012) | −0.070 (−0.096 to −0.043) |
| England, women | 0.005 (0.007) | 0.048 (0.031 to 0.064) | 0.001 (0.010) | −0.057 (−0.079 to −0.036) |
| Spain, women | −0.002 (0.005) | 0.053 (0.040 to 0.066) | −0.008 (0.007) | −0.055 (−0.071 to −0.040) |
| Italy, women | 0.004 (0.008) | 0.050 (0.030 to 0.069) | −0.021 (0.014) | −0.074 (−0.102 to −0.046) |
| Norway, women | 0.011 (0.007) | 0.045 (0.030 to 0.060) | 0.001 (0.009) | −0.050 (−0.070 to −0.030) |
| Switzerland, women | 0.002 (0.008) | 0.045 (0.026 to 0.064) | −0.009 (0.013) | −0.065 (−0.091 to −0.039) |
| Sweden, women | 0.018 (0.006) | 0.050 (0.036 to 0.064) | −0.008 (0.009) | −0.060 (−0.080 to −0.041) |
| France, women | −0.003 (0.010) | 0.040 (0.022 to 0.059) | −0.016 (0.012) | −0.060 (−0.084 to −0.035) |
| United States, women | −0.001 (0.008) | 0.048 (0.031 to 0.066) | −0.017 (0.010) | −0.079 (−0.100 to −0.057) |
| Japan, men | 0.014 (0.004) | 0.077 (0.065 to 0.089) | 0.002 (0.006) | −0.063 (−0.078 to −0.048) |
| Argentina, men | −0.004 (0.006) | 0.056 (0.044 to 0.068) | −0.007 (0.007) | −0.071 (−0.086 to −0.056) |
| Mexico, men | −0.015 (0.006) | 0.035 (0.021 to 0.048) | −0.007 (0.009) | −0.067 (−0.084 to −0.049) |
| Australia, men | 0.005 (0.008) | 0.064 (0.043 to 0.086) | −0.013 (0.011) | −0.083 (−0.110 to −0.057) |
| United States, men | 0.002 (0.005) | 0.040 (0.030 to 0.051) | −0.012 (0.006) | −0.067 (−0.080 to −0.054) |
| South Korea, men | 0.005 (0.007) | 0.103 (0.082 to 0.124) | −0.003 (0.009) | −0.072 (−0.097 to −0.046) |
| Australia, women, Australian rules (2023) | −0.012 (0.010) | 0.079 (0.058 to 0.100) | −0.002 (0.011) | −0.010 (−0.035 to 0.016) |
| France, men, rugby union (Top 14) | 0.001 (0.005) | 0.063 (0.051 to 0.076) | 0.007 (0.005) | −0.053 (−0.065 to −0.041) |

**Keeping loaded starters on.** In the eight European men's leagues the substitution model was refitted under each substitution limit (Table S16). Without the pandemic seasons, the substitution coefficient ran from −0.45 to −0.29. Over fourteen days, it ran from −0.42 to −0.30. Every interval lay below zero. Over four days, every estimate stayed below zero, from −0.58 to −0.14. But two of the 16 intervals reached zero.

With load in categories, starters with a full match or more had lower odds of substitution than rested starters (odds ratios 0.48 to 0.62). Starters with only part of a match had higher odds (1.81 to 2.33). If the size of the load set the odds, they would change in step from none to part of a match to a full match. Instead, they were highest after part of a match. Across starters, the pattern follows standing rather than the size of the load. With rest split by its cause, starters absent from the club's previous match had higher odds of

substitution than those the calendar rested. The pooled odds ratios were 1.31 and 1.16 under three and five substitutions. Loaded starters had lower odds than those the calendar rested (0.62 and 0.53; Table S16).

**Goalkeepers, position and the same player.** Without goalkeepers, the pooled odds ratio of the eight leagues moved from 0.70 to 0.73 under three substitutions and from 0.69 to 0.71 under five. With outfield position added, it moved to 0.74 and 0.72. Every estimate stayed below one. Without goalkeepers, whose minutes almost never change, the within-player slope of minutes changed by at most 0.001. Every interval stayed below its starters' gradient.

Within the same player, the odds ratio was 0.90 (95% CI 0.89 to 0.92) under three substitutions and 0.86 (0.84 to 0.87) under five. It was below one in all 16 league samples. Conditional logistic regression removes the bias of one intercept per player (Section S3). It gave odds ratios of 0.90 (0.89 to 0.92) and 0.86 (0.84 to 0.87). So after a heavy week, the same player is kept on too. But the link is weaker than across starters, where regular starters are set against rotation players.

With load in categories, a full match or more gave odds ratios of 0.85 and 0.77 against rested weeks within the same player. Across starters, it gave 0.58 and 0.52. Part of a match gave 1.09 and 1.13 within the same player, against 2.12 and 1.98 across starters. Most of the categorical pattern is therefore standing. Within the same player, weeks absent from the club's previous match also went with higher odds of substitution than weeks the calendar rested (1.21 and 1.20; Table S17). In an absent week, as in a heavy week, the club played in the window. Yet the same player's odds were higher in absent weeks and lower in heavy ones. So the same player's lower odds after a heavy week are not the calendar alone.

**Within the same team sheet, by period and with the score.** The comparison is among the starters of one club in one match who are still on the field at a period's start. Among them, each extra full match went with an odds ratio of substitution of 0.41 (95% CI 0.37 to 0.45) under three substitutions and 0.37 (0.33 to 0.41) under five. Without goalkeepers and with position, the odds ratios were 0.47 and 0.41 (Table S18). Comparing teammates, who share the schedule and the score, makes the association stronger.

The stronger association measures standing within a match. After a gap longer than the window, every starter of the club carries no load. With no difference in load between teammates, those team-matches drop out of the fit. In the others, loaded starters are compared with part-loaded teammates and with teammates absent from the club's previous match. So the team-sheet odds ratio mostly sets regular starters against the rotation starters of the same club in the same match. The within-player odds ratios (0.90 and 0.86) measure the effect of load instead.

The substitution model across starters was also fitted by period and with the score (Table S18). Loaded starters came off less often in every part of the match. The gap was largest between the 60th and the 75th minutes, where most substitutions are made. Under three substitutions, the odds ratios were 0.76 to the 60th minute, 0.65 from the 61st to the 75th and 0.73 after it. Under five, they were 0.72, 0.64 and 0.73. Adding the score at each period's start left the pooled odds ratio of the eight leagues unchanged, at 0.70 and 0.69. Starters of a leading team were substituted a little more often (1.06 and 1.13). At the 75th minute, 88% to 91% of loaded starters were still on the field (Figure S5). So were 78% to 85% of those the calendar rested and 74% to 80% of those absent from the club's previous match.

**Table S16.** The substitution model in the eight European men's leagues under each variation. The first columns give the substitution coefficient per extra full match (SE). With load in categories, the odds ratios are against rested starters. With rest split by its cause, they are against starters the calendar rested. Absent means absent from the club's previous match.

| League | Limit | Seven days | Four days | Fourteen days | No pandemic seasons | OR, full match | OR, part | OR, full match v calendar | OR, part v calendar | OR, absent v calendar |
|---|---|---|---|---|---|---|---|---|---|---|
| Spain | three | −0.320 (0.032) | −0.221 (0.058) | −0.310 (0.024) | −0.291 (0.037) | 0.60 | 2.02 | 0.62 | 2.11 | 1.18 |
| France | three | −0.414 (0.032) | −0.363 (0.055) | −0.377 (0.023) | −0.446 (0.036) | 0.54 | 2.21 | 0.57 | 2.35 | 1.28 |
| England | three | −0.310 (0.023) | −0.239 (0.034) | −0.316 (0.018) | −0.326 (0.028) | 0.61 | 2.21 | 0.65 | 2.35 | 1.36 |
| Italy | three | −0.335 (0.032) | −0.368 (0.063) | −0.319 (0.023) | −0.337 (0.036) | 0.62 | 1.92 | 0.65 | 2.03 | 1.23 |
| Germany | three | −0.438 (0.037) | −0.580 (0.087) | −0.423 (0.025) | −0.450 (0.044) | 0.54 | 1.99 | 0.59 | 2.20 | 1.49 |
| Netherlands | three | −0.323 (0.034) | −0.304 (0.079) | −0.348 (0.025) | −0.342 (0.040) | 0.60 | 2.08 | 0.64 | 2.22 | 1.52 |
| Portugal | three | −0.338 (0.036) | −0.136 (0.081) | −0.298 (0.026) | −0.354 (0.042) | 0.57 | 2.23 | 0.59 | 2.32 | 1.29 |
| Turkey | three | −0.360 (0.034) | −0.181 (0.240) | −0.358 (0.024) | −0.337 (0.040) | 0.57 | 2.33 | 0.59 | 2.41 | 1.20 |
| Spain | five | −0.369 (0.022) | −0.330 (0.035) | −0.344 (0.016) | −0.393 (0.025) | 0.52 | 1.91 | 0.53 | 1.95 | 1.11 |
| France | five | −0.364 (0.022) | −0.302 (0.041) | −0.384 (0.017) | −0.355 (0.025) | 0.51 | 2.08 | 0.53 | 2.16 | 1.24 |
| England | five | −0.425 (0.025) | −0.378 (0.037) | −0.397 (0.019) | −0.428 (0.025) | 0.51 | 2.05 | 0.52 | 2.13 | 1.30 |
| Italy | five | −0.378 (0.021) | −0.345 (0.038) | −0.347 (0.015) | −0.392 (0.025) | 0.55 | 1.81 | 0.57 | 1.88 | 1.19 |
| Germany | five | −0.397 (0.023) | −0.399 (0.054) | −0.367 (0.018) | −0.396 (0.027) | 0.51 | 1.97 | 0.53 | 2.04 | 1.20 |
| Netherlands | five | −0.340 (0.021) | −0.222 (0.044) | −0.362 (0.017) | −0.344 (0.023) | 0.53 | 1.96 | 0.54 | 2.00 | 1.24 |
| Portugal | five | −0.358 (0.021) | −0.299 (0.055) | −0.347 (0.016) | −0.346 (0.024) | 0.48 | 2.08 | 0.48 | 2.08 | 1.00 |
| Turkey | five | −0.349 (0.020) | −0.324 (0.034) | −0.309 (0.015) | −0.337 (0.023) | 0.52 | 1.99 | 0.52 | 2.00 | 1.04 |

**Table S17.** The substitution model in the eight European men's leagues. The first columns give the substitution coefficient per extra full match (SE) with all starters. Then come its standard error clustered by player and team-match, and the coefficient without goalkeepers and with outfield position. The coefficient is also given within the same player, with one intercept per player and by conditional logistic regression. Within the same player, odds ratios compare a full match or more and part of a match with rested weeks. With rest split by its cause, the conditional fit compares three kinds of week with weeks the calendar rested. They are a full match or more, part of a match, and absence from the club's previous match. Goalkeepers were 9% of starts in every league.

| League | Limit | All starters (SE) | SE, two-way clustered | Without goalkeepers (SE) | With position (SE) | Within the same player (SE) | Within, conditional (SE) | Within, OR full match | Within, OR part | Within, OR full v calendar | Within, OR part v calendar | Within, OR absent v calendar |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Spain | three | −0.320 (0.032) | 0.027 | −0.279 (0.032) | −0.279 (0.032) | −0.072 (0.033) | −0.070 (0.032) | 0.87 | 1.03 | 0.90 | 1.06 | 1.15 |
| France | three | −0.414 (0.032) | 0.027 | −0.388 (0.031) | −0.362 (0.032) | −0.135 (0.034) | −0.133 (0.034) | 0.81 | 1.08 | 0.85 | 1.13 | 1.26 |
| England | three | −0.310 (0.023) | 0.021 | −0.276 (0.023) | −0.254 (0.024) | −0.094 (0.023) | −0.093 (0.022) | 0.84 | 1.13 | 0.88 | 1.19 | 1.29 |
| Italy | three | −0.335 (0.032) | 0.027 | −0.299 (0.031) | −0.302 (0.032) | −0.123 (0.033) | −0.121 (0.032) | 0.87 | 1.02 | 0.90 | 1.06 | 1.16 |
| Germany | three | −0.438 (0.037) | 0.033 | −0.396 (0.037) | −0.358 (0.037) | −0.129 (0.039) | −0.127 (0.038) | 0.85 | 1.09 | 0.88 | 1.13 | 1.19 |
| Netherlands | three | −0.323 (0.034) | 0.027 | −0.302 (0.034) | −0.288 (0.035) | −0.089 (0.038) | −0.087 (0.038) | 0.86 | 1.12 | 0.89 | 1.15 | 1.26 |
| Portugal | three | −0.338 (0.036) | 0.027 | −0.306 (0.036) | −0.268 (0.038) | −0.102 (0.043) | −0.100 (0.041) | 0.87 | 1.10 | 0.89 | 1.13 | 1.16 |
| Turkey | three | −0.360 (0.034) | 0.026 | −0.335 (0.034) | −0.316 (0.036) | −0.100 (0.040) | −0.098 (0.039) | 0.85 | 1.18 | 0.87 | 1.20 | 1.15 |
| Spain | five | −0.369 (0.022) | 0.020 | −0.320 (0.020) | −0.322 (0.020) | −0.150 (0.021) | −0.148 (0.020) | 0.78 | 1.09 | 0.80 | 1.12 | 1.16 |
| France | five | −0.364 (0.022) | 0.021 | −0.326 (0.021) | −0.299 (0.021) | −0.123 (0.022) | −0.121 (0.022) | 0.80 | 1.18 | 0.84 | 1.22 | 1.30 |
| England | five | −0.425 (0.025) | 0.024 | −0.406 (0.025) | −0.400 (0.026) | −0.200 (0.027) | −0.197 (0.027) | 0.74 | 1.14 | 0.78 | 1.19 | 1.43 |
| Italy | five | −0.378 (0.021) | 0.019 | −0.334 (0.019) | −0.333 (0.019) | −0.163 (0.020) | −0.161 (0.019) | 0.79 | 1.12 | 0.82 | 1.16 | 1.23 |
| Germany | five | −0.397 (0.023) | 0.021 | −0.357 (0.023) | −0.343 (0.023) | −0.160 (0.025) | −0.158 (0.025) | 0.78 | 1.12 | 0.80 | 1.16 | 1.20 |
| Netherlands | five | −0.340 (0.021) | 0.018 | −0.316 (0.021) | −0.300 (0.022) | −0.155 (0.023) | −0.153 (0.023) | 0.78 | 1.13 | 0.80 | 1.16 | 1.25 |
| Portugal | five | −0.358 (0.021) | 0.019 | −0.321 (0.021) | −0.298 (0.022) | −0.131 (0.025) | −0.128 (0.024) | 0.77 | 1.16 | 0.78 | 1.18 | 1.09 |
| Turkey | five | −0.349 (0.020) | 0.019 | −0.329 (0.020) | −0.322 (0.020) | −0.177 (0.021) | −0.175 (0.021) | 0.74 | 1.13 | 0.75 | 1.13 | 1.06 |

**Table S18.** The substitution model within the same team sheet, by period and with the score, in the eight European men's leagues. The first two columns give the substitution coefficient within the team sheet (SE), by conditional logistic regression stratified by team-match and period. They use all starters, and then outfield starters with their position. Odds ratios within the team sheet compare a full match or more and part of a match with none. The further columns come from the substitution model across starters. They give the odds ratio per extra full match in each period, and then the substitution coefficient with the score at each period's start. The last two give the odds ratios of leading and trailing against level.

| League | Limit | Within the team sheet (SE) | Team sheet, no goalkeepers (SE) | Team sheet, OR full | Team sheet, OR part | OR, to 60th min. | OR, 61st to 75th | OR, after 75th | With the score (SE) | OR, leading | OR, trailing |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Spain | three | −0.718 (0.043) | −0.641 (0.055) | 0.51 | 1.72 | 0.78 | 0.66 | 0.78 | −0.321 (0.032) | 1.02 | 0.94 |
| France | three | −0.940 (0.056) | −0.812 (0.057) | 0.43 | 1.78 | 0.72 | 0.60 | 0.69 | −0.414 (0.032) | 1.01 | 0.98 |
| England | three | −0.885 (0.045) | −0.739 (0.048) | 0.44 | 1.62 | 0.82 | 0.70 | 0.72 | −0.309 (0.023) | 1.07 | 1.06 |
| Italy | three | −0.752 (0.054) | −0.667 (0.050) | 0.52 | 1.61 | 0.75 | 0.63 | 0.78 | −0.332 (0.032) | 1.11 | 0.98 |
| Germany | three | −0.980 (0.072) | −0.800 (0.077) | 0.39 | 1.41 | 0.71 | 0.62 | 0.63 | −0.437 (0.037) | 1.12 | 1.00 |
| Netherlands | three | −1.042 (0.063) | −0.927 (0.064) | 0.41 | 1.41 | 0.77 | 0.67 | 0.75 | −0.323 (0.034) | 1.08 | 1.09 |
| Portugal | three | −0.942 (0.061) | −0.733 (0.067) | 0.47 | 1.80 | 0.70 | 0.70 | 0.74 | −0.339 (0.036) | 1.05 | 0.95 |
| Turkey | three | −0.909 (0.055) | −0.785 (0.061) | 0.50 | 1.99 | 0.74 | 0.63 | 0.73 | −0.360 (0.034) | 1.07 | 0.93 |
| Spain | five | −0.901 (0.037) | −0.802 (0.041) | 0.44 | 1.76 | 0.73 | 0.64 | 0.73 | −0.371 (0.022) | 1.12 | 1.01 |
| France | five | −1.038 (0.044) | −0.885 (0.043) | 0.42 | 1.85 | 0.75 | 0.63 | 0.74 | −0.368 (0.022) | 1.08 | 1.19 |
| England | five | −1.221 (0.054) | −1.143 (0.056) | 0.37 | 1.67 | 0.74 | 0.61 | 0.66 | −0.427 (0.025) | 1.15 | 1.23 |
| Italy | five | −0.910 (0.036) | −0.814 (0.038) | 0.45 | 1.59 | 0.69 | 0.63 | 0.75 | −0.378 (0.021) | 1.11 | 1.04 |
| Germany | five | −0.998 (0.037) | −0.864 (0.040) | 0.42 | 1.74 | 0.70 | 0.65 | 0.68 | −0.398 (0.023) | 1.23 | 1.17 |
| Netherlands | five | −1.148 (0.050) | −1.004 (0.052) | 0.42 | 1.68 | 0.76 | 0.64 | 0.76 | −0.341 (0.021) | 1.17 | 1.13 |
| Portugal | five | −0.966 (0.036) | −0.796 (0.040) | 0.47 | 2.15 | 0.66 | 0.67 | 0.75 | −0.356 (0.021) | 1.07 | 1.10 |
| Turkey | five | −0.838 (0.038) | −0.767 (0.040) | 0.49 | 2.00 | 0.71 | 0.64 | 0.77 | −0.350 (0.020) | 1.14 | 1.02 |

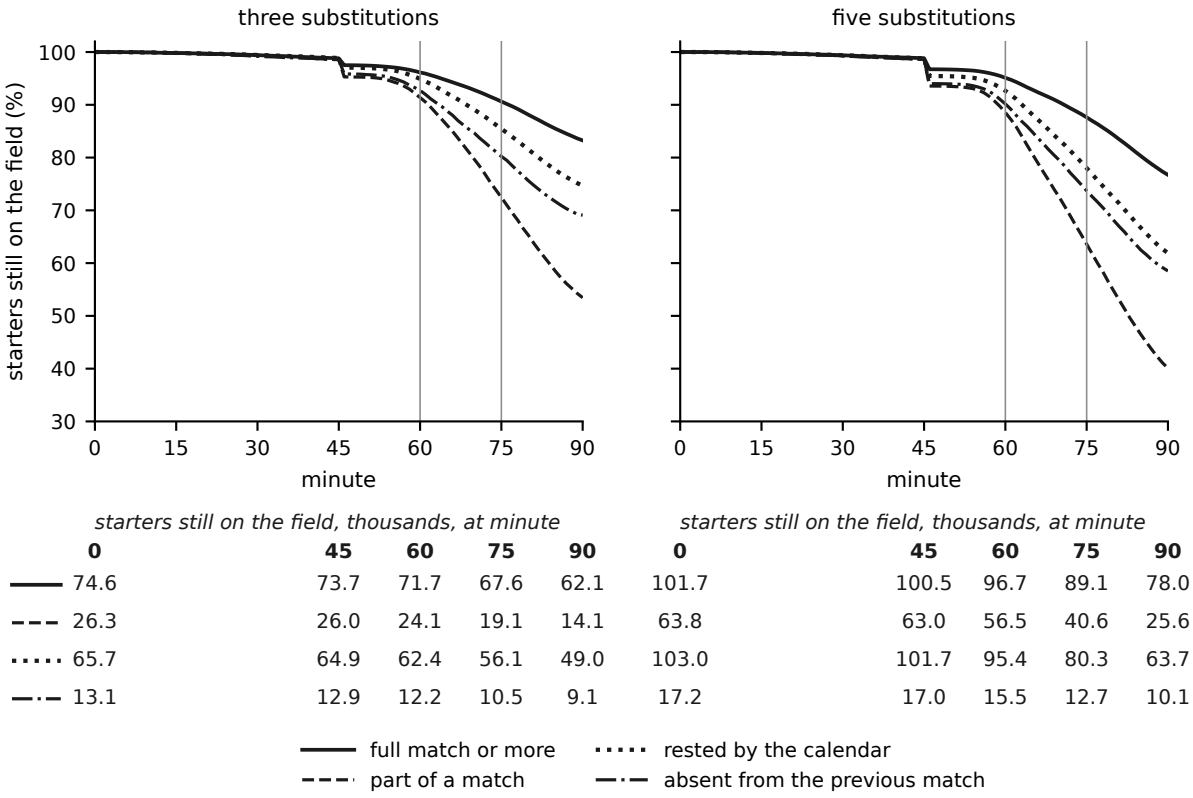



**Figure S5.** Starters after a full match stayed on the field longest. Starters still on the field in the eight European men's leagues, by recent match load: a full match or more in the previous week, part of a match, and none. None is split into starters the fixture calendar rested and starters absent from the club's previous match. A starter substituted or sent off counts as on the field up to that minute. The vertical lines mark the periods of the substitution model. Under each panel, each group has one row, in the line style of its curve. The row gives, in thousands, the group's starters still on the field. The minute is printed above each column: kick-off, half-time and the 60th, 75th and 90th minutes. The curves count every starter, goalkeepers included. Goalkeepers are almost never substituted. They make a larger share of loaded starts than of the others, so they lift the full-match curve. The substitution model without them keeps the result (Table S17).

**What the within-player estimator can see.** Recorded minutes have a ceiling: they stop at the full match. Recent match load is the player's own earlier minutes. So the load is an earlier value of the outcome, a lagged outcome. The within-player slope removes each player's average from both. That biases the slope downwards when each player has few starts. This is the dynamic-panel bias (Nickell, Econometrica 1981).

A plasmode simulation measured both the ceiling's effect and this bias. Such a simulation keeps the real data and simulates only the part under study. For each sample, it kept the real dates, roles and substitutes' minutes. It regenerated starters' minutes with a known within-player effect and rebuilt recent load from the simulated history. It then applied the paper's estimator. To measure the bias, it ran 50 times for each setting. For the calibration of Figure S6, it ran 20 times at each true effect. The calibration shows how much of a true effect the estimator recovers. The simulation's settings were fitted to each sample. They were the share of starts that last the full match, the mean log minutes of the other starts, and the spread of players' own averages.

With no true effect, the estimator returned $-0.006$ to $-0.001$ on recorded minutes, and $-0.016$ to $-0.001$ without the ceiling. The true effect acts on the minutes a coach would give a starter, before the full match caps them. A true effect of 0.03 on those minutes appeared as 0.005 to 0.010 in recorded minutes. The calibration ran over true effects from a fall of five per cent in minutes per extra full match to a rise of ten per cent. Over that range, the estimate rose 0.29 to 0.38 for every unit of true effect in football (Figure S6). So in football the estimator recovered about a third of a true effect. It recovered 0.47 of it in rugby union. In women's Australian rules, whose load is the share played, it recovered 0.83, because few starts there last to the final whistle (Table S19).

The smallest effect the design would detect lies 2.8 standard errors from the estimate under no effect. That distance gives a two-sided test at 5 per cent with 80 per cent power. It measures the design's sensitivity, not a power calculation after the fact. The effect has a size on each of two scales: recorded minutes, and the minutes

a coach would give. The smallest detectable within-player slope on recorded minutes, the scale of the starters' gradients, was 0.006 to 0.024. In the eight European men's leagues, it was 0.006 to 0.007, or 16% to 22% of their starters' gradients. In the other men's leagues, it was 0.010 to 0.021 (20% to 57%). In women's football, it was 0.014 to 0.024 (35% to 123%). In women's Australian rules and men's rugby union, it was 0.012 and 0.011 (16% and 27%).

In the women's samples of Italy, Switzerland and France, a within-player slope as large as three-quarters of the whole starters' gradient could go undetected. In the women's sample of France, one as large as the whole gradient could go undetected. The smallest detectable effect on the minutes a coach would give, the scale of the simulation's true effect, was 0.015 to 0.067. In the eight European men's leagues, it was 0.015 to 0.020 (Table S19).

The eight European men's leagues are the best measured samples. There, the within-player slope was 11% to 31% of the starters' gradient on the same appearances. With each sample's simulated bias removed, it was 16% to 37%. Little of the starters' gradient lies within the player, and most of it lies between players. Across all 25 samples, each slope was also taken with its own simulated bias removed. Every within-player interval still stayed below its starters' gradient. The starts per repeat starter ran from 9 to 38. The within-player slopes rose with them, as the bias predicts (Spearman correlation 0.60, 95% CI 0.27 to 0.81). Sex, source and league differ between those samples too.

Load from the club's fixture list is the club's matches in the previous seven days. It is set by the schedule rather than the player's own minutes, so it carries no lagged outcome and no ceiling on the load. It was available in 16 samples. It counts every club competition in the eight European men's leagues, and league matches elsewhere. On that load, the within-player slope pooled to −0.004 per extra club match (95% CI −0.006 to −0.002). The starters' gradient pooled to −0.003 (−0.006 to −0.001). So a congested week leaves the same player's minutes, and starters' minutes, almost unchanged. The within-player slope is small but not null. Its sign is the one fatigue or rotation would give: fewer minutes after a congested week.

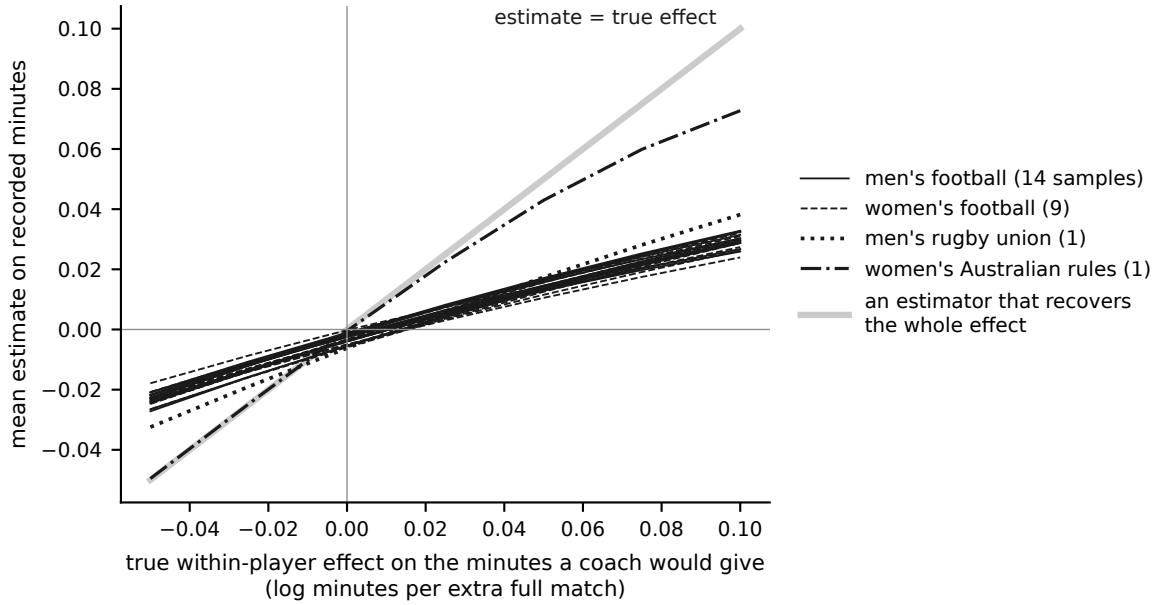



**Figure S6.** Recorded minutes stop at the full match, so the within-player estimator recovers a fraction of a true effect. In women's Australian rules, whose load is the share of the match played, it recovers most of it. Each line is one sample's calibration: its mean estimate on recorded minutes against the true within-player effect on the minutes a coach would give. Each sample was simulated 20 times at each of seven true effects. The smallest within-player slope the design detects is 0.006 to 0.024 on recorded minutes, the scale of the starters' gradients (Table S19).

**Table S19.** The within-player estimator, sample by sample. Starts to the final whistle is the share of starts that lasted the full match. The other percentages are of the starters' gradient on the same appearances. The slope's percentage is given as estimated and with the simulated bias removed. The simulation's bias with no true effect and its estimate of a true effect of 0.03 are on recorded minutes. The share of a true effect recovered is the slope of the calibration in Figure S6. The smallest detectable slope on recorded minutes is 2.8 standard errors of the observed slope. It is also given as a percentage of the starters' gradient. The smallest detectable true effect is on the minutes a coach would give (Figure S6). The last column is the within-player slope on the club's fixture list, where clubs are recorded.

| League sample | Starts per repeat starter | Starts to the final whistle (%) | Within-player slope | Of the starters' gradient (%) | Bias removed (%) | Simulated bias, no true effect | A true effect of 0.03, as estimated | Share of a true effect recovered | Detectable, recorded minutes | Of the starters' gradient (%) | Detectable, a coach's minutes | On the club's fixture list (SE) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Spain, men | 38 | 60 | 0.004 | 15 | 20 | −0.002 | 0.010 | 0.37 | 0.006 | 20 | 0.016 | −0.005 (0.002) |
| France, men | 33 | 63 | 0.003 | 11 | 16 | −0.001 | 0.009 | 0.34 | 0.006 | 22 | 0.017 | −0.008 (0.002) |
| England, men | 32 | 66 | 0.007 | 23 | 29 | −0.002 | 0.008 | 0.31 | 0.006 | 20 | 0.020 | 0.000 (0.002) |
| Italy, men | 37 | 60 | 0.008 | 24 | 28 | −0.002 | 0.010 | 0.37 | 0.006 | 16 | 0.015 | −0.006 (0.002) |
| Germany, men | 35 | 62 | 0.005 | 15 | 19 | −0.001 | 0.010 | 0.35 | 0.007 | 21 | 0.018 | −0.005 (0.002) |
| Netherlands, men | 31 | 64 | 0.007 | 24 | 31 | −0.002 | 0.008 | 0.34 | 0.006 | 21 | 0.017 | −0.004 (0.002) |
| Portugal, men | 26 | 61 | 0.011 | 31 | 37 | −0.002 | 0.009 | 0.37 | 0.007 | 19 | 0.019 | −0.001 (0.002) |
| Turkey, men | 30 | 64 | 0.007 | 24 | 31 | −0.002 | 0.009 | 0.34 | 0.006 | 22 | 0.019 | −0.001 (0.002) |
| Germany, women | 12 | 62 | −0.006 | −22 | −9 | −0.004 | 0.007 | 0.34 | 0.018 | 69 | 0.053 | – |
| England, women | 12 | 65 | 0.007 | 21 | 34 | −0.004 | 0.006 | 0.33 | 0.019 | 57 | 0.056 | – |
| Spain, women | 16 | 62 | −0.005 | −12 | −1 | −0.004 | 0.007 | 0.36 | 0.014 | 36 | 0.038 | – |
| Italy, women | 10 | 61 | −0.002 | −8 | 7 | −0.005 | 0.007 | 0.37 | 0.023 | 76 | 0.062 | – |
| Norway, women | 15 | 67 | 0.006 | 20 | 27 | −0.002 | 0.007 | 0.29 | 0.015 | 53 | 0.052 | – |
| Switzerland, women | 10 | 64 | 0.000 | −1 | 12 | −0.004 | 0.007 | 0.33 | 0.022 | 83 | 0.067 | – |
| Sweden, women | 14 | 67 | 0.016 | 36 | 45 | −0.004 | 0.005 | 0.31 | 0.016 | 35 | 0.050 | – |
| France, women | 12 | 59 | −0.008 | −42 | −31 | −0.002 | 0.009 | 0.38 | 0.024 | 123 | 0.063 | −0.011 (0.008) |
| United States, women | 13 | 61 | −0.006 | −19 | −3 | −0.005 | 0.006 | 0.36 | 0.019 | 58 | 0.052 | −0.011 (0.006) |
| Japan, men | 17 | 61 | 0.010 | 18 | 22 | −0.003 | 0.009 | 0.36 | 0.011 | 20 | 0.031 | 0.008 (0.004) |
| Argentina, men | 12 | 60 | −0.009 | −29 | −12 | −0.005 | 0.006 | 0.37 | 0.014 | 49 | 0.038 | −0.005 (0.004) |
| Mexico, men | 14 | 60 | −0.011 | −46 | −24 | −0.005 | 0.007 | 0.38 | 0.013 | 57 | 0.035 | −0.007 (0.004) |
| Australia, men | 13 | 63 | 0.000 | 0 | 12 | −0.006 | 0.005 | 0.35 | 0.021 | 42 | 0.059 | 0.002 (0.007) |
| United States, men | 16 | 64 | 0.000 | 0 | 11 | −0.004 | 0.007 | 0.34 | 0.010 | 28 | 0.028 | −0.008 (0.003) |
| South Korea, men | 14 | 60 | 0.003 | 4 | 12 | −0.005 | 0.006 | 0.36 | 0.016 | 21 | 0.044 | −0.001 (0.005) |
| Australia, women, Australian rules (2023) | 9 | 3 | −0.006 | −9 | −8 | −0.001 | 0.027 | 0.83 | 0.012 | 16 | 0.015 | – |
| France, men, rugby union (Top 14) | 20 | 49 | 0.002 | 4 | 16 | −0.005 | 0.009 | 0.47 | 0.011 | 27 | 0.024 | – |

**The 27 league-seasons and the earlier seven.** An earlier version of the injury comparison used seven league-seasons. They were those with near-complete reasons in a harvest of match reports: the 2022/23 and 2023/24 seasons of the Premier League and the Bundesliga, and the 2023/24 seasons of La Liga, Serie A and Ligue 1. The provider's snapshot holds near-complete reasons for more league-seasons. So the comparison was rerun on the 27 that allowed five substitutions throughout. The same scripts ran it, under a plan written before any of them was compared (Section S3). In the seven, 359 ratios had an interval wholly on one side of one. In the 27, 326 of them lie on the same side, and 33 no longer do. None is reversed. Most of those no longer clear compare a starter's weeks within the same starter, pooled over seasons. Two further readings kept the rerun's pattern. Within the same starter and season, the interval for heavy weeks stayed wholly above one against every rested week. Across starters, per starter hour, the interval of the loaded-to-rested ratio stayed wholly below one. The first reading added the league-seasons that allowed three substitutions for all or part of the season, for 34 league-season strata. Its two ratios were 1.14 (1.07 to 1.20) and 0.84 (0.79 to 0.88). The second left out the four 2020/21 seasons played without crowds, for 23 league-seasons. Its two ratios were 1.09 (1.03 to 1.17) and 0.81 (0.76 to 0.86).

**The injury comparison examined.** Table S20 first gives Figure 2's comparisons on both outcomes, with the part-loaded ratios beside them. It then repeats the comparison under each check, on every player. A team that has made its last permitted substitution cannot replace an injured player. Loaded starters are the ones still on the field late in matches. So late in a match, the outcome is most likely to miss a loaded starter's injury. One check therefore kept only events before the 70th minute and ended the time at risk there. The loaded-to-rested ratio was then 0.93 per starter hour (95% CI 0.87 to 1.00) and 0.95 per recorded hour among all players (0.88 to 1.02). In team-matches with a substitution left, the two ratios were 0.77 and 0.80.

Adding up to 15 minutes to every appearance that lasted to the final whistle moved the per-hour ratio among all players to at most 0.90. Among starters, it stayed at 0.81 to 0.82. With first-half added time given too, the ratio was 0.88 to 0.90 among all players and 0.81 to 0.82 among starters. So per hour, loaded players had fewer injury-coded substitutions under every one of these checks. Only with the events before the 70th minute did the intervals reach one. Poisson models with standard errors clustered by player gave 1.18 per appearance and 0.87 per recorded hour among all players. Among starters, they gave 0.86 and 0.82. On the outfield players of Figure 2, Poisson models agreed too, and counting every club match changed the ratios little.

Part-loaded starters had higher rates than rested starters (Table S9). Part of a match can be a return from injury. So one check left out the 7 or 28 days after an injury-coded substitution (2654 and 13 582 appearances removed). The part-loaded starters' excess held, with ratios of 1.18 and 1.18.

Further checks of the reasons, the pauses and the outcome kept the pattern of Figure 2. Under every check, the per-appearance ratio lay above the per-hour ratio. The ratio per starter hour lay below the same starters' one. Only the sizes moved. Reasons were given for at least 99 per cent of substitutions in 18 league-seasons. In them, the ratios were 1.12 per appearance (1.05 to 1.20) and 0.83 per recorded hour (0.77 to 0.88). Per starter hour, the ratio was 0.79 (0.73 to 0.84), and within the same starters 0.99 (0.92 to 1.06). One bound counted every substitution without a reason as injury-coded. It gave 1.15, 0.85 and 0.80 (0.76 to 0.84), in the same order, and 0.99 within the same starters (0.94 to 1.05).

Leaving out the first league match after each pause gave 1.18, 0.85 and 0.80, and 0.97 within the same starters. The narrower outcome, an injury-coded substitution followed by a recorded absence, gave 1.29 (1.19 to 1.41), 0.95 and 0.93, and 1.14 within the same starters (1.04 to 1.25).

With rest split by its cause, loaded starters' rate per starter hour was 0.87 times that of starters rested by the calendar (95% CI 0.82 to 0.92). It was 0.64 times that of starters absent from the club's previous match (0.58 to 0.70). Against starters rested by the calendar, absent starters had 1.36 times the rate (1.24 to 1.49) and part-loaded starters 1.25 times (1.17 to 1.32). Stratified by league-season and by whether the club played in the window, loaded starters' ratio against all rested starters was 0.64 (0.58 to 0.70). A loaded starter's club always played in the window, and where the club played, every rested starter was absent. So this ratio is the same as the ratio against absent starters alone. Within a team-match, too, every rested starter was absent. Stratified by team-match, the ratio was 0.65 (0.59 to 0.72).

The weeks the calendar rested differ by the gap since the club's previous league match (Figure S7). After a gap of 8 to 13 days since the club's previous league match, starters the calendar rested had 23.9 injury-coded substitutions per 1000 starter hours (47 334 starts). After a break of 14 to 60 days, the international windows and winter breaks, they had 22.1 (21 913 starts). After a longer gap, or with no previous league match on file, they had 20.9 (5614 starts). Most were a season's first match. So the calendar's reference mixes longer-gap weeks with long breaks, and the long breaks carry the lower rate.

Across all starters, the rate was flat from three to eight days since the club's previous league match (22.8 to 25.1). It was slightly lower, with overlapping intervals, after nine to 13 days (22.3) and after 14 days or more (21.8). No previous league match was on file for 206 starts.

But earlier per-hour studies of congestion found higher rates after short gaps. They found fewer muscle

injuries with at least six days between matches (Bengtsson and colleagues, British Journal of Sports Medicine 2018). They also found a higher injury rate in weeks with two matches (Dupont and colleagues, American Journal of Sports Medicine 2010). A higher rate was also found in the matches of a congested period (Dellal and colleagues, British Journal of Sports Medicine 2015). This journal's congestion studies set short gaps against every longer gap (Bengtsson and colleagues, British Journal of Sports Medicine 2013; Carling and colleagues, British Journal of Sports Medicine 2016). That reference includes long breaks, so it holds the same mix as the calendar's weeks here. The outcome differs too: here it is an injury-coded substitution, not a time-loss injury. So a flat rate across short gaps does not show that congestion is safe. The same breaks bear on the first limitation of the main text. The calendar's reference weeks include the international windows and winter breaks, which the gap bands separate.

A loaded starter's club always played within the window. So in a comparison with loaded starters, the gap cannot be a stratum. Instead, the weeks the calendar rested are taken band by band as the reference. Loaded starters' ratio against the weeks after 8 to 13 days was 0.84 (0.79 to 0.90), per starter hour and by Mantel–Haenszel strata of league-season. Part-loaded starters' ratio was 1.21 (1.13 to 1.29). Absent starters' ratio was 1.32 (1.20 to 1.45). Against the breaks of 14 to 60 days, the three ratios were 0.90 (0.83 to 0.99), 1.31 and 1.43 (Table S20). So heavy weeks look safest against starters who missed the club's previous match: the returning, the dropped and the least trusted. Against the calendar's longer-gap weeks, the difference is smaller, but heavy weeks still look safer.

For the comparisons within the same starter, the archive's deposits give the events and the starters or player-seasons each fit uses. Raw counts against every rested week give 0.84 for heavy weeks and 1.14 for part-loaded ones. The fits give 1.00 and 1.00. The raw counts mix starters who start in different kinds of week. So the gap between the raw ratios and the fitted ones is selection among starters.

Within the same starters, heavy weeks against all the weeks the calendar rested gave 1.03 (0.97 to 1.10), and weeks absent from the previous match 1.20 (1.09 to 1.32). With the gap band in the model, the reference is the weeks after 8 to 13 days. Against them, heavy weeks gave 0.99 (0.93 to 1.06), part-loaded weeks 0.99 and absent weeks 1.16. Weeks after breaks of 14 to 60 days gave 0.91 (0.83 to 1.00), and longer gaps 0.82 (0.69 to 0.97). With the longer breaks left out instead, heavy weeks gave 0.99 (0.92 to 1.06). So, pooled over seasons, the same starters' heavy weeks showed no clear difference from the calendar's longer-gap weeks. Their weeks after long breaks carried lower rates, as after a recovery. A comparison against every rested week therefore mixes in a recovery contrast at the long end of the calendar. Congestion studies seek the contrast at the short end. The check without the first league match after each pause gave the same reading (0.97).

Absent starters were split by the provider's injury histories (Section S3). Those with no recorded injury had 30.2 injury-coded substitutions per 1000 starter hours (10 408 starts). Those injured at the time of the club's previous match had 33.1 (1704 starts). Those back from a recorded injury within 28 days had 39.8 (1107 starts), the highest rate of any group. Loaded starters against absent starters with no recorded injury gave a ratio of 0.66 (0.60 to 0.74). Absent starters with no recorded injury against the calendar's weeks gave 1.31 (1.18 to 1.45). Starters back from injury against the calendar's weeks gave 1.69 (1.30 to 2.20). So heavy weeks look safer against absent starters even without the returning players, who are few. But the histories record long injuries better than short ones. So a starter with no recorded injury may still have had a minor one.

The within-starter comparison was also fitted under its design's own tests (Section S3). Each test is set against every rested week. With strata by player and league-season, heavy weeks gave 1.11 (1.04 to 1.18; 4797 player-seasons). The other three tests pool the seasons, as the first fit did. That fit gave 1.00 (Table S20). With only each starter's first injury-coded substitution counted, and every start kept, heavy weeks gave 1.03 (0.94 to 1.13; 2695 events). Without the 28 days after an injury-coded substitution, they gave 1.00 (0.94 to 1.06). Without the 84 days, they gave 1.00 (0.94 to 1.07).

The same-season fit was then set against the same starter's longer-gap weeks in the same league-season, on the same player-seasons. This comparison was saved to the project's repository before it was run. By then, the two fits it completes were known: the same-season fit against every rested week, and the pooled fit against longer-gap weeks. Heavy weeks gave 1.07 (1.00 to 1.15). That is no clear difference from them, because the interval's lower end is 0.997, just below one. The plan named this middle reading "no different". Since 15 October 2026 it is called "no clear difference", and the rule is unchanged. In this comparison, part-loaded weeks gave 0.92 (0.86 to 0.99) and absent weeks 1.08 (0.97 to 1.20). Weeks after breaks of 14 to 60 days gave 0.89 (0.81 to 0.98), and longer gaps 0.72 (0.60 to 0.85). So every rested week, as a reference, includes the weeks after long breaks, whose rates were lower.

**The main text's population, and three checks of the advised comparison.** The rows above count every starter. The plan for the injury checks set a rule (Section S10). If any refit fell outside the interval it re-checked, the refit without goalkeepers would become the primary estimate. Of the 48 refits, 11 fell outside. So the main text and Figures 1 to 3 give the comparisons on outfield players, and the 19 231 starts by goalkeepers are left out. On them, against the same starter's longer-gap weeks in the same league-season, heavy weeks gave

1.06 (0.99 to 1.13), part-loaded weeks 0.92 (0.85 to 0.99), weeks after breaks of 14 to 60 days 0.89 and longer gaps 0.72. Three checks of that comparison were planned before they ran, but after the results above were known. The plan is in the archive. Each check was fitted on outfield players and on every player.

- **The narrower outcome**, an injury-coded substitution followed by a recorded absence. Heavy weeks gave 1.16 (1.03 to 1.29) on outfield players and 1.18 (1.06 to 1.32) on every player: riskier. Part-loaded weeks gave 0.88 (0.78 to 0.99). Across outfield starters, against every rested week, the narrower outcome gave 0.96 (0.88 to 1.05): no clear difference.
- **The reference after a completed match.** A starter who completed the club's previous league match did not come off early with a knock. So this check kept only such starts in the longer-gap weeks, and heavy weeks were set against longer-gap weeks after a completed match. Heavy weeks gave 0.99 (0.91 to 1.09) on outfield players and 1.01 (0.92 to 1.10) on every player: no clear difference. Longer-gap weeks after a match not completed gave 0.89 (0.80 to 0.99) against those after a completed one.
- **Every match counted.** Recent load and the calendar were read from every match the snapshot records. These are league, cup, continental and super cup matches, and the national-team tournaments the snapshot holds. Heavy weeks then gave 1.07 (0.99 to 1.15) on outfield players: no clear difference. On every player they gave 1.08 (1.01 to 1.16): riskier.

So the comparison the main text advises shows no clear difference on injury-coded substitutions, and reads riskier on the narrower outcome. When both weeks follow a completed match, it shows no clear difference.

**The outcome, the exposure and the estimator, checked.** A further set of checks was planned before it ran, but after the results above were known. Its plan is in the archive too. Figure 2 of the main text draws each same-season estimate, and Table S20 gives their counts. Over the 12 versions within the same starter, injury-coded substitutions gave 0.98 to 1.10. The nine versions on the narrower outcome gave 1.13 to 1.25. With injury-coded substitutions, only one version was clearly higher (1.10): the comparison against every rested week in the same season. All nine versions on the narrower outcome were clearly higher. A single club sees only 10 to 16 injury-coded substitutions a season. That is too few for this comparison.

- **When the substitutions run out.** After the team's fifth substitution, the last one permitted, an injured player could not be replaced. So this check ended every player's time at risk at that minute, and counted an injury-coded substitution only up to it. Heavy weeks then gave 1.06 (0.99 to 1.13): no clear difference. On the narrower outcome they gave 1.16 (1.03 to 1.29): riskier. Across starters, against longer-gap weeks, they gave 0.88 (0.82 to 0.94) and 0.95 (0.85 to 1.05). Without the cut, the two were 0.88 and 0.94.
- **Pause weeks.** International players played for their countries in the pauses of 9 to 60 days in the five leagues together: the international windows and tournaments. So each club's first league match after such a pause left the comparison. Heavy weeks then gave 1.06 (0.98 to 1.14): no clear difference. On the narrower outcome they gave 1.16 (1.03 to 1.31): riskier. With every match counted as well, they gave 1.07 and 1.23.
- **The days since the previous match and the minutes in it.** Table S21 puts two terms in one model: the club's days since its previous match in any competition, and the starter's minutes in that match. The model uses outfield starters, with every match counted. It leaves out 40 starts with no previous club match on file, or with one under two days earlier. On the narrower outcome, gaps of two to three and of four to five days carried higher rates than six to seven days. Within the same starter, two to three days carried 40% more (1.40, 1.23 to 1.59). That held within the same starter and across starters. On injury-coded substitutions, those gaps showed no clear difference (1.05 for two to three days, within the same starter). Within the same starter, rates were lower after none or part of the previous match than after a full one. Across starters, on injury-coded substitutions, the same groups had higher rates. So for the previous match's minutes, as for load, the comparison across starters and the one within each starter point opposite ways.
- **Under-exposure.** The idea that too little match play raises risk (Moreno-Pérez and colleagues, Biology of Sport 2021) predicts more risk in a starter's own part-loaded weeks. This study tested the idea only in that narrowest form, a single part-loaded week. In that form, part-loaded weeks showed less risk than longer-gap weeks in the same season, not more (0.92, 0.85 to 0.99).
- **The coaches' response where the calendar sets the gap.** This check modelled a substitution not coded as injury, over three periods of the match. It used outfield starters who played the whole of their club's previous match in any competition (79 002 starts). A gap of two to four days was set against a

gap of five to seven days. Within the same starter and league-season, it gave an odds ratio of 1.06 (1.01 to 1.12). Across starters, it gave 1.04 (0.99 to 1.08). A gap of eight to thirteen days gave 0.97 (0.92 to 1.02) and 0.98 (0.94 to 1.02). So where the calendar set the gap, coaches took the same starter off slightly sooner after a short one.

- **The design's own tests on the same-season fit.** The first test counted only each starter's first injury-coded substitution, and kept every start. Heavy weeks then gave 1.08 (0.96 to 1.22): no clear difference. On the narrower outcome they gave 1.25 (1.07 to 1.45): riskier. Without the 28 days after an injury-coded substitution, they gave 1.04 and, on the narrower outcome, 1.15. Without the 84 days, they gave 1.04 and 1.15.

- **The estimators in a simulation.** A simulation kept every outfield start of the advised comparison and drew each start's event anew (Table S22). It assumed selection between players. It also assumed a knock: a higher risk at the next start for a starter who did not complete the club's previous league match. In the simulation, heavy weeks had no effect. With selection alone, the comparison across starters read them as 0.93 on average. Its interval held the true ratio in 48% of replicates. The advised comparison read 1.00 (92%). With selection kept, a knock that raised the next start's risk by a tenth moved the advised comparison to 0.96 (81%). One that raised it by a quarter moved it to 0.91 (25%). The comparison against longer-gap weeks after a completed match stayed at 1.00 and 1.01. So a knock would put the advised comparison below the comparison after a completed match. In the data the order was the reverse: 0.99 after a completed match, against the advised comparison's 1.06. So the data give no sign of such a knock.

The hybrid model fits the comparison within starters and the comparison across starters in one model (Section S3). In it, the same starter's heavy weeks carried a rate ratio of 1.00 (0.94 to 1.06). Across starters, a starter loaded in every start against one loaded in none gave 0.07 (0.05 to 0.09). The within-starter ratio was 15.0 times the across-starter one (11.0 to 20.3). The across term rests on each starter's share of loaded starts. That share is taken over all of a starter's starts, including the weeks after the starter's injuries. So it borrows from the outcome it is compared with (Section S3).

The model was also fitted with the share taken over earlier starts only, a prospective share. On the 189 030 starts with at least five earlier ones, the across-starter ratio was then 0.27 (0.19 to 0.37). The within-starter ratio was 3.4 times it (2.5 to 4.7). With the share over all starts, on the same starts, the two were 0.04 and 26.4. On the 169 679 starts with at least 10 earlier ones, the prospective share gave an across-starter ratio of 0.19 (0.12 to 0.28). The within-starter ratio was 4.9 times it (3.3 to 7.3). With the share over all starts, the two were 0.03 and 29.3 (Table S23). On outfield players, with the prospective share, the within-starter ratio was 2.6 times the across-starter ratio (1.9 to 3.6) with five earlier starts. With ten, it was 3.4 times (2.2 to 5.2).

The definition of the share alone moves the across term several-fold (0.04 against 0.27 on the same starts). The direction holds, so selection among starters is real. But the gap's size is about 3.4, not 15.0. The prospective within term (0.91) sets a start against the starter's own past, rather than against the starter's average. So it answers a different question, and the conditional Poisson fit stays the within-starter estimate.

The crude rates by share band show the same direction. The larger a starter's share of loaded starts, the lower the rate. Starters loaded in at most a fifth of their starts had 35 injury-coded substitutions per 1000 starter hours. Those loaded in a fifth to two-fifths, two-fifths to three-fifths and three-fifths to four-fifths had 27, 18 and 5. Too few starters were loaded in more than four-fifths of their starts to show. The lowest band held 2167 starters with 20 starts each on average, the shortest histories. With each start placed by the share of the starter's earlier starts instead, the four bands had 30, 26, 21 and 9: the same direction, less steep.

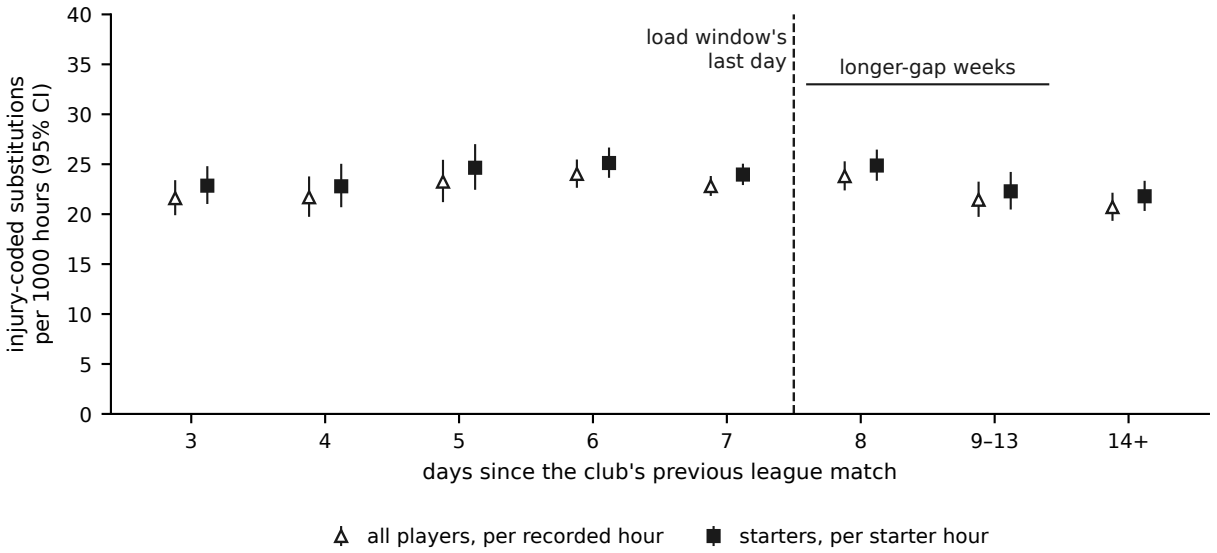



**Figure S7.** Injury-coded substitutions per 1000 hours by days since the club's previous league match, in the 27 league-seasons of the injury comparison, on every player, goalkeepers included; Figure 1C gives outfield players. Filled squares are per starter hour among starters, open triangles per recorded hour among all players, with exact 95% CIs. The bracket marks the longer-gap weeks. The club's calendar, not the team sheet, sets the days since the club's previous league match. Congestion studies seek the effect of this exposure. The dashed line marks the load window's last day. The last band holds every gap of 14 days or more, including the start of each season. The 206 starts with no previous league match on file are not drawn. The outcome is an injury-coded substitution, not a time-loss injury, so a flat line across short gaps does not show that congestion is safe.

**Table S20.** The injury comparison under each check: ratios, with 95% intervals, for a full match or more and for part of a match against a reference group. Ratios across players are Mantel–Haenszel ratios. Ratios within the same starter come from conditional Poisson regression. The first rows are Figure 2's comparisons on outfield players, on injury-coded substitutions and then on those followed by a recorded absence. They add the part-loaded ratios, which the figure leaves out. The rows after them count every player, goalkeepers included. The reference is players with no load and the strata are league-seasons, unless the row names others. The calendar's weeks are the reference band by band, by the days since the club's previous league match. Absent players with no recorded injury are a reference of their own. Added time at the end is given to every appearance that lasted to the final whistle. Rows with added time in both halves also give first-half added time to every appearance on the field at half-time. Their labels give the first half's minutes first. The last rows compare the same starters' own weeks by conditional Poisson regression, per starter hour. Their references are every rested week, the calendar's weeks, and the weeks after 8 to 13 days with longer breaks as terms or left out. Further rows apply the design's own tests (Section S3).

| Analysis | Players | Denominator | Full match or more against the reference (95% CI) | Part of a match against the reference (95% CI) |
|---|---|---|---|---|
| Figure 2: across, all players, per appearance, v every rested week | outfield players | per appearance | 1.26 (1.19 to 1.33) | 1.03 (0.97 to 1.09) |
| Figure 2: across, all players, per recorded hour, v every rested week | outfield players | per recorded hour | 0.91 (0.86 to 0.97) | 1.07 (1.01 to 1.13) |
| Figure 2: across, starters, per appearance, v every rested week | outfield starters | per appearance | 0.90 (0.85 to 0.95) | 1.04 (0.98 to 1.10) |
| Figure 2: across, starters, per starter hour, v every rested week | outfield starters | per starter hour | 0.86 (0.81 to 0.91) | 1.09 (1.03 to 1.16) |
| Figure 2: across, starters, per starter hour, v longer-gap weeks | outfield starters | per starter hour | 0.88 (0.82 to 0.94) | 1.11 (1.04 to 1.19) |
| Figure 2: within the same starter, same season, v longer-gap weeks | outfield starters | per starter hour | 1.06 (0.99 to 1.13) | 0.92 (0.85 to 0.99) |
| Figure 2: within the same starter, seasons pooled, v longer-gap weeks | outfield starters | per starter hour | 0.98 (0.92 to 1.05) | 0.98 (0.91 to 1.05) |
| Figure 2: within the same starter, same season, v every rested week | outfield starters | per starter hour | 1.10 (1.03 to 1.17) | 0.95 (0.89 to 1.01) |
| Figure 2: within the same starter, seasons pooled, v every rested week | outfield starters | per starter hour | 0.99 (0.93 to 1.05) | 0.99 (0.93 to 1.06) |
| Figure 2: within the same starter, every match counted | outfield starters | per starter hour | 1.07 (0.99 to 1.15) | 0.90 (0.83 to 0.97) |
| Figure 2: within the same starter, first match after a pause left out | outfield starters | per starter hour | 1.06 (0.98 to 1.14) | 0.92 (0.85 to 0.99) |

| Analysis | Players | Denominator | Full match or more against the reference (95% CI) | Part of a match against the reference (95% CI) |
|---|---|---|---|---|
| Figure 2: within the same starter, every match, first after a pause out | outfield starters | per starter hour | 1.07 (0.99 to 1.16) | 0.90 (0.83 to 0.98) |
| Figure 2: within the same starter, each starter's first event only | outfield starters | per starter hour | 1.08 (0.96 to 1.22) | 0.91 (0.80 to 1.02) |
| Figure 2: within the same starter, 28 days after an injury left out | outfield starters | per starter hour | 1.04 (0.97 to 1.12) | 0.94 (0.87 to 1.02) |
| Figure 2: within the same starter, 84 days after an injury left out | outfield starters | per starter hour | 1.04 (0.96 to 1.13) | 0.94 (0.86 to 1.02) |
| Figure 2: within the same starter, time at risk ended at the fifth substitution | outfield starters | per starter hour | 1.06 (0.99 to 1.13) | 0.92 (0.85 to 0.99) |
| Figure 2: within the same starter, v longer-gap weeks after a completed match | outfield starters | per starter hour | 0.99 (0.91 to 1.09) | 0.85 (0.78 to 0.94) |
| Figure 2, followed by a recorded absence: across, all players, per appearance, v every rested week | outfield players | per appearance | 1.36 (1.25 to 1.49) | 0.99 (0.90 to 1.08) |
| Figure 2, followed by a recorded absence: across, all players, per recorded hour, v every rested week | outfield players | per recorded hour | 0.99 (0.90 to 1.08) | 1.02 (0.93 to 1.12) |
| Figure 2, followed by a recorded absence: across, starters, per appearance, v every rested week | outfield starters | per appearance | 1.00 (0.92 to 1.10) | 1.00 (0.91 to 1.10) |
| Figure 2, followed by a recorded absence: across, starters, per starter hour, v every rested week | outfield starters | per starter hour | 0.96 (0.88 to 1.05) | 1.05 (0.95 to 1.16) |
| Figure 2, followed by a recorded absence: across, starters, per starter hour, v longer-gap weeks | outfield starters | per starter hour | 0.94 (0.85 to 1.05) | – |
| Figure 2, followed by a recorded absence: within the same starter, same season, v longer-gap weeks | outfield starters | per starter hour | 1.16 (1.03 to 1.29) | 0.88 (0.78 to 0.99) |
| Figure 2, followed by a recorded absence: within the same starter, seasons pooled, v every rested week | outfield starters | per starter hour | 1.13 (1.02 to 1.24) | 0.99 (0.89 to 1.09) |
| Figure 2, followed by a recorded absence: within the same starter, every match counted | outfield starters | per starter hour | 1.23 (1.09 to 1.39) | 0.88 (0.77 to 1.01) |
| Figure 2, followed by a recorded absence: within the same starter, first match after a pause left out | outfield starters | per starter hour | 1.16 (1.03 to 1.31) | 0.87 (0.76 to 0.98) |
| Figure 2, followed by a recorded absence: within the same starter, every match, first after a pause out | outfield starters | per starter hour | 1.23 (1.08 to 1.41) | 0.88 (0.76 to 1.02) |
| Figure 2, followed by a recorded absence: within the same starter, each starter's first event only | outfield starters | per starter hour | 1.25 (1.07 to 1.45) | 0.92 (0.78 to 1.08) |
| Figure 2, followed by a recorded absence: within the same starter, 28 days after an injury left out | outfield starters | per starter hour | 1.15 (1.03 to 1.30) | 0.90 (0.79 to 1.02) |
| Figure 2, followed by a recorded absence: within the same starter, 84 days after an injury left out | outfield starters | per starter hour | 1.15 (1.01 to 1.31) | 0.93 (0.81 to 1.07) |
| Figure 2, followed by a recorded absence: within the same starter, time at risk ended at the fifth substitution | outfield starters | per starter hour | 1.16 (1.03 to 1.29) | 0.88 (0.78 to 1.00) |
| As analysed | all players | per appearance | 1.18 (1.12 to 1.25) | 1.07 (1.02 to 1.14) |
| As analysed | all players | per recorded hour | 0.87 (0.82 to 0.92) | 1.15 (1.09 to 1.22) |
| As analysed | starters | per appearance | 0.86 (0.81 to 0.91) | 1.12 (1.06 to 1.19) |
| As analysed | starters | per starter hour | 0.82 (0.78 to 0.87) | 1.18 (1.12 to 1.25) |
| Reference: rested by the calendar | all players | per appearance | 1.13 (1.07 to 1.20) | 1.03 (0.97 to 1.09) |
| Reference: rested by the calendar | all players | per recorded hour | 0.90 (0.85 to 0.95) | 1.19 (1.12 to 1.26) |
| Reference: rested by the calendar | starters | per appearance | 0.90 (0.85 to 0.95) | 1.18 (1.11 to 1.25) |
| Reference: rested by the calendar | starters | per starter hour | 0.87 (0.82 to 0.92) | 1.25 (1.17 to 1.32) |
| Reference: the calendar, after 8 to 13 days | all players | per appearance | 1.09 (1.02 to 1.16) | 0.99 (0.93 to 1.06) |
| Reference: the calendar, after 8 to 13 days | all players | per recorded hour | 0.87 (0.81 to 0.93) | 1.15 (1.08 to 1.23) |
| Reference: the calendar, after 8 to 13 days | starters | per appearance | 0.87 (0.82 to 0.93) | 1.14 (1.07 to 1.22) |
| Reference: the calendar, after 8 to 13 days | starters | per starter hour | 0.84 (0.79 to 0.90) | 1.21 (1.13 to 1.29) |

| Analysis | Players | Denominator | Full match or more against the reference (95% CI) | Part of a match against the reference (95% CI) |
|---|---|---|---|---|
| Reference: the calendar, after 14 to 60 days | all players | per appearance | 1.19 (1.09 to 1.29) | 1.09 (0.99 to 1.18) |
| Reference: the calendar, after 14 to 60 days | all players | per recorded hour | 0.94 (0.86 to 1.03) | 1.26 (1.15 to 1.38) |
| Reference: the calendar, after 14 to 60 days | starters | per appearance | 0.94 (0.86 to 1.02) | 1.24 (1.13 to 1.35) |
| Reference: the calendar, after 14 to 60 days | starters | per starter hour | 0.90 (0.83 to 0.99) | 1.31 (1.20 to 1.44) |
| Reference: the calendar, after more than 60 days or none on file | all players | per appearance | 1.26 (1.07 to 1.47) | 1.15 (0.98 to 1.34) |
| Reference: the calendar, after more than 60 days or none on file | all players | per recorded hour | 1.00 (0.85 to 1.17) | 1.33 (1.14 to 1.56) |
| Reference: the calendar, after more than 60 days or none on file | starters | per appearance | 0.99 (0.84 to 1.16) | 1.30 (1.11 to 1.53) |
| Reference: the calendar, after more than 60 days or none on file | starters | per starter hour | 0.95 (0.81 to 1.12) | 1.39 (1.18 to 1.63) |
| Reference: the calendar, after 14 days or more | all players | per appearance | 1.20 (1.11 to 1.30) | 1.10 (1.02 to 1.19) |
| Reference: the calendar, after 14 days or more | all players | per recorded hour | 0.95 (0.88 to 1.03) | 1.28 (1.18 to 1.39) |
| Reference: the calendar, after 14 days or more | starters | per appearance | 0.95 (0.88 to 1.03) | 1.25 (1.16 to 1.36) |
| Reference: the calendar, after 14 days or more | starters | per starter hour | 0.92 (0.84 to 0.99) | 1.33 (1.23 to 1.45) |
| Reference: absent from the previous match | all players | per appearance | 1.39 (1.27 to 1.52) | 1.26 (1.15 to 1.38) |
| Reference: absent from the previous match | all players | per recorded hour | 0.75 (0.69 to 0.82) | 0.99 (0.91 to 1.09) |
| Reference: absent from the previous match | starters | per appearance | 0.68 (0.62 to 0.74) | 0.88 (0.80 to 0.97) |
| Reference: absent from the previous match | starters | per starter hour | 0.64 (0.58 to 0.70) | 0.91 (0.83 to 1.01) |
| Reference: absent, with no recorded injury | all players | per appearance | 1.45 (1.31 to 1.60) | 1.32 (1.19 to 1.46) |
| Reference: absent, with no recorded injury | all players | per recorded hour | 0.79 (0.71 to 0.87) | 1.04 (0.94 to 1.15) |
| Reference: absent, with no recorded injury | starters | per appearance | 0.70 (0.63 to 0.78) | 0.91 (0.82 to 1.01) |
| Reference: absent, with no recorded injury | starters | per starter hour | 0.66 (0.60 to 0.74) | 0.95 (0.85 to 1.05) |
| Strata: league-season and calendar | all players | per appearance | 1.39 (1.27 to 1.52) | 1.26 (1.15 to 1.38) |
| Strata: league-season and calendar | all players | per recorded hour | 0.75 (0.69 to 0.82) | 0.99 (0.91 to 1.09) |
| Strata: league-season and calendar | starters | per appearance | 0.68 (0.62 to 0.74) | 0.88 (0.80 to 0.97) |
| Strata: league-season and calendar | starters | per starter hour | 0.64 (0.58 to 0.70) | 0.92 (0.83 to 1.01) |
| Strata: team-match | all players | per appearance | 1.51 (1.37 to 1.66) | 1.29 (1.17 to 1.42) |
| Strata: team-match | all players | per recorded hour | 0.85 (0.78 to 0.94) | 1.01 (0.91 to 1.11) |
| Strata: team-match | starters | per appearance | 0.68 (0.61 to 0.75) | 0.86 (0.78 to 0.96) |
| Strata: team-match | starters | per starter hour | 0.65 (0.59 to 0.72) | 0.89 (0.81 to 0.99) |
| Before the 70th minute | all players | per appearance | 1.08 (1.01 to 1.16) | 1.11 (1.03 to 1.19) |
| Before the 70th minute | all players | per recorded hour | 0.95 (0.88 to 1.02) | 1.18 (1.09 to 1.27) |
| Before the 70th minute | starters | per appearance | 0.95 (0.88 to 1.01) | 1.17 (1.09 to 1.26) |
| Before the 70th minute | starters | per starter hour | 0.93 (0.87 to 1.00) | 1.19 (1.11 to 1.28) |
| A substitution left | all players | per appearance | 1.05 (0.96 to 1.15) | 1.18 (1.07 to 1.30) |
| A substitution left | all players | per recorded hour | 0.80 (0.73 to 0.88) | 1.29 (1.17 to 1.42) |
| A substitution left | starters | per appearance | 0.79 (0.72 to 0.87) | 1.27 (1.15 to 1.40) |
| A substitution left | starters | per starter hour | 0.77 (0.69 to 0.84) | 1.33 (1.21 to 1.47) |
| Without 7 days after an injury | all players | per appearance | 1.18 (1.12 to 1.25) | 1.06 (1.00 to 1.12) |
| Without 7 days after an injury | all players | per recorded hour | 0.87 (0.82 to 0.92) | 1.14 (1.08 to 1.21) |
| Without 7 days after an injury | starters | per appearance | 0.86 (0.81 to 0.90) | 1.11 (1.05 to 1.18) |
| Without 7 days after an injury | starters | per starter hour | 0.82 (0.78 to 0.87) | 1.18 (1.11 to 1.25) |
| Without 28 days after an injury | all players | per appearance | 1.18 (1.12 to 1.25) | 1.06 (1.00 to 1.13) |
| Without 28 days after an injury | all players | per recorded hour | 0.87 (0.82 to 0.92) | 1.15 (1.08 to 1.22) |
| Without 28 days after an injury | starters | per appearance | 0.85 (0.80 to 0.90) | 1.12 (1.05 to 1.19) |
| Without 28 days after an injury | starters | per starter hour | 0.82 (0.77 to 0.87) | 1.18 (1.11 to 1.26) |
| Without the first match after a pause | all players | per appearance | 1.18 (1.12 to 1.25) | 1.07 (1.01 to 1.14) |
| Without the first match after a pause | all players | per recorded hour | 0.85 (0.80 to 0.90) | 1.13 (1.06 to 1.20) |
| Without the first match after a pause | starters | per appearance | 0.83 (0.79 to 0.88) | 1.09 (1.02 to 1.16) |
| Without the first match after a pause | starters | per starter hour | 0.80 (0.75 to 0.85) | 1.15 (1.08 to 1.22) |
| Reasons 99% complete or more | all players | per appearance | 1.12 (1.05 to 1.20) | 1.05 (0.98 to 1.13) |

| Analysis | Players | Denominator | Full match or more against the reference (95% CI) | Part of a match against the reference (95% CI) |
|---|---|---|---|---|
| Reasons 99% complete or more | all players | per recorded hour | 0.83 (0.77 to 0.88) | 1.13 (1.05 to 1.21) |
| Reasons 99% complete or more | starters | per appearance | 0.82 (0.77 to 0.88) | 1.11 (1.03 to 1.19) |
| Reasons 99% complete or more | starters | per starter hour | 0.79 (0.73 to 0.84) | 1.17 (1.09 to 1.25) |
| Unrecorded reasons as injuries | all players | per appearance | 1.15 (1.09 to 1.21) | 1.16 (1.10 to 1.22) |
| Unrecorded reasons as injuries | all players | per recorded hour | 0.85 (0.80 to 0.89) | 1.24 (1.18 to 1.31) |
| Unrecorded reasons as injuries | starters | per appearance | 0.83 (0.79 to 0.87) | 1.21 (1.15 to 1.28) |
| Unrecorded reasons as injuries | starters | per starter hour | 0.80 (0.76 to 0.84) | 1.28 (1.22 to 1.35) |
| Followed by a recorded absence | all players | per appearance | 1.29 (1.19 to 1.41) | 1.02 (0.93 to 1.12) |
| Followed by a recorded absence | all players | per recorded hour | 0.95 (0.87 to 1.04) | 1.09 (1.00 to 1.20) |
| Followed by a recorded absence | starters | per appearance | 0.97 (0.89 to 1.06) | 1.07 (0.97 to 1.18) |
| Followed by a recorded absence | starters | per starter hour | 0.93 (0.85 to 1.02) | 1.13 (1.02 to 1.24) |
| 2 minutes added to the first half, 5 to the second | all players | per recorded hour | 0.88 (0.83 to 0.93) | 1.16 (1.09 to 1.23) |
| 2 minutes added to the first half, 5 to the second | starters | per starter hour | 0.82 (0.77 to 0.86) | 1.19 (1.13 to 1.27) |
| 4 minutes added to the first half, 10 to the second | all players | per recorded hour | 0.89 (0.84 to 0.94) | 1.16 (1.10 to 1.23) |
| 4 minutes added to the first half, 10 to the second | starters | per starter hour | 0.81 (0.77 to 0.86) | 1.21 (1.14 to 1.28) |
| 6 minutes added to the first half, 15 to the second | all players | per recorded hour | 0.90 (0.85 to 0.95) | 1.17 (1.10 to 1.24) |
| 6 minutes added to the first half, 15 to the second | starters | per starter hour | 0.81 (0.76 to 0.86) | 1.22 (1.15 to 1.29) |
| 5 minutes added at the end | all players | per recorded hour | 0.88 (0.83 to 0.93) | 1.16 (1.10 to 1.23) |
| 5 minutes added at the end | starters | per starter hour | 0.82 (0.77 to 0.86) | 1.20 (1.13 to 1.27) |
| 10 minutes added at the end | all players | per recorded hour | 0.89 (0.84 to 0.94) | 1.17 (1.10 to 1.23) |
| 10 minutes added at the end | starters | per starter hour | 0.81 (0.77 to 0.86) | 1.21 (1.14 to 1.28) |
| 15 minutes added at the end | all players | per recorded hour | 0.90 (0.85 to 0.95) | 1.17 (1.11 to 1.24) |
| 15 minutes added at the end | starters | per starter hour | 0.81 (0.76 to 0.85) | 1.22 (1.15 to 1.30) |
| Same starters: as analysed | starters | per starter hour | 1.00 (0.94 to 1.06) | 1.00 (0.94 to 1.06) |
| Same starters: reference, the calendar | starters | per starter hour | 1.03 (0.97 to 1.10) | 1.03 (0.96 to 1.09) |
| Same starters: reference, the calendar after 8 to 13 days, longer breaks as terms | starters | per starter hour | 0.99 (0.93 to 1.06) | 0.99 (0.92 to 1.06) |
| Same starters: reference, the calendar after 8 to 13 days, longer breaks left out | starters | per starter hour | 0.99 (0.92 to 1.06) | 0.99 (0.92 to 1.06) |
| Same starters: first injury-coded substitution only | starters | per starter hour | 1.03 (0.94 to 1.13) | 0.92 (0.83 to 1.01) |
| Same starters: strata by player and league-season | starters | per starter hour | 1.11 (1.04 to 1.18) | 0.95 (0.90 to 1.02) |
| Same starters: without 28 days after an injury | starters | per starter hour | 1.00 (0.94 to 1.06) | 1.02 (0.95 to 1.08) |
| Same starters: without 84 days after an injury | starters | per starter hour | 1.00 (0.94 to 1.07) | 1.03 (0.96 to 1.10) |
| Same starters: reasons 99% complete or more | starters | per starter hour | 0.99 (0.92 to 1.06) | 0.97 (0.90 to 1.04) |
| Same starters: unrecorded reasons as injuries | starters | per starter hour | 0.99 (0.94 to 1.05) | 1.05 (0.99 to 1.11) |
| Same starters: without the first match after a pause | starters | per starter hour | 0.97 (0.91 to 1.03) | 0.97 (0.91 to 1.03) |
| Same starters: followed by a recorded absence | starters | per starter hour | 1.14 (1.04 to 1.25) | 0.99 (0.90 to 1.10) |

**Table S21.** The days since the club's previous match beside the starter's minutes in it, as terms of one model, on outfield starters with every match counted: rate ratios with 95% CIs. The within columns compare the same starter's weeks in the same league-season, by conditional Poisson regression. The across columns compare starters, by Poisson regression with league-season terms. That model uses the log of starter hours as its offset, with standard errors clustered by player. The references are a gap of six to seven days since the club's previous match in any competition, and 90 minutes or more in that match. Starts are each band's starts; 40 starts with no previous club match on file, or one under two days earlier, are left out. The last row gives the events and the starters each fit uses. Each ratio reads as safer, riskier or no clear difference by the paper's rule.

| Term | Starts | Within, injury-coded | Within, with a recorded absence | Across, injury-coded | Across, with a recorded absence |
|---|---|---|---|---|---|
| 2 to 3 days since the previous match | 33 003 | 1.05 (0.97 to 1.13) | 1.40 (1.23 to 1.59) | 0.95 (0.88 to 1.03) | 1.32 (1.17 to 1.49) |
| 4 to 5 days | 38 288 | 1.02 (0.95 to 1.09) | 1.19 (1.05 to 1.33) | 0.98 (0.91 to 1.05) | 1.21 (1.08 to 1.35) |
| 8 to 13 days | 31 830 | 1.02 (0.95 to 1.10) | 1.06 (0.94 to 1.21) | 1.01 (0.94 to 1.08) | 1.06 (0.94 to 1.20) |
| 14 days or more | 21 380 | 0.85 (0.78 to 0.93) | 0.89 (0.77 to 1.03) | 0.88 (0.81 to 0.96) | 1.01 (0.87 to 1.16) |
| none of the previous match | 26 358 | 0.93 (0.86 to 1.01) | 0.70 (0.61 to 0.81) | 1.35 (1.25 to 1.45) | 1.05 (0.93 to 1.19) |
| 1 to 59 minutes in it | 34 609 | 0.83 (0.77 to 0.90) | 0.65 (0.58 to 0.74) | 1.30 (1.21 to 1.40) | 1.10 (0.98 to 1.24) |
| 60 to 89 minutes in it | 42 838 | 0.89 (0.83 to 0.95) | 0.88 (0.78 to 0.99) | 1.25 (1.17 to 1.34) | 1.13 (1.02 to 1.26) |
| Events (starters) | – | 6709 (4671) | 2537 (2153) | 6709 (4590) | 2537 (4590) |

**Table S22.** The estimators in a simulation. Each setting was simulated 100 times, and each run is a replicate. A replicate keeps every outfield start of the advised comparison and draws its event anew. Each start keeps its load group, calendar band, starter hours, player and league-season, and whether its starter completed the club's previous league match. The settings are assumptions, not estimates. The true ratio is the ratio a heavy week carries in the simulation. The knock is the factor by which not completing the club's previous league match raises the next start's risk. The selection is the fall in a player's level, in standard deviations, per standard deviation of the player's share of loaded starts. Across starters is the Mantel–Haenszel ratio per starter hour against longer-gap weeks. Within, longer-gap weeks is the advised comparison. Within, after a completed match sets heavy weeks against longer-gap weeks after a completed match. Mean is the geometric mean of the estimates. Against the truth is that mean's ratio to the true ratio. Coverage is the share of replicates whose 95% CI holds the true ratio. The last three columns give the shares read safer, no clear difference and riskier.

| True ratio | Knock | Selection | Estimator | Mean | Against the truth | Coverage, % | Safer, % | No clear difference, % | Riskier, % |
|---|---|---|---|---|---|---|---|---|---|
| 1.00 | 1.00 | 0.0 | across starters | 1.00 | 1.00 | 94 | 3 | 94 | 3 |
| | | | within, longer-gap weeks | 1.00 | 1.00 | 97 | 2 | 97 | 1 |
| | | | within, after a completed match | 1.00 | 1.00 | 93 | 4 | 93 | 3 |
| 1.00 | 1.00 | 0.3 | across starters | 0.93 | 0.93 | 48 | 52 | 48 | 0 |
| | | | within, longer-gap weeks | 1.00 | 1.00 | 92 | 2 | 92 | 6 |
| | | | within, after a completed match | 1.00 | 1.00 | 96 | 0 | 96 | 4 |
| 1.00 | 1.10 | 0.0 | across starters | 0.96 | 0.96 | 81 | 19 | 81 | 0 |
| | | | within, longer-gap weeks | 0.97 | 0.97 | 84 | 16 | 84 | 0 |
| | | | within, after a completed match | 1.01 | 1.01 | 96 | 0 | 96 | 4 |
| 1.00 | 1.10 | 0.3 | across starters | 0.89 | 0.89 | 7 | 93 | 7 | 0 |
| | | | within, longer-gap weeks | 0.96 | 0.96 | 81 | 19 | 81 | 0 |
| | | | within, after a completed match | 1.00 | 1.00 | 93 | 4 | 93 | 3 |
| 1.00 | 1.25 | 0.0 | across starters | 0.91 | 0.91 | 22 | 78 | 22 | 0 |
| | | | within, longer-gap weeks | 0.92 | 0.92 | 33 | 67 | 33 | 0 |
| | | | within, after a completed match | 1.01 | 1.01 | 97 | 1 | 97 | 2 |
| 1.00 | 1.25 | 0.3 | across starters | 0.84 | 0.84 | 1 | 99 | 1 | 0 |
| | | | within, longer-gap weeks | 0.91 | 0.91 | 25 | 75 | 25 | 0 |
| | | | within, after a completed match | 1.01 | 1.01 | 97 | 0 | 97 | 3 |
| 1.10 | 1.00 | 0.0 | across starters | 1.10 | 1.00 | 94 | 0 | 22 | 78 |
| | | | within, longer-gap weeks | 1.10 | 1.00 | 95 | 0 | 27 | 73 |
| | | | within, after a completed match | 1.10 | 1.00 | 96 | 0 | 41 | 59 |
| 1.10 | 1.00 | 0.3 | across starters | 1.02 | 0.93 | 41 | 0 | 88 | 12 |
| | | | within, longer-gap weeks | 1.10 | 1.00 | 95 | 0 | 24 | 76 |
| | | | within, after a completed match | 1.10 | 1.00 | 98 | 0 | 44 | 56 |
| 1.10 | 1.10 | 0.0 | across starters | 1.04 | 0.95 | 62 | 0 | 78 | 22 |
| | | | within, longer-gap weeks | 1.05 | 0.95 | 75 | 0 | 78 | 22 |
| | | | within, after a completed match | 1.10 | 1.00 | 98 | 0 | 45 | 55 |
| 1.10 | 1.10 | 0.3 | across starters | 0.98 | 0.89 | 5 | 6 | 94 | 0 |
| | | | within, longer-gap weeks | 1.06 | 0.96 | 86 | 0 | 70 | 30 |
| | | | within, after a completed match | 1.10 | 1.00 | 100 | 0 | 40 | 60 |
| 1.10 | 1.25 | 0.0 | across starters | 0.99 | 0.90 | 10 | 4 | 96 | 0 |
| | | | within, longer-gap weeks | 1.00 | 0.91 | 24 | 1 | 95 | 4 |
| | | | within, after a completed match | 1.11 | 1.01 | 96 | 0 | 39 | 61 |
| 1.10 | 1.25 | 0.3 | across starters | 0.92 | 0.83 | 0 | 68 | 32 | 0 |
| | | | within, longer-gap weeks | 1.00 | 0.91 | 22 | 1 | 97 | 2 |
| | | | within, after a completed match | 1.12 | 1.01 | 94 | 0 | 37 | 63 |

# S7 Intervals and heterogeneity

Table S23 gives every pooled estimate. The fixed pool weights each sample by the inverse of its variance, so that more precise samples count more. It is a check: the script stops unless this pool matches the value first deposited. The paper reports the random-effects pool instead. That pool allows each sample its own true value around a common level. It is estimated by restricted maximum likelihood, REML in Table S23. Its interval is the Hartung–Knapp–Sidik–Jonkman interval (IntHout and colleagues, BMC Medical Research Methodology 2014). Table S23 calls it the Hartung–Knapp interval. That interval uses the $t$ distribution with $k-1$ degrees of freedom, where $k$ is the number of samples pooled. It does not understate the uncertainty when few samples are pooled. Earlier versions of the paper printed DerSimonian–Laird pools, another random-effects method. They are in the archive. The paper also reports the 95% prediction interval for a new league, with $k-2$ degrees of freedom. The standard deviation between the samples' true values, with chance removed, is $\tau$. The share of the spread between their estimates that chance does not explain is $I^2$. The within-player slope is pooled over football for the main estimate, over men's and women's football apart, and over every sport as corroboration.

**Table S23.** Every pooled estimate: the fixed pool, the REML pool with its Hartung–Knapp interval and its prediction interval, the number of samples, $I^2$ and $\tau$. Substitution-model rows are in log-odds per extra full match, and the next start in log-odds. The slopes and gradients are in log minutes. The injury-coded rate is per starter appearance. The rows without a prediction interval are single estimates, with Wald intervals. They are in log rate ratios or log odds.

| Quantity | Fixed | REML | 95% CI (Hartung–Knapp) | 95% prediction interval | $k$ | $I^2$ | $\tau$ |
|---|---|---|---|---|---|---|---|
| Substitution model, fifteen study samples, three substitutions | −0.350 | −0.353 | −0.376 to −0.331 | −0.410 to −0.297 | 15 | 0.21 | 0.024 |
| Substitution model, worldwide survey, three substitutions | −0.362 | −0.363 | −0.382 to −0.344 | −0.466 to −0.260 | 49 | 0.66 | 0.050 |
| Substitution model, survey without this study's leagues, three | −0.359 | −0.360 | −0.380 to −0.340 | −0.464 to −0.256 | 45 | 0.67 | 0.051 |
| Substitution model, worldwide survey, five substitutions | −0.387 | −0.385 | −0.399 to −0.370 | −0.500 to −0.269 | 91 | 0.74 | 0.058 |
| Substitution model, survey without this study's leagues, five | −0.386 | −0.384 | −0.399 to −0.369 | −0.499 to −0.269 | 86 | 0.73 | 0.057 |
| Injury-coded substitutions per starter appearance | 0.0320 | 0.0323 | 0.0309 to 0.0336 | 0.0263 to 0.0382 | 27 | 0.68 | 0.0028 |
| Next start after an early substitution, by league | −2.117 | −2.133 | −2.224 to −2.042 | −2.473 to −1.794 | 14 | 0.91 | 0.150 |
| Within-player slope, 25 samples | 0.005 | 0.003 | 0.000 to 0.006 | −0.007 to 0.013 | 25 | 0.59 | 0.005 |
| Within-player slope, football, 23 samples | 0.005 | 0.004 | 0.001 to 0.006 | −0.006 to 0.013 | 23 | 0.57 | 0.005 |
| Within-player slope, men's football | 0.005 | 0.005 | 0.001 to 0.008 | −0.005 to 0.014 | 14 | 0.62 | 0.004 |
| Within-player slope, women's football | 0.001 | 0.001 | −0.006 to 0.007 | −0.014 to 0.016 | 9 | 0.42 | 0.006 |
| Substitution model, eight European leagues, all starters, three | −0.348 | −0.352 | −0.391 to −0.313 | −0.442 to −0.262 | 8 | 0.52 | 0.033 |
| Substitution model, eight European leagues, two-way clustered, three | −0.348 | −0.352 | −0.390 to −0.314 | −0.449 to −0.256 | 8 | 0.64 | 0.036 |
| Substitution model, eight European leagues, without goalkeepers, three | −0.316 | −0.320 | −0.359 to −0.281 | −0.413 to −0.227 | 8 | 0.55 | 0.034 |
| Substitution model, eight European leagues, with position, three | −0.297 | −0.301 | −0.334 to −0.267 | −0.374 to −0.227 | 8 | 0.39 | 0.026 |
| Substitution model, eight European leagues, within the same player, three | −0.104 | −0.104 | −0.122 to −0.086 | −0.122 to −0.085 | 8 | 0.00 | 0.000 |
| Substitution model, part of a match against none, across starters, three | 0.751 | 0.751 | 0.695 to 0.806 | 0.624 to 0.877 | 8 | 0.49 | 0.046 |
| Substitution model, a full match or more against none, across starters, three | −0.537 | −0.541 | −0.587 to −0.495 | −0.645 to −0.437 | 8 | 0.46 | 0.038 |
| Substitution model, part of a match against none, within the same player, three | 0.089 | 0.088 | 0.049 to 0.127 | 0.023 to 0.153 | 8 | 0.17 | 0.021 |
| Substitution model, a full match or more against none, within the same player, three | −0.164 | −0.164 | −0.183 to −0.144 | −0.184 to −0.143 | 8 | 0.00 | 0.000 |
| Substitution model, eight European leagues, within the same player, conditional, three | −0.102 | −0.102 | −0.120 to −0.084 | −0.120 to −0.084 | 8 | 0.00 | 0.000 |
| Substitution model, eight European leagues, within the same team sheet, three | −0.872 | −0.890 | −0.983 to −0.797 | −1.147 to −0.634 | 8 | 0.77 | 0.097 |
| Substitution model, eight European leagues, team sheet, outfield, position, three | −0.751 | −0.758 | −0.833 to −0.683 | −0.941 to −0.575 | 8 | 0.57 | 0.068 |
| Substitution model, eight European leagues, with the score, three | −0.348 | −0.352 | −0.390 to −0.313 | −0.442 to −0.261 | 8 | 0.52 | 0.033 |
| Substitution model, part of a match against calendar rest, across starters, three | 0.810 | 0.809 | 0.759 to 0.859 | 0.712 to 0.907 | 8 | 0.31 | 0.034 |
| Substitution model, a full match or more against calendar rest, across starters, three | −0.480 | −0.484 | −0.529 to −0.439 | −0.584 to −0.385 | 8 | 0.43 | 0.036 |
| Substitution model, absent from the previous match against calendar rest, across starters, three | 0.268 | 0.270 | 0.191 to 0.348 | 0.080 to 0.459 | 8 | 0.58 | 0.070 |
| Substitution model, part of a match against calendar rest, within the same player, three | 0.126 | 0.125 | 0.086 to 0.163 | 0.060 to 0.189 | 8 | 0.12 | 0.021 |
| Substitution model, a full match or more against calendar rest, within the same player, three | −0.124 | −0.124 | −0.138 to −0.109 | −0.139 to −0.109 | 8 | 0.00 | 0.000 |

| Quantity | Fixed | REML | 95% CI (Hartung–Knapp) | 95% prediction interval | $k$ | $I^2$ | $\tau$ |
|---|---|---|---|---|---|---|---|
| Substitution model, absent from the previous match against calendar rest, within the same player, three | 0.188 | 0.188 | 0.144 to 0.231 | 0.143 to 0.232 | 8 | 0.00 | 0.000 |
| Substitution model, to the 60th minute, three | −0.277 | −0.280 | −0.326 to −0.233 | −0.356 to −0.204 | 8 | 0.00 | 0.024 |
| Substitution model, 61st to 75th minute, three | −0.422 | −0.427 | −0.475 to −0.379 | −0.530 to −0.324 | 8 | 0.36 | 0.037 |
| Substitution model, after the 75th minute, three | −0.322 | −0.321 | −0.378 to −0.264 | −0.451 to −0.191 | 8 | 0.51 | 0.048 |
| Substitution model, eight European leagues, all starters, five | −0.370 | −0.370 | −0.392 to −0.349 | −0.407 to −0.333 | 8 | 0.28 | 0.012 |
| Substitution model, eight European leagues, two-way clustered, five | −0.369 | −0.370 | −0.392 to −0.348 | −0.412 to −0.328 | 8 | 0.37 | 0.014 |
| Substitution model, eight European leagues, without goalkeepers, five | −0.335 | −0.336 | −0.359 to −0.313 | −0.376 to −0.296 | 8 | 0.36 | 0.013 |
| Substitution model, eight European leagues, with position, five | −0.324 | −0.325 | −0.352 to −0.299 | −0.382 to −0.269 | 8 | 0.49 | 0.020 |
| Substitution model, eight European leagues, within the same player, five | −0.156 | −0.156 | −0.175 to −0.137 | −0.176 to −0.136 | 8 | 0.02 | 0.000 |
| Substitution model, part of a match against none, across starters, five | 0.678 | 0.681 | 0.641 to 0.721 | 0.586 to 0.775 | 8 | 0.51 | 0.035 |
| Substitution model, a full match or more against none, across starters, five | −0.661 | −0.662 | −0.697 to −0.627 | −0.740 to −0.583 | 8 | 0.45 | 0.028 |
| Substitution model, part of a match against none, within the same player, five | 0.123 | 0.123 | 0.102 to 0.144 | 0.099 to 0.147 | 8 | 0.00 | 0.005 |
| Substitution model, a full match or more against none, within the same player, five | −0.258 | −0.258 | −0.281 to −0.236 | −0.281 to −0.235 | 8 | 0.00 | 0.000 |
| Substitution model, eight European leagues, within the same player, conditional, five | −0.154 | −0.154 | −0.173 to −0.135 | −0.174 to −0.134 | 8 | 0.04 | 0.000 |
| Substitution model, eight European leagues, within the same team sheet, five | −0.975 | −0.999 | −1.106 to −0.892 | −1.310 to −0.687 | 8 | 0.87 | 0.119 |
| Substitution model, eight European leagues, team sheet, outfield, position, five | −0.859 | −0.880 | −0.985 to −0.775 | −1.181 to −0.579 | 8 | 0.85 | 0.115 |
| Substitution model, eight European leagues, with the score, five | −0.371 | −0.372 | −0.394 to −0.349 | −0.411 to −0.332 | 8 | 0.31 | 0.013 |
| Substitution model, part of a match against calendar rest, across starters, five | 0.703 | 0.704 | 0.666 to 0.743 | 0.623 to 0.786 | 8 | 0.39 | 0.029 |
| Substitution model, a full match or more against calendar rest, across starters, five | −0.637 | −0.638 | −0.680 to −0.596 | −0.745 to −0.531 | 8 | 0.62 | 0.040 |
| Substitution model, absent from the previous match against calendar rest, across starters, five | 0.137 | 0.146 | 0.069 to 0.223 | −0.062 to 0.353 | 8 | 0.74 | 0.078 |
| Substitution model, part of a match against calendar rest, within the same player, five | 0.150 | 0.150 | 0.128 to 0.172 | 0.119 to 0.180 | 8 | 0.01 | 0.008 |
| Substitution model, a full match or more against calendar rest, within the same player, five | −0.226 | −0.227 | −0.255 to −0.199 | −0.279 to −0.175 | 8 | 0.25 | 0.018 |
| Substitution model, absent from the previous match against calendar rest, within the same player, five | 0.174 | 0.185 | 0.108 to 0.263 | −0.018 to 0.388 | 8 | 0.71 | 0.076 |
| Substitution model, to the 60th minute, five | −0.335 | −0.334 | −0.374 to −0.295 | −0.407 to −0.261 | 8 | 0.27 | 0.025 |
| Substitution model, 61st to 75th minute, five | −0.450 | −0.450 | −0.473 to −0.427 | −0.474 to −0.426 | 8 | 0.00 | 0.000 |
| Substitution model, after the 75th minute, five | −0.313 | −0.316 | −0.361 to −0.271 | −0.429 to −0.202 | 8 | 0.64 | 0.042 |
| Within-player slope on the club's fixture list | −0.004 | −0.004 | −0.006 to −0.002 | −0.009 to 0.002 | 16 | 0.47 | 0.002 |
| Starters' gradient on the club's fixture list | −0.004 | −0.003 | −0.006 to −0.001 | −0.010 to 0.003 | 16 | 0.44 | 0.003 |
| Injury, same starter, heavy against light weeks (log) | | −0.001 | −0.059 to 0.058 | | 1 | | |
| Injury, across starters, always against never loaded (log) | | −2.707 | −3.010 to −2.404 | | 1 | | |
| Injury, within over across starters (log) | | 2.706 | 2.400 to 3.012 | | 1 | | |
| Injury, same starter, share over all starts, starts with 5 earlier or more (log) | | 0.004 | −0.056 to 0.064 | | 1 | | |
| Injury, across starters, share over all starts, starts with 5 earlier or more (log) | | −3.271 | −3.713 to −2.828 | | 1 | | |
| Injury, within over across, share over all starts, starts with 5 earlier or more (log) | | 3.275 | 2.830 to 3.720 | | 1 | | |
| Injury, same starter, share over earlier starts, starts with 5 earlier or more (log) | | −0.091 | −0.151 to −0.032 | | 1 | | |
| Injury, across starters, share over earlier starts, starts with 5 earlier or more (log) | | −1.326 | −1.654 to −0.998 | | 1 | | |
| Injury, within over across, share over earlier starts, starts with 5 earlier or more (log) | | 1.234 | 0.915 to 1.554 | | 1 | | |
| Injury, same starter, share over all starts, starts with 10 earlier or more (log) | | −0.011 | −0.073 to 0.052 | | 1 | | |
| Injury, across starters, share over all starts, starts with 10 earlier or more (log) | | −3.389 | −3.920 to −2.859 | | 1 | | |
| Injury, within over across, share over all starts, starts with 10 earlier or more (log) | | 3.379 | 2.846 to 3.911 | | 1 | | |
| Injury, same starter, share over earlier starts, starts with 10 earlier or more (log) | | −0.095 | −0.157 to −0.032 | | 1 | | |

| Quantity | Fixed | REML | 95% CI (Hartung–Knapp) | 95% prediction interval | $k$ | $I^2$ | $\tau$ |
|---|---|---|---|---|---|---|---|
| Injury, across starters, share over earlier starts, starts with 10 earlier or more (log) | | −1.680 | −2.094 to −1.267 | | 1 | | |
| Injury, within over across, share over earlier starts, starts with 10 earlier or more (log) | | 1.586 | 1.180 to 1.992 | | 1 | | |
| Next start within the same player, conditional (log odds) | | −2.064 | −2.092 to −2.036 | | 1 | | |
| Substitution model, survey with doubled minimum sizes, three | −0.361 | −0.362 | −0.381 to −0.342 | −0.465 to −0.259 | 42 | 0.68 | 0.050 |
| Substitution model, survey with doubled minimum sizes, five | −0.388 | −0.388 | −0.403 to −0.373 | −0.502 to −0.274 | 80 | 0.75 | 0.057 |
| Substitution model, survey without samples above their limit, three | −0.369 | −0.370 | −0.389 to −0.351 | −0.462 to −0.277 | 43 | 0.61 | 0.045 |
| Substitution model, survey without samples above their limit, five | −0.387 | −0.385 | −0.399 to −0.370 | −0.500 to −0.269 | 91 | 0.74 | 0.058 |
| Substitution model, survey without youth-rule samples, three | −0.357 | −0.358 | −0.381 to −0.335 | −0.467 to −0.249 | 37 | 0.67 | 0.053 |
| Substitution model, survey without youth-rule samples, five | −0.378 | −0.375 | −0.391 to −0.359 | −0.474 to −0.277 | 64 | 0.67 | 0.049 |

# S8 The worldwide survey, league by league

Figure S8 draws every fitted survey sample whose substitution limit could be placed. Table S24 lists every fitted sample, ordered by substitution limit and estimate. Unplaced samples hold the seasons whose mean substitutions per team and match fell between the readings for three and for five (Section S2). The same table is in the archive as a CSV file. Figure S9 sets each estimate against its precision. The spread of the survey is present among its largest samples too. But many samples narrow a pooled interval and leave a bias they all share (Bradley and colleagues, Nature 2021). The survey's samples all share one model. So the model's form was also checked against the observed fall in substitutions in the study's own samples (Table S4).

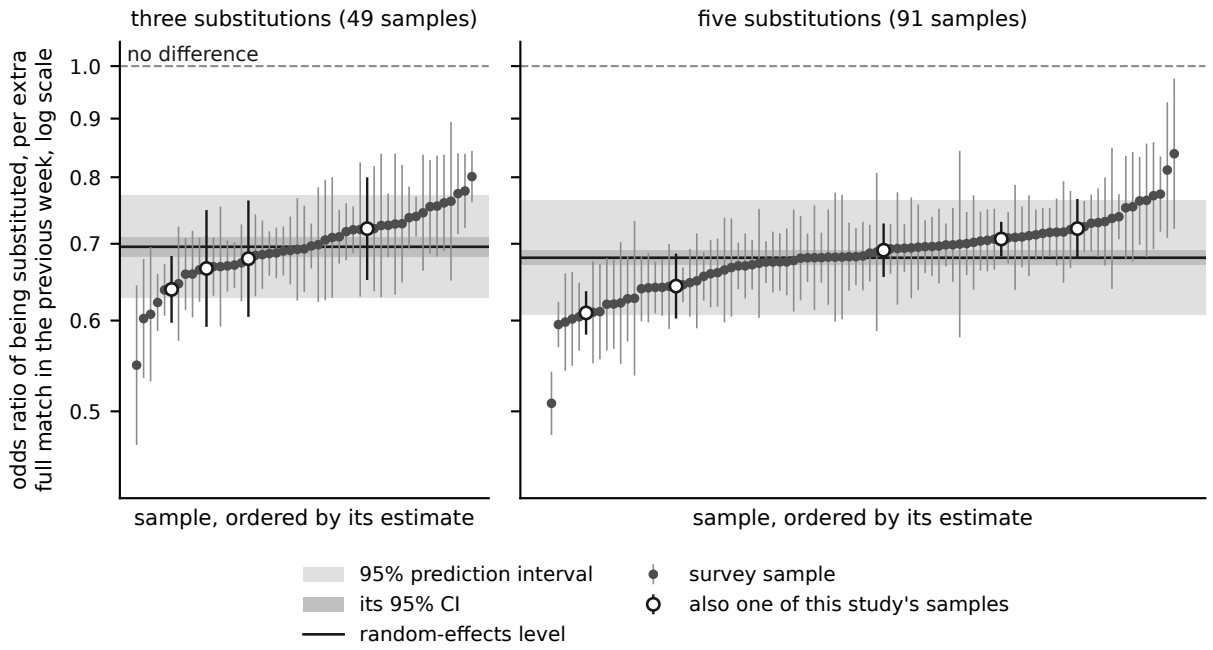



**Figure S8.** Coaches keep loaded starters on, in every men's first-tier league measured. The odds ratio of being substituted per extra full match in the previous week, with its 95% CI, by substitution limit. It is drawn for every men's first-tier sample of the worldwide survey whose limit could be placed. Each such sample met the survey's minimum sizes. Values below one mean that loaded starters are kept on. Open circles mark leagues also among this study's own samples. The lowest drawn estimates are South Korea's K League 1 under five substitutions and Vietnam's V.League 1 under three (Table S24). In the K League, a rule tied substitutions to young players. The solid line is the random-effects level, fitted by restricted maximum likelihood. The darker band is its Hartung–Knapp 95% CI, and the lighter band its 95% prediction interval. The prediction interval lies wholly below one under both limits. So a new league's own value would also be expected to lie below one. On the log-odds scale, the spread between leagues, $\tau$, was 0.050 under three substitutions and 0.058 under five. The 22 unplaced samples are not drawn. Their intervals too lie wholly below one: the highest upper end of any survey interval was 0.998, in an unplaced sample from Oman. The eight European men's leagues are not in the survey. There, with load in categories, a full match or more gave odds ratios of 0.48 to 0.62 against rested starters. But starters with part of a match were taken off more often, with odds ratios of 1.81 to 2.33 (Table S16). The survey's model treats every position alike, and goalkeepers are almost never substituted. In the eight European men's leagues, leaving out goalkeepers moved the odds ratio from 0.70 to 0.73 under three substitutions and from 0.69 to 0.71 under five.

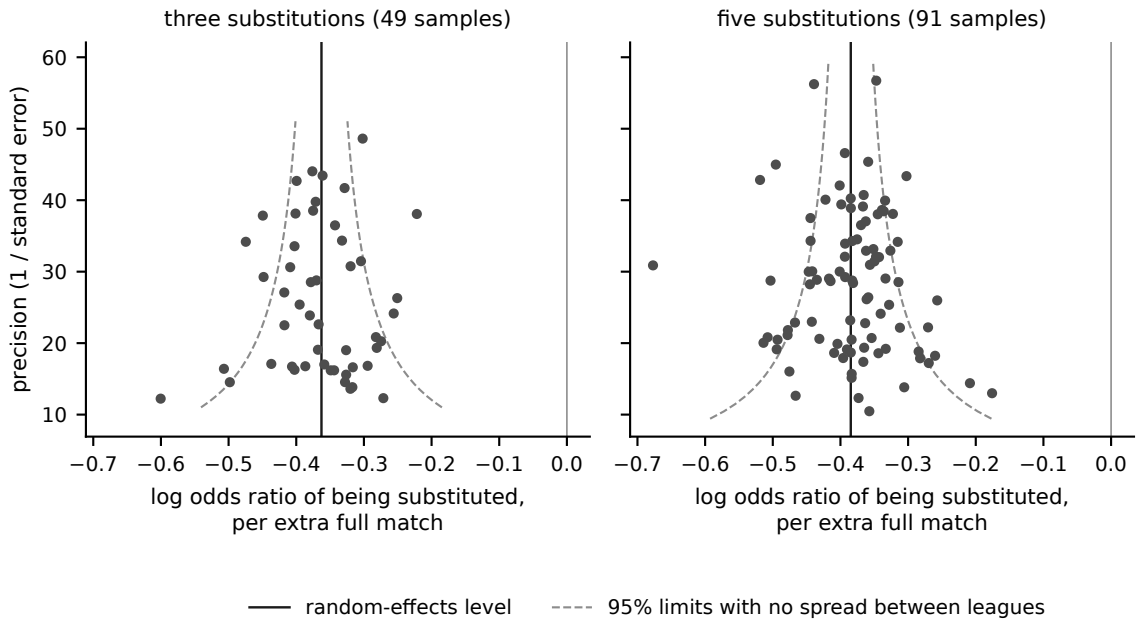



**Figure S9.** The spread of the worldwide survey is present among its largest samples too. Each sample's estimate against its precision, by substitution limit, on one log-odds scale. The solid line is the random-effects level, and the thin line at zero marks no difference. The dashed lines show where 95% of samples of each precision would fall if every league shared that level.

**Table S24.** The worldwide survey, one row per league and limit. The coefficient is the log odds ratio of substitution per extra full match, and SE is its standard error. Above the limit is the share of team-matches that used more substitutes than the limit read from the league's own data. Unplaced samples have no limit, so this column shows a dash for them. Youth rule counts seasons under a rule tied to players' age. Its letters give what the rule tied: the substitutions allowed (A), a starting place (B), a place on the pitch (C) or a quota of minutes (D). Section S2 gives the rules.

| Country | League | Limit | Seasons | Starts | Coefficient | SE | Above the limit (%) | Youth rule |
|---|---|---|---|---|---|---|---|---|
| India | Indian Super League | unplaced | 1 | 2398 | −0.728 | 0.092 | – | |
| Singapore | Singapore Premier League | unplaced | 3 | 7964 | −0.452 | 0.050 | – | 2 (C) |
| Hong Kong | Hong Kong Premier League | unplaced | 1 | 2376 | −0.448 | 0.100 | – | |
| Australia | A-League Men | unplaced | 1 | 3344 | −0.442 | 0.064 | – | |
| Qatar | Qatar Stars League | unplaced | 3 | 8668 | −0.424 | 0.048 | – | |
| Albania | Kategoria Superiore | unplaced | 5 | 19 228 | −0.413 | 0.032 | – | |
| Iceland | Besta Deild | unplaced | 2 | 5522 | −0.409 | 0.054 | – | |
| Saudi Arabia | Pro League | unplaced | 6 | 35 508 | −0.397 | 0.023 | – | |
| Ireland | League of Ireland Premier Division | unplaced | 1 | 3894 | −0.391 | 0.059 | – | |
| Slovakia | Nike Liga | unplaced | 2 | 5654 | −0.369 | 0.050 | – | |
| Senegal | Ligue 1 | unplaced | 1 | 5104 | −0.369 | 0.053 | – | |
| Estonia | Premium Liiga | unplaced | 2 | 7810 | −0.364 | 0.039 | – | |
| Nigeria | Nigeria Professional Football League | unplaced | 1 | 7854 | −0.363 | 0.049 | – | |
| Norway | Eliteserien | unplaced | 2 | 10 516 | −0.320 | 0.037 | – | |
| Lithuania | Toplyga | unplaced | 1 | 3850 | −0.317 | 0.068 | – | |
| Northern Ireland | Premiership | unplaced | 2 | 8580 | −0.315 | 0.041 | – | |
| Bulgaria | Efbet Liga | unplaced | 2 | 9548 | −0.310 | 0.040 | – | |
| Cyprus | Cyprus League | unplaced | 1 | 3938 | −0.306 | 0.065 | – | |
| Armenia | Premier League | unplaced | 2 | 4334 | −0.286 | 0.057 | – | 2 (A) |
| South Africa | Betway Premiership | unplaced | 1 | 5258 | −0.234 | 0.043 | – | |
| Wales | Cymru Premier | unplaced | 1 | 4202 | −0.230 | 0.056 | – | |
| Oman | Oman Jindal League | unplaced | 1 | 3344 | −0.156 | 0.079 | – | |
| South Korea | K League 1 | five | 5 | 22 814 | −0.677 | 0.032 | 0.1 | 5 (A) |
| Indonesia | Super League | five | 6 | 33 616 | −0.519 | 0.023 | 0.2 | 3 (C, D) |
| Taiwan | Taiwan Football Premier League | five | 4 | 6776 | −0.514 | 0.050 | 1.0 | |
| China | Chinese Super League | five | 2 | 7656 | −0.508 | 0.048 | 1.0 | 1 (C) |

| Country | League | Limit | Seasons | Starts | Coefficient | SE | Above the limit (%) | Youth rule |
|---|---|---|---|---|---|---|---|---|
| Vietnam | V.League 1 | five | 4 | 14 630 | −0.503 | 0.035 | 0.0 | |
| Japan | J1 League | five | 6 | 38 544 | −0.496 | 0.022 | 1.1 | |
| Paraguay | Primera Division Clausura | five | 2 | 5698 | −0.494 | 0.052 | 0.8 | |
| Canada | Canadian Premier League | five | 5 | 10 054 | −0.493 | 0.049 | 0.5 | 5 (D) |
| Latvia | Virsliga | five | 2 | 7832 | −0.478 | 0.047 | 0.0 | |
| Montenegro | Meridianbet 1. CFL | five | 2 | 7876 | −0.478 | 0.046 | 0.0 | 2 (C) |
| Albania | Kategoria Superiore | five | 1 | 3960 | −0.475 | 0.062 | 0.0 | |
| Paraguay | Primera Division Apertura | five | 3 | 8272 | −0.467 | 0.044 | 0.7 | |
| Jordan | Jordanian Pro League | five | 1 | 2794 | −0.466 | 0.079 | 0.0 | |
| Georgia | Erovnuli Liga | five | 5 | 15 576 | −0.447 | 0.033 | 0.0 | |
| Panama | Liga Panamena de Futbol Apertura | five | 6 | 10 604 | −0.445 | 0.035 | 0.1 | 5 (D) |
| Thailand | Thai League | five | 6 | 26 466 | −0.444 | 0.027 | 0.5 | |
| Finland | Veikkausliiga | five | 6 | 17 380 | −0.444 | 0.029 | 0.0 | |
| Jamaica | Jamaica Premier League | five | 1 | 5786 | −0.442 | 0.044 | 1.1 | |
| Australia | A-League Men | five | 4 | 13 552 | −0.442 | 0.033 | 0.6 | |
| United States | Major League Soccer | five | 5 | 52 998 | −0.439 | 0.018 | 0.7 | |
| Sweden | Allsvenskan | five | 3 | 15 488 | −0.435 | 0.035 | 0.2 | |
| Guatemala | National League Apertura | five | 3 | 7040 | −0.431 | 0.049 | 0.8 | 3 (D) |
| Israel | Ligat Haal | five | 6 | 23 760 | −0.422 | 0.025 | 0.0 | |
| Panama | Liga Panamena de Futbol Clausura | five | 7 | 13 464 | −0.417 | 0.034 | 0.0 | 5 (D) |
| Argentina | Torneo Apertura | five | 3 | 10 780 | −0.415 | 0.035 | 0.1 | |
| Honduras | Liga Nacional Apertura | five | 2 | 3850 | −0.409 | 0.054 | 0.0 | 2 (D) |
| Costa Rica | Primera Division Apertura | five | 2 | 4774 | −0.404 | 0.050 | 1.6 | 2 (D) |
| Romania | SuperLiga | five | 6 | 31 020 | −0.401 | 0.024 | 0.0 | 6 (B, C) |
| Guatemala | National League Clausura | five | 4 | 10 054 | −0.401 | 0.033 | 0.8 | 4 (D) |
| North Macedonia | Prva Makedonska Fudbalska Liga | five | 5 | 20 790 | −0.399 | 0.025 | 0.2 | 5 (C) |
| Colombia | Liga Dimayor Finalizacion | five | 1 | 4202 | −0.396 | 0.056 | 0.5 | |
| Poland | Ekstraklasa | five | 6 | 38 720 | −0.393 | 0.021 | 0.1 | 5 (C, D) |
| Slovenia | Prva Liga | five | 5 | 18 986 | −0.393 | 0.031 | 0.0 | |
| Belarus | Vysheyshaya Liga | five | 4 | 20 328 | −0.393 | 0.029 | 0.1 | 2 (C) |
| Ireland | League of Ireland Premier Division | five | 3 | 11 770 | −0.393 | 0.034 | 0.5 | |
| Myanmar | M League | five | 2 | 5742 | −0.390 | 0.052 | 0.0 | |
| Uzbekistan | Ozbekiston Superligasi | five | 2 | 8888 | −0.386 | 0.043 | 0.0 | |
| Bulgaria | Efbet Liga | five | 4 | 24 882 | −0.385 | 0.025 | 0.0 | |
| Cambodia | Cambodian Premier League | five | 2 | 6490 | −0.385 | 0.054 | 0.0 | |
| Serbia | SuperLiga | five | 5 | 26 114 | −0.385 | 0.026 | 0.1 | 5 (C) |
| Iraq | Iraq Stars League | five | 1 | 7612 | −0.384 | 0.049 | 0.0 | |
| Israel | Ligat Haal Relegation Round | five | 6 | 3674 | −0.384 | 0.066 | 0.0 | |
| Venezuela | Liga FUTVE Apertura | five | 3 | 4136 | −0.383 | 0.064 | 0.8 | |
| Armenia | Premier League | five | 5 | 17 204 | −0.383 | 0.035 | 0.0 | |
| Croatia | HNL | five | 6 | 23 738 | −0.383 | 0.029 | 0.0 | |
| India | Indian Super League | five | 5 | 13 002 | −0.381 | 0.035 | 0.1 | |
| Peru | Liga 1 Clausura | five | 6 | 17 512 | −0.375 | 0.029 | 0.0 | 2 (D) |
| Venezuela | Liga FUTVE Clausura | five | 2 | 2134 | −0.373 | 0.081 | 0.5 | |
| Mexico | Liga MX Clausura | five | 6 | 19 734 | −0.370 | 0.027 | 0.5 | 2 (D) |
| Austria | Bundesliga | five | 6 | 25 124 | −0.367 | 0.026 | 0.0 | |
| Argentina | Torneo Clausura | five | 2 | 5610 | −0.366 | 0.058 | 0.4 | |
| Hungary | Nemzeti Bajnoksag | five | 6 | 25 916 | −0.366 | 0.025 | 0.0 | |
| Bolivia | Division Profesional Clausura | five | 1 | 4092 | −0.365 | 0.052 | 0.0 | 1 (C) |
| Lithuania | Toplyga | five | 2 | 7898 | −0.363 | 0.044 | 0.0 | |
| Mexico | Liga MX Apertura | five | 6 | 19 844 | −0.363 | 0.027 | 0.3 | 2 (D) |
| El Salvador | Primera Division Clausura | five | 4 | 10 714 | −0.362 | 0.030 | 0.0 | 2 (C, D) |
| Costa Rica | Primera Division Clausura | five | 3 | 7546 | −0.361 | 0.038 | 0.4 | 3 (D) |
| Czechia | First League | five | 6 | 32 956 | −0.359 | 0.022 | 0.9 | |
| Colombia | Liga Dimayor Apertura | five | 2 | 8492 | −0.359 | 0.038 | 0.0 | |
| Qatar | Qatar Stars League | five | 1 | 2882 | −0.357 | 0.095 | 0.0 | |
| Kazakhstan | Premier Liga | five | 4 | 15 334 | −0.356 | 0.032 | 0.0 | |

| Country | League | Limit | Seasons | Starts | Coefficient | SE | Above the limit (%) | Youth rule |
|---|---|---|---|---|---|---|---|---|
| El Salvador | Primera Division Apertura | five | 2 | 5082 | −0.354 | 0.048 | 0.0 | 2 (C, D) |
| Uruguay | Liga AUF Clausura | five | 6 | 15 576 | −0.351 | 0.030 | 0.0 | |
| Uruguay | Liga AUF Apertura | five | 6 | 15 422 | −0.350 | 0.032 | 0.1 | |
| Norway | Eliteserien | five | 3 | 15 664 | −0.347 | 0.031 | 0.0 | |
| Brazil | Campeonato Brasileiro Serie A | five | 6 | 49 720 | −0.347 | 0.018 | 0.4 | |
| Chile | Liga de Primera | five | 5 | 26 840 | −0.345 | 0.026 | 0.2 | 5 (D) |
| Estonia | Premium Liiga | five | 1 | 3960 | −0.344 | 0.054 | 0.0 | |
| Peru | Liga 1 Apertura | five | 7 | 19 734 | −0.343 | 0.031 | 0.1 | 2 (D) |
| Kosovo | Superliga e Kosoves | five | 2 | 7766 | −0.341 | 0.041 | 0.0 | |
| Switzerland | Super League | five | 6 | 26 818 | −0.339 | 0.026 | 0.0 | |
| Bosnia and Herzegovina | Premijer Liga Bosne i Hercegovine | five | 6 | 25 146 | −0.336 | 0.026 | 0.0 | 2 (C) |
| Iran | Persian Gulf Pro League | five | 5 | 24 508 | −0.334 | 0.025 | 0.0 | |
| Azerbaijan | Premyer Liqa | five | 4 | 16 126 | −0.333 | 0.034 | 0.0 | |
| Hong Kong | Hong Kong Premier League | five | 4 | 7062 | −0.333 | 0.052 | 0.0 | |
| United Arab Emirates | UAE Pro League | five | 3 | 11 418 | −0.328 | 0.039 | 0.0 | |
| Argentina | Professional Football League 23/24 | five | 2 | 16 500 | −0.326 | 0.030 | 0.1 | |
| Egypt | Premier League | five | 4 | 20 944 | −0.322 | 0.026 | 0.0 | |
| Cyprus | Cyprus League | five | 5 | 19 844 | −0.315 | 0.029 | 0.0 | |
| Malaysia | Malaysia Super League | five | 4 | 13 552 | −0.314 | 0.035 | 0.1 | |
| Philippines | Philippines Football League | five | 4 | 7084 | −0.312 | 0.045 | 1.4 | |
| Slovakia | Nike Liga | five | 1 | 2860 | −0.306 | 0.072 | 0.0 | |
| South Africa | Betway Premiership | five | 5 | 25 564 | −0.302 | 0.023 | 0.0 | |
| Puerto Rico | Liga Puerto Rico Clausura | five | 2 | 4224 | −0.284 | 0.053 | 0.5 | |
| Uruguay | Liga AUF Intermedio | five | 5 | 5060 | −0.282 | 0.056 | 0.0 | |
| Bolivia | Division Profesional | five | 2 | 5368 | −0.271 | 0.045 | 0.2 | 1 (B, C) |
| Ecuador | Ligapro Serie A | five | 2 | 5456 | −0.269 | 0.058 | 0.0 | |
| Israel | Ligat Haal Championship Round | five | 6 | 3916 | −0.260 | 0.055 | 0.0 | |
| Algeria | Ligue 1 | five | 2 | 9526 | −0.257 | 0.038 | 0.0 | |
| Moldova | Super Liga | five | 3 | 3630 | −0.209 | 0.069 | 0.0 | |
| Puerto Rico | Liga Puerto Rico Apertura | five | 2 | 2024 | −0.176 | 0.077 | 0.0 | |
| Vietnam | V.League 1 | three | 1 | 3850 | −0.600 | 0.082 | 0.0 | |
| Slovakia | Nike Liga | three | 2 | 5698 | −0.507 | 0.061 | 0.0 | |
| Singapore | Singapore Premier League | three | 2 | 4180 | −0.498 | 0.069 | 0.0 | 2 (C) |
| Israel | Ligat Haal | three | 8 | 31 746 | −0.475 | 0.029 | 0.0 | |
| South Korea | K League 1 | three | 6 | 24 354 | −0.449 | 0.026 | 0.0 | 6 (A, B) |
| Australia | A-League Men | three | 7 | 20 416 | −0.448 | 0.034 | 0.0 | |
| Panama | Liga Panamena de Futbol Clausura | three | 2 | 3938 | −0.437 | 0.059 | 0.0 | |
| Finland | Veikkausliiga | three | 4 | 15 906 | −0.418 | 0.037 | 0.0 | |
| United States | Major League Soccer | three | 1 | 8910 | −0.417 | 0.044 | 0.0 | |
| Kazakhstan | Premier Liga | three | 6 | 22 902 | −0.409 | 0.033 | 0.0 | 3 (C) |
| Mexico | Liga MX Clausura | three | 2 | 5324 | −0.406 | 0.060 | 0.0 | 2 (D) |
| North Macedonia | Prva Makedonska Fudbalska Liga | three | 5 | 17 732 | −0.403 | 0.030 | 0.0 | 2 (C) |
| Belarus | Vysheyshaya Liga | three | 1 | 5236 | −0.403 | 0.061 | 0.0 | |
| Sweden | Allsvenskan | three | 6 | 31 504 | −0.401 | 0.026 | 0.0 | |
| Norway | Eliteserien | three | 8 | 41 778 | −0.399 | 0.023 | 0.0 | |
| China | Chinese Super League | three | 3 | 15 114 | −0.395 | 0.039 | 0.0 | 1 (B, C) |
| Japan | J1 League | three | 1 | 6666 | −0.387 | 0.060 | 0.0 | |
| Armenia | Premier League | three | 4 | 10 274 | −0.380 | 0.042 | 13.7 | 1 (A) |
| Croatia | HNL | three | 5 | 19 646 | −0.378 | 0.035 | 3.0 | |
| Iran | Persian Gulf Pro League | three | 9 | 51 040 | −0.376 | 0.023 | 0.0 | |
| Serbia | SuperLiga | three | 5 | 29 546 | −0.375 | 0.026 | 1.8 | |
| Hungary | Nemzeti Bajnoksag | three | 8 | 37 378 | −0.371 | 0.025 | 2.2 | |
| Nigeria | Nigeria Professional Football League | three | 2 | 15 576 | −0.370 | 0.035 | 42.4 | |
| Bosnia and Herzegovina | Premijer Liga Bosne i Hercegovine | three | 2 | 7150 | −0.368 | 0.052 | 0.0 | |
| Peru | Liga 1 Apertura | three | 3 | 8448 | −0.367 | 0.044 | 0.0 | 3 (D) |
| Poland | Ekstraklasa | three | 6 | 37 400 | −0.361 | 0.023 | 0.0 | 1 (C) |
| India | Indian Super League | three | 3 | 5676 | −0.359 | 0.059 | 0.0 | |
| Ireland | League of Ireland Premier Division | three | 1 | 3916 | −0.349 | 0.062 | 0.0 | |
| Saudi Arabia | Pro League | three | 1 | 5280 | −0.344 | 0.062 | 0.0 | |

| Country | League | Limit | Seasons | Starts | Coefficient | SE | Above the limit (%) | Youth rule |
|---|---|---|---|---|---|---|---|---|
| Austria | Bundesliga | three | 8 | 31 086 | −0.343 | 0.027 | 3.5 | |
| Switzerland | Super League | three | 6 | 23 320 | −0.332 | 0.029 | 0.0 | |
| Romania | SuperLiga | three | 8 | 38 830 | −0.329 | 0.024 | 0.0 | 4 (B, C) |
| Albania | Kategoria Superiore | three | 1 | 3916 | −0.328 | 0.069 | 13.5 | |
| Argentina | Professional Football League 23/24 | three | 1 | 7128 | −0.326 | 0.053 | 0.3 | |
| Paraguay | Primera Division Apertura | three | 2 | 4774 | −0.326 | 0.064 | 0.0 | |
| Paraguay | Primera Division Clausura | three | 2 | 3410 | −0.320 | 0.073 | 0.0 | |
| Czechia | First League | three | 4 | 20 790 | −0.320 | 0.033 | 3.5 | |
| Georgia | Erovnuli Liga | three | 1 | 3806 | −0.317 | 0.072 | 0.0 | |
| South Africa | Betway Premiership | three | 1 | 5236 | −0.316 | 0.060 | 0.0 | |
| Mexico | Liga MX Apertura | three | 4 | 13 662 | −0.304 | 0.032 | 0.0 | 2 (D) |
| Bulgaria | Efbet Liga | three | 9 | 43 846 | −0.302 | 0.021 | 1.2 | |
| Northern Ireland | Premiership | three | 1 | 4180 | −0.295 | 0.059 | 44.5 | |
| Estonia | Premium Liiga | three | 3 | 10 450 | −0.282 | 0.048 | 0.0 | |
| Azerbaijan | Premyer Liqa | three | 4 | 9042 | −0.281 | 0.052 | 0.0 | |
| Qatar | Qatar Stars League | three | 3 | 8426 | −0.275 | 0.049 | 22.2 | |
| Israel | Ligat Haal Championship Round | three | 4 | 2552 | −0.271 | 0.081 | 0.0 | |
| Peru | Liga 1 Clausura | three | 3 | 8536 | −0.256 | 0.041 | 0.0 | 3 (D) |
| Cyprus | Cyprus League | three | 4 | 13 596 | −0.251 | 0.038 | 0.0 | |
| Wales | Cymru Premier | three | 6 | 24 640 | −0.222 | 0.026 | 17.8 | |

# S9 Other leagues and sports

The study's measurements were repeated beyond football, in every league of a panel. The panel is a set of leagues in other team sports. It holds the men's Australian Football League (AFL) and leagues of basketball, ice hockey, American football, rugby league, baseball and cricket. The panel was built to take our earlier preprint (Ricou and Mahony, arXiv:2608.11228v3) beyond football. Its rules for admitting a league were written for that work, not for this study. These analyses were planned on 28 September 2026, before any of their results was read. But the first author's nine-sport preprint (arXiv:2609.05584v1) had already reported related results for some of these sports. The analyses are exploratory: none is a pre-set criterion. The plan, and every later change with its reason, is in the archive.

**The records.** The panel holds one row per player per game. Time played is in the sport's own unit. It is minutes in basketball, ice hockey, rugby league and Australian rules, and snaps in American football. In baseball it is outs for a pitcher and plate appearances for a batter. In cricket it is balls bowled or faced. Recent match load is the time played in the seven days before the game, counted in full games of the sport.

**Who starts, and within the same player.** Where the panel names the starters, the gradient was split exactly as in football (Section S3). The within-player slope was fitted among starters, or among all appearances where no starter is named. Intervals come from 1000 bootstrap draws over players.

**Groups whose time is fixed.** In a group whose starters play the whole game, a starter's time cannot rise with load. The whole gradient among that group's players is then due to who starts.

**When loaded starters come off.** A starter's first exit is the first time the starter leaves the field in a game. The model is a discrete-time hazard, like the substitution model of Section S3. It gives the odds of the first exit in each twelfth of the regulation game, or in each inning for a pitcher. These odds are for starters still on, given recent match load. A starter still on at the end of regulation counts as staying on.

The time of each exit comes from each league's own play-by-play, from the NRL's match feed, or, for a pitcher, from the outs recorded. Play-by-play for the WNBA, the G League, South Korea's KBL, Italy's Serie A and France's Betclic ÉLITE was gathered for this study. The regular seasons 2023-24 to 2025-26 of Spain's Liga ACB and 2021-22 to 2025-26 of Serie A lie outside the panel. The ACB's minutes were rebuilt from its plays, and Serie A's come from its box scores. Both were held to the panel's check that each side's minutes add up to five players for the whole game. Every team-game of both passed it. Their loads follow the panel's rule.

**Injury within the same player.** Where the panel records injuries, a conditional Poisson model linked each player's injury rate to the same player's recent match load, per extra full game. The time played was the exposure, the time at risk. Beside it, a Poisson model with season terms compared different players.

**Results.** Where players are substituted and the starters are named, who starts explained 31% to 95% of the gradient (Figure S10 and Table S25). In cricket, nearly every appearance is a start. Who starts explained at most 0.5% of the gradient there. That gradient comes from differences between players, such as their place in the batting order. Table S25 has one row per player group: a league, or its batters, pitchers or bowlers. In all 54 groups, the within-player interval lay wholly below the gradient on the same rows. Outside cricket, though, the within-player slope was 10% to 57% of that gradient. So in those sports a player's own heavy weeks do go with more time played. But they go with far less than the comparison across players suggests.

Loaded starters stayed on longer in 15 of the 15 competitions whose records time the first exit (Figure S12 and Table S26). On injury, by point estimates, heavy weeks looked safer across players than within the same player in 9 of the 10 player groups with an injury record (Figure S11 and Table S27). The B.LEAGUE was the exception. Football showed the same, and there the two intervals lay apart. So by point estimates, with football counted, heavy weeks looked safer across players than within the same player in 10 of 11 groups. The WNBA is the one women's league with an injury record. There the ratio within the same player was 0.95 (95% CI 0.86 to 1.06): no clear difference.

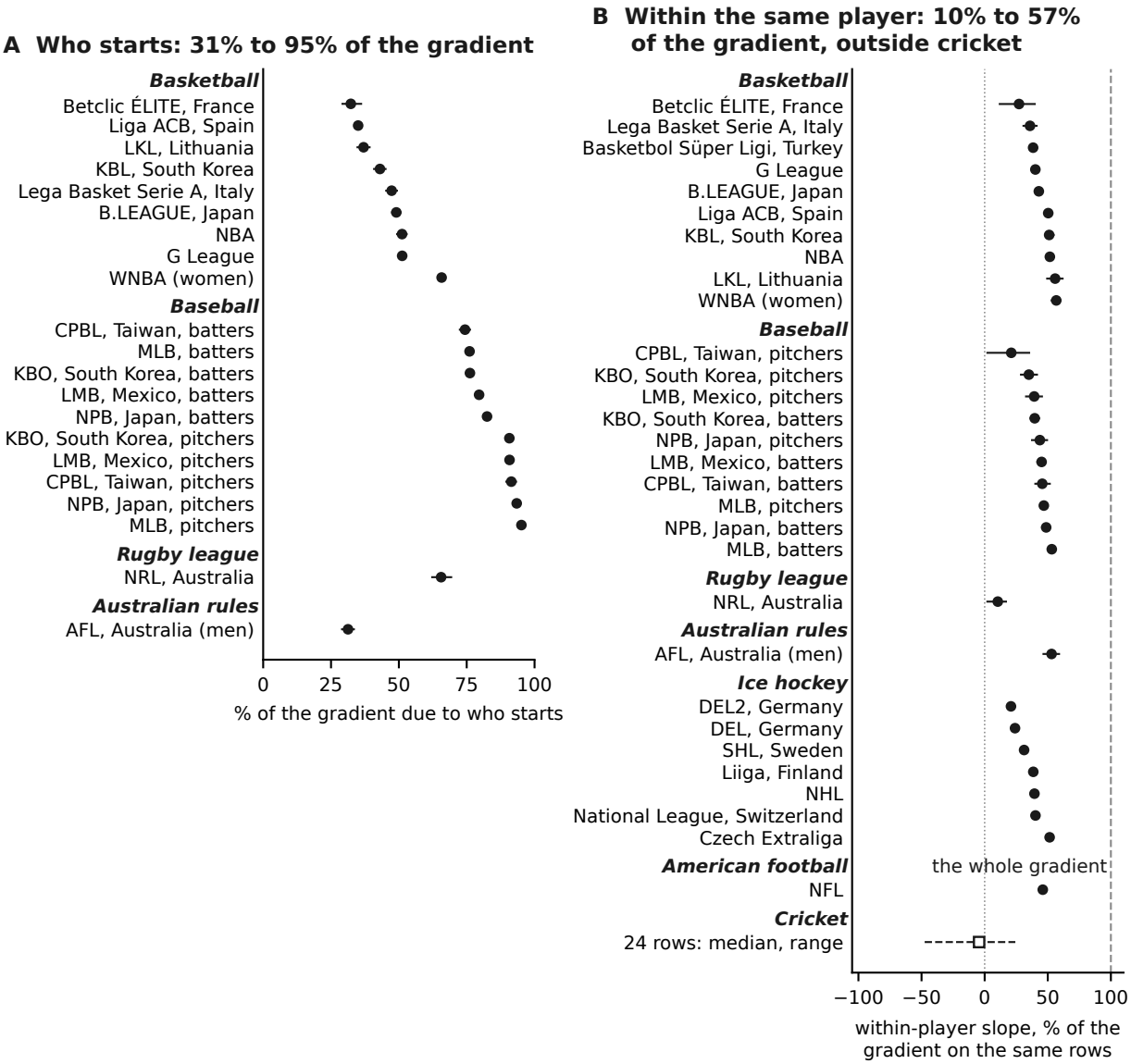



**Figure S10.** Who starts and within the same player, in the other leagues. (A) The share of the gradient due to who starts, with its 95% CI, where players are substituted and the starters are named. In cricket nearly every appearance is a start, and who starts explained at most 0.5% (Table S25). (B) The within-player slope as a percentage of the gradient on the same rows, with its 95% CI. The dashed line at 100% marks where a player's own heavy weeks would raise time played as much as the comparison across players does. The dotted line marks zero. The cricket rows are drawn as one row, an open square at their median. A dashed line runs from the lowest of them to the highest. It is a range, not an interval. In several cricket leagues the batters' slopes fall below zero. Table S25 gives every row.

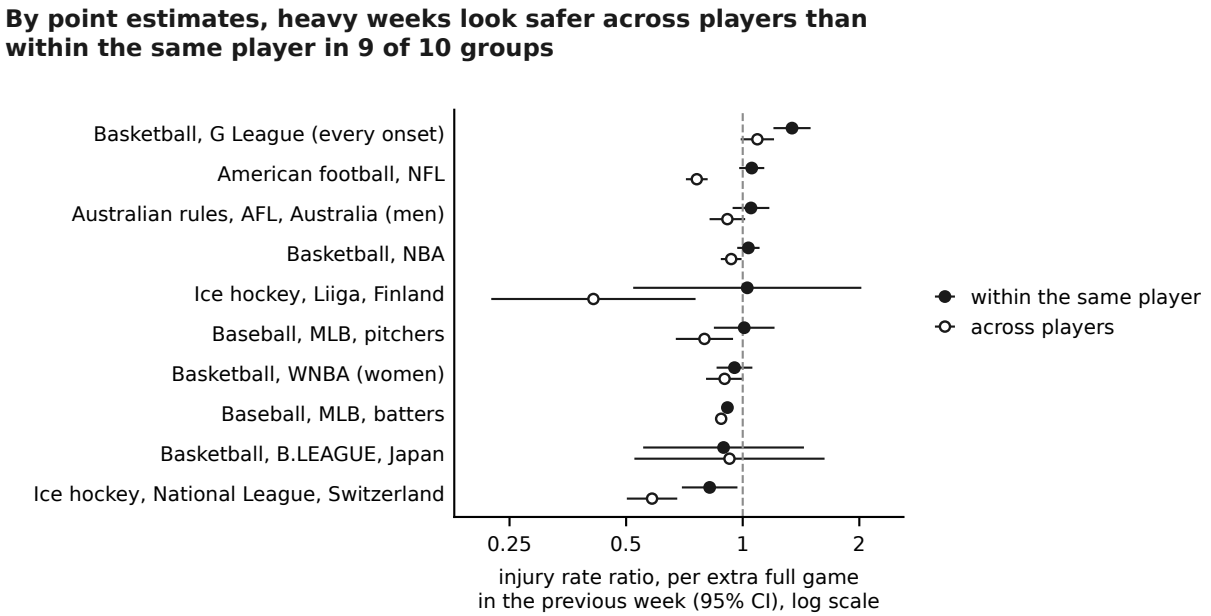



**Figure S11.** Injury within the same player and across players, in the other leagues with injury records, sorted by the within-player ratio. Each rate ratio is per extra full game in the previous week, with its 95% CI. Filled circles are within the same player, from conditional Poisson regression. Open circles are across players, from Poisson regression with season terms. By point estimates, heavy weeks look safer across players than within the same player in 9 of the 10 groups. But in most groups the two intervals overlap. The G League's row counts every onset, as registered. Table S27 gives every value. It also gives the G League without its likely moves to the NBA. These are injuries followed within 14 days by an NBA game of the same player (Section S11).

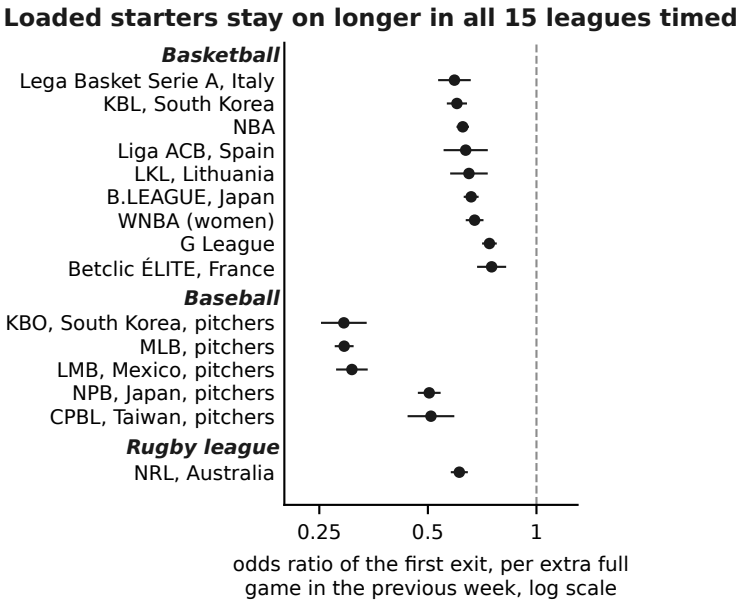



**Figure S12.** Loaded starters stay on longer in each of the other leagues whose records time a starter's first exit. The odds ratio of the first exit is per extra full game in the previous week, with its player-clustered 95% CI. The dashed line marks one. Below it, loaded starters stay on longer. Table S26 gives every value and its records.

**Table S25.** Who starts and within the same player, in the other leagues. The gradient is the unadjusted slope of log time played on recent match load, as in Table S5. It is fitted among all players who appeared, in the sport's own unit. The share from who starts is the per cent of the gradient due to who starts, from the exact split of Section S3. It is given where the starters are named. The within-player slope is fitted among starters, or among all appearances where no starter is named (a). The last column is the gradient on the same rows as the within-player slope. Intervals are from 1000 bootstrap draws over players.

| Competition | Time played | Seasons | Appearances | Gradient | From who starts, % (95% CI) | Within-player slope (95% CI) | Gradient on the same rows (95% CI) |
|---|---|---|---|---|---|---|---|
| *Basketball* | | | | | | | |
| B.LEAGUE, Japan | minutes | 10 | 130 305 | 0.621 | 49.1 (47.1 to 51.0) | 0.054 (0.048 to 0.060) | 0.126 (0.115 to 0.136) |
| Basketbol Süper Ligi, Turkey | minutes | 13 | 66 461 | 0.714 | – | 0.274 (0.245 to 0.303)[a] | 0.714 (0.665 to 0.764) |
| Betclic ÉLITE, France | minutes | 3 | 16 428 | 0.464 | 32.3 (28.9 to 36.5) | 0.025 (0.009 to 0.040) | 0.091 (0.071 to 0.110) |
| G League | minutes | 13 | 143 931 | 0.364 | 51.2 (50.0 to 52.3) | 0.027 (0.023 to 0.029) | 0.066 (0.062 to 0.071) |
| KBL, South Korea | minutes | 13 | 76 480 | 0.684 | 43.0 (40.5 to 45.5) | 0.133 (0.119 to 0.149) | 0.261 (0.238 to 0.283) |
| LKL, Lithuania | minutes | 20 | 69 736 | 0.375 | 37.0 (34.4 to 39.6) | 0.047 (0.039 to 0.054) | 0.084 (0.074 to 0.093) |
| Lega Basket Serie A, Italy | minutes | 13 | 69 380 | 0.597 | 47.3 (45.0 to 49.7) | 0.038 (0.031 to 0.045) | 0.105 (0.095 to 0.115) |
| Liga ACB, Spain | minutes | 17 | 114 933 | 0.574 | 35.0 (33.3 to 36.7) | 0.074 (0.067 to 0.082) | 0.147 (0.137 to 0.157) |
| NBA | minutes | 10 | 254 462 | 0.537 | 51.2 (49.0 to 53.2) | 0.054 (0.049 to 0.058) | 0.104 (0.097 to 0.111) |
| WNBA (women) | minutes | 29 | 119 367 | 0.551 | 65.8 (64.1 to 67.4) | 0.054 (0.049 to 0.059) | 0.095 (0.089 to 0.102) |
| *Baseball* | | | | | | | |
| CPBL, Taiwan, batters | plate appearances | 8 | 50 828 | 0.115 | 74.4 (72.1 to 76.6) | 0.017 (0.014 to 0.019) | 0.037 (0.033 to 0.040) |
| CPBL, Taiwan, pitchers | outs | 8 | 19 446 | 1.382 | 91.4 (89.2 to 93.5) | 0.032 (0.002 to 0.061) | 0.152 (0.121 to 0.187) |
| KBO, South Korea, batters | plate appearances | 7 | 113 599 | 0.127 | 76.2 (74.6 to 77.8) | 0.016 (0.014 to 0.018) | 0.040 (0.037 to 0.043) |
| KBO, South Korea, pitchers | outs | 7 | 45 855 | 1.739 | 90.7 (89.3 to 92.0) | 0.120 (0.090 to 0.151) | 0.344 (0.302 to 0.388) |
| LMB, Mexico, batters | plate appearances | 8 | 135 069 | 0.096 | 79.6 (78.1 to 80.9) | 0.011 (0.010 to 0.012) | 0.025 (0.024 to 0.026) |
| LMB, Mexico, pitchers | outs | 8 | 63 519 | 1.504 | 90.8 (89.3 to 92.2) | 0.127 (0.097 to 0.159) | 0.324 (0.292 to 0.360) |
| MLB, batters | plate appearances | 16 | 777 211 | 0.088 | 76.0 (75.1 to 77.0) | 0.013 (0.013 to 0.014) | 0.025 (0.025 to 0.026) |
| MLB, pitchers | outs | 16 | 313 454 | 2.152 | 95.2 (94.7 to 95.6) | 0.135 (0.125 to 0.145) | 0.287 (0.270 to 0.305) |
| NPB, Japan, batters | plate appearances | 10 | 181 413 | 0.140 | 82.5 (81.0 to 84.0) | 0.017 (0.016 to 0.018) | 0.035 (0.033 to 0.038) |
| NPB, Japan, pitchers | outs | 10 | 69 022 | 1.344 | 93.4 (92.4 to 94.5) | 0.074 (0.060 to 0.089) | 0.170 (0.152 to 0.189) |
| *Rugby league* | | | | | | | |
| NRL, Australia | minutes | 4 | 27 741 | 0.292 | 65.6 (61.9 to 69.6) | 0.009 (0.001 to 0.017) | 0.087 (0.071 to 0.102) |
| *Australian rules* | | | | | | | |
| AFL, Australia (men) | minutes | 9 | 80 888 | 0.072 | 31.3 (28.7 to 33.8) | 0.016 (0.012 to 0.019) | 0.029 (0.025 to 0.034) |
| *Ice hockey* | | | | | | | |
| Czech Extraliga | minutes | 12 | 180 837 | 0.541 | – | 0.278 (0.262 to 0.295)[a] | 0.541 (0.518 to 0.565) |
| DEL, Germany | minutes | 3 | 43 535 | 0.730 | – | 0.176 (0.156 to 0.199)[a] | 0.730 (0.662 to 0.805) |
| DEL2, Germany | minutes | 1 | 14 257 | 0.902 | – | 0.188 (0.159 to 0.223)[a] | 0.902 (0.783 to 1.029) |
| Liiga, Finland | minutes | 4 | 77 568 | 0.420 | – | 0.162 (0.149 to 0.178)[a] | 0.420 (0.397 to 0.446) |
| NHL | minutes | 17 | 730 501 | 0.513 | – | 0.202 (0.195 to 0.209)[a] | 0.513 (0.496 to 0.528) |
| SHL, Sweden | minutes | 3 | 46 621 | 0.645 | – | 0.201 (0.174 to 0.230)[a] | 0.645 (0.594 to 0.691) |
| National League, Switzerland | minutes | 9 | 128 056 | 0.573 | – | 0.231 (0.211 to 0.251)[a] | 0.573 (0.536 to 0.611) |
| *American football* | | | | | | | |
| NFL | snaps | 13 | 307 095 | 1.008 | – | 0.464 (0.449 to 0.481)[a] | 1.008 (0.987 to 1.031) |
| *Cricket* | | | | | | | |
| Bangladesh Premier League, batters | balls faced | 10 | 6598 | 0.095 | −0.1 (−0.2 to 0.0) | −0.006 (−0.019 to 0.006) | 0.095 (0.081 to 0.111) |
| Bangladesh Premier League, bowlers | balls bowled | 10 | 5066 | 0.099 | – | 0.024 (0.010 to 0.038)[a] | 0.099 (0.079 to 0.119) |
| Big Bash League, batters | balls faced | 11 | 8423 | 0.124 | 0.1 (0.0 to 0.4) | 0.005 (−0.007 to 0.018) | 0.124 (0.106 to 0.143) |
| Big Bash League, bowlers | balls bowled | 11 | 6365 | 0.153 | 0.3 (0.0 to 0.8) | 0.031 (0.016 to 0.048) | 0.153 (0.123 to 0.184) |
| CSA T20 Challenge, batters | balls faced | 3 | 1173 | 0.097 | 0.3 (0.0 to 1.1) | −0.030 (−0.071 to 0.008) | 0.097 (0.067 to 0.127) |
| CSA T20 Challenge, bowlers | balls bowled | 3 | 902 | 0.105 | – | −0.001 (−0.032 to 0.031)[a] | 0.105 (0.071 to 0.137) |
| Caribbean Premier League, batters | balls faced | 11 | 5488 | 0.087 | 0.1 (0.0 to 0.2) | −0.009 (−0.022 to 0.002) | 0.087 (0.072 to 0.102) |
| Caribbean Premier League, bowlers | balls bowled | 11 | 4198 | 0.100 | – | 0.008 (−0.007 to 0.023)[a] | 0.100 (0.079 to 0.121) |
| ILT20, batters | balls faced | 4 | 2101 | 0.087 | 0.4 (−0.6 to 2.1) | −0.034 (−0.056 to −0.013) | 0.088 (0.066 to 0.111) |
| ILT20, bowlers | balls bowled | 4 | 1614 | 0.111 | 0.0 (−0.4 to 1.1) | −0.005 (−0.027 to 0.015) | 0.115 (0.087 to 0.142) |
| IPL, batters | balls faced | 19 | 18 763 | 0.123 | 0.0 (−0.2 to 0.5) | 0.007 (−0.001 to 0.015) | 0.124 (0.111 to 0.137) |
| IPL, bowlers | balls bowled | 19 | 14 699 | 0.135 | 0.4 (−0.1 to 1.1) | 0.025 (0.016 to 0.033) | 0.135 (0.116 to 0.154) |
| Lanka Premier League, batters | balls faced | 6 | 2321 | 0.074 | 0.0 (−0.3 to 0.8) | −0.035 (−0.055 to −0.018) | 0.074 (0.054 to 0.093) |
| Lanka Premier League, bowlers | balls bowled | 6 | 1762 | 0.105 | 0.5 (−0.3 to 2.2) | −0.006 (−0.031 to 0.019) | 0.104 (0.076 to 0.131) |
| Pakistan Super League, batters | balls faced | 10 | 4908 | 0.108 | 0.1 (0.0 to 0.4) | −0.013 (−0.028 to 0.000) | 0.108 (0.091 to 0.124) |
| Pakistan Super League, bowlers | balls bowled | 10 | 3647 | 0.106 | – | 0.018 (0.004 to 0.033)[a] | 0.106 (0.083 to 0.132) |
| SA20, batters | balls faced | 4 | 2019 | 0.098 | – | −0.042 (−0.066 to −0.021)[a] | 0.098 (0.070 to 0.124) |
| SA20, bowlers | balls bowled | 4 | 1489 | 0.074 | – | −0.024 (−0.049 to −0.001)[a] | 0.074 (0.043 to 0.101) |
| Super Smash, batters | balls faced | 6 | 2861 | 0.112 | – | −0.005 (−0.029 to 0.018)[a] | 0.112 (0.084 to 0.139) |
| Super Smash, bowlers | balls bowled | 6 | 2107 | 0.086 | 0.1 (0.0 to 0.5) | 0.014 (−0.010 to 0.038) | 0.086 (0.054 to 0.118) |
| The Hundred, men, batters | balls faced | 6 | 3066 | 0.109 | – | −0.005 (−0.023 to 0.011)[a] | 0.109 (0.092 to 0.129) |
| The Hundred, men, bowlers | balls bowled | 6 | 2376 | 0.105 | – | −0.002 (−0.022 to 0.017)[a] | 0.105 (0.075 to 0.136) |
| Women's Premier League, batters | balls faced | 4 | 1364 | 0.120 | 0.1 (0.0 to 0.2) | −0.007 (−0.027 to 0.012) | 0.120 (0.095 to 0.145) |

| Competition | Time played | Seasons | Appearances | Gradient | From who starts, % (95% CI) | Within-player slope (95% CI) | Gradient on the same rows (95% CI) |
|---|---|---|---|---|---|---|---|
| Women's Premier League, bowlers | balls bowled | 4 | 1104 | 0.146 | – | 0.016 (−0.017 to 0.045)[a] | 0.146 (0.108 to 0.183) |

**Table S26.** When loaded starters come off, in the other leagues. The odds ratio is for a starter's first exit, per extra full game in the previous week. It comes from a discrete-time hazard, as in the substitution model. The hazard is the chance of a first exit in each of twelve parts of the regulation game, or in each inning for pitchers. It counts only starters still on. Intervals are clustered by player. Below one means that loaded starters stay on longer. The records column names the source of each exit time and the seasons read.

| Competition | Records | Starter-games | Starters | Exits | Odds ratio per extra full game (95% CI) |
|---|---|---|---|---|---|
| B.LEAGUE, Japan | B.LEAGUE play-by-play, 6 seasons | 42 150 | 598 | 42 027 | 0.66 (0.63 to 0.69) |
| Betclic ÉLITE, France | the Betclic ÉLITE's play-by-play, 3 seasons | 8207 | 419 | 8179 | 0.75 (0.68 to 0.82) |
| G League | G League play-by-play, 3 seasons | 16 000 | 1019 | 15 678 | 0.74 (0.71 to 0.78) |
| KBL, South Korea | KBL text play-by-play, 13 seasons | 37 476 | 552 | 37 476 | 0.60 (0.56 to 0.64) |
| LKL, Lithuania | the LKL's own play-by-play, 5 seasons | 4765 | 357 | 4762 | 0.65 (0.58 to 0.73) |
| Lega Basket Serie A, Italy | LBA box scores and play-by-play, regular seasons 2021-22 to 2025-26 (outside the panel); 0 games failed the panel's check | 11 703 | 527 | 11 700 | 0.59 (0.53 to 0.66) |
| Liga ACB, Spain | the ACB's plays, regular seasons 2023-24 to 2025-26 (outside the panel); 0 games failed the panel's check | 9180 | 472 | 9179 | 0.64 (0.55 to 0.73) |
| NBA | NBA play-by-play, 3 seasons | 36 900 | 830 | 35 693 | 0.62 (0.60 to 0.65) |
| WNBA (women) | WNBA play-by-play, 3 seasons | 6470 | 237 | 6259 | 0.67 (0.64 to 0.71) |
| CPBL, Taiwan, pitchers | the panel's outs recorded | 4678 | 227 | 4624 | 0.51 (0.44 to 0.59) |
| KBO, South Korea, pitchers | the panel's outs recorded | 10 072 | 380 | 10 029 | 0.29 (0.25 to 0.34) |
| LMB, Mexico, pitchers | the panel's outs recorded | 13 294 | 872 | 13 254 | 0.31 (0.28 to 0.34) |
| MLB, pitchers | the panel's outs recorded | 77 748 | 1616 | 76 552 | 0.29 (0.28 to 0.31) |
| NPB, Japan, pitchers | the panel's outs recorded | 16 884 | 563 | 16 256 | 0.50 (0.47 to 0.54) |
| NRL, Australia | NRL match feed, interchanges | 21 255 | 644 | 7633 | 0.61 (0.58 to 0.65) |

**Table S27.** Injury within the same player and across players, in the other leagues (Figure S11). Each rate ratio is per extra full game in the previous week, with the time played as the exposure. Within the same player it comes from conditional Poisson regression, and across players from Poisson regression with season terms. Both use starters where the starters are named, and all appearances otherwise. An injury is the panel's record (Section S12). A game missed with injury counts at the player's game before it. A placement on an injured list counts at its own game. The G League's row counts every onset, as registered. Without its likely moves to the NBA, its within-player ratio was 1.33 (95% CI 1.18 to 1.49; Section S11).

| Competition | Players compared | Injuries | Appearances | Players | Within the same player (95% CI) | Across players (95% CI) |
|---|---|---|---|---|---|---|
| B.LEAGUE, Japan | starters | 27 | 59 640 | 779 | 0.89 (0.55 to 1.44) | 0.92 (0.53 to 1.63) |
| G League | starters | 758 | 67 504 | 2353 | 1.34 (1.20 to 1.50) | 1.09 (0.99 to 1.20) |
| NBA | starters | 1856 | 114 924 | 1027 | 1.03 (0.97 to 1.11) | 0.93 (0.88 to 0.99) |
| WNBA (women) | starters | 599 | 58 358 | 748 | 0.95 (0.86 to 1.06) | 0.90 (0.80 to 1.00) |
| MLB, batters | starters | 3109 | 669 554 | 2354 | 0.91 (0.90 to 0.93) | 0.88 (0.86 to 0.89) |
| MLB, pitchers | starters | 1634 | 77 685 | 1616 | 1.01 (0.84 to 1.21) | 0.80 (0.67 to 0.94) |
| AFL, Australia (men) | starters | 1824 | 40 578 | 1253 | 1.05 (0.94 to 1.17) | 0.91 (0.82 to 1.01) |
| Liiga, Finland | all appearances | 139 | 73 790 | 916 | 1.03 (0.52 to 2.02) | 0.41 (0.22 to 0.76) |
| National League, Switzerland | all appearances | 1807 | 115 274 | 1016 | 0.82 (0.70 to 0.97) | 0.58 (0.50 to 0.68) |
| NFL | all appearances | 4613 | 213 450 | 6580 | 1.06 (0.98 to 1.14) | 0.76 (0.71 to 0.81) |

# S10 Three checks planned before they were run

Two independent reviews of the injury comparison asked for three further checks. Their definitions, and how each result would be read, were written into a plan before any of them was computed. The plan is in the archive. The script refuses to run if the plan's fixed part has changed since. Every check first reproduces the deposited comparison it re-checks.

**Two matches in the week against one.** A heavy week is a full match or more in the seven days before. For a regular starter that is usually one match. Among starts after at least one league match in the window, starts after two or more were compared with starts after exactly one. They were compared within the same starter (conditional Poisson) and across starters (Mantel–Haenszel by league-season). Within the same starter,

the ratio was 0.97 (95% CI 0.85 to 1.11): no clear difference. It rests on 8056 starts after two matches or more, with 245 injury-coded substitutions. Counted by minutes, two full matches or more against one to two gave 0.96 (0.77 to 1.21) within the same starter, also no clear difference. Table S28 gives the comparisons across starters.

**Season phase, venue and the opponent.** The within-starter model had no term for the phase of the season, the venue or the opponent. Three such terms were added: each start's third of its club's league season, whether the club played at home, and the opponent's points per league match over the season. They were added each alone and then all three together. Every start had all three terms. Against longer-gap weeks, the same starter's heavy weeks gave 0.99 (0.93 to 1.06) with all three, as they did without them. Against every rested week they gave 1.00 (0.94 to 1.06). Both read no clear difference.

**Goalkeepers and position.** Goalkeepers are almost never substituted and seldom injured, yet they entered every comparison across players. So the comparisons were refitted 48 times, without goalkeepers and with position as terms or as strata. Every refit of the starters' heavy weeks stayed inside the interval of the estimate it re-checks. But 11 of the 48 refits fell outside theirs. Among all players per appearance, loaded against rested, the ratio without goalkeepers was 1.26 (1.19 to 1.33), against 1.18 with them. The others were part-loaded contrasts, whose excess shrank: per starter hour, 1.09 without goalkeepers against 1.18. The plan was written for the seven league-seasons of the earlier comparison (Section S6). It says that once any refit falls outside its interval, the refits without goalkeepers become the primary estimates. The main text and Figures 1 to 3 follow it: their comparisons of heavy and light weeks leave out goalkeepers. Table S20 and Section S6 give the comparisons with goalkeepers. Table S28 gives the starters' refits per starter hour. Against longer-gap weeks, the comparison across outfield starters was 0.88 (0.82 to 0.94): safer. The comparison within the same outfield starter was 0.98 (0.92 to 1.05): no clear difference.

**A check added after them.** The hybrid model's across-starter term is a comparison across players too, so the model was refitted on outfield starters. This was not planned. The within-starter ratio was 2.6 times the across-starter ratio (1.9 to 3.6), against 3.4 times with goalkeepers. The outfield ratio's interval lay wholly above one. Table S28 gives each estimate beside the one it re-checks.

**Table S28.** The three checks planned before they were run, and the one added after them, per starter hour, beside the estimates they re-check. Within the same starter, the ratios come from conditional Poisson regression. Across starters, they are Mantel–Haenszel ratios stratified by league-season, and by position too where the row says so, or Poisson regression with position terms. The hybrid model takes each starter's share of loaded starts from the starter's earlier starts, on starts with at least five earlier ones. Within over across is the within-starter rate ratio divided by the across-starter one. The deposit holds every refit, all players and per appearance included.

| Comparison, per starter hour | Ratio (95% CI) | Size |
|---|---|---|
| Two league matches or more in the week, v one, within the same starter | 0.97 (0.85 to 1.11) | 8056 v 115 531 starts |
| Two league matches or more in the week, v one, across starters | 0.96 (0.84 to 1.09) | 8056 v 115 531 starts |
| Two full matches or more, v one to two, within the same starter | 0.96 (0.77 to 1.21) | 3484 v 71 226 starts |
| Two full matches or more, v one to two, across starters | 0.81 (0.65 to 1.01) | 3484 v 71 226 starts |
| Same starter v longer-gap weeks, as fitted | 0.99 (0.93 to 1.06) | 6859 events |
| Same starter v longer-gap weeks, adjusted for season phase | 0.99 (0.93 to 1.06) | 6859 events |
| Same starter v longer-gap weeks, adjusted for venue | 0.99 (0.93 to 1.06) | 6859 events |
| Same starter v longer-gap weeks, adjusted for opponent's points per match | 0.99 (0.93 to 1.06) | 6859 events |
| Same starter v longer-gap weeks, adjusted for all three | 0.99 (0.93 to 1.06) | 6859 events |
| Same starter v every rested week, as fitted | 1.00 (0.94 to 1.06) | 6859 events |
| Same starter v every rested week, adjusted for season phase | 1.00 (0.94 to 1.06) | 6859 events |
| Same starter v every rested week, adjusted for venue | 1.00 (0.94 to 1.06) | 6859 events |
| Same starter v every rested week, adjusted for opponent's points per match | 1.00 (0.94 to 1.06) | 6859 events |
| Same starter v every rested week, adjusted for all three | 1.00 (0.94 to 1.06) | 6859 events |
| Across starters v every rested week, as fitted | 0.82 (0.78 to 0.87) | all positions |
| Across starters v every rested week, without goalkeepers | 0.86 (0.81 to 0.91) | outfield |
| Across starters v every rested week, with position terms | 0.86 (0.81 to 0.91) | outfield |
| Across starters v every rested week, stratified by position | 0.86 (0.81 to 0.91) | outfield |
| Across starters v longer-gap weeks, as fitted | 0.84 (0.79 to 0.90) | all positions |
| Across starters v longer-gap weeks, without goalkeepers | 0.88 (0.82 to 0.94) | outfield |
| Across starters v longer-gap weeks, with position terms | 0.87 (0.81 to 0.93) | outfield |
| Across starters v longer-gap weeks, stratified by position | 0.88 (0.82 to 0.94) | outfield |
| Same starter v longer-gap weeks, outfield starters | 0.98 (0.92 to 1.05) | 6710 events |
| Hybrid model, within the starter: all starters | 0.91 (0.86 to 0.97) | 189 030 starts |
| Hybrid model, within the starter: outfield (not planned) | 0.92 (0.87 to 0.98) | 171 598 starts |
| Hybrid model, across starters: all starters | 0.27 (0.19 to 0.37) | 189 030 starts |
| Hybrid model, across starters: outfield (not planned) | 0.35 (0.25 to 0.49) | 171 598 starts |
| Hybrid model, within over across: all starters | 3.44 (2.50 to 4.73) | 189 030 starts |
| Hybrid model, within over across: outfield (not planned) | 2.62 (1.88 to 3.64) | 171 598 starts |

## S11 Who the player was against recent load

We split the differences in injury rate between who the player was and recent match load. The split covers every league with an injury record: the European men's football leagues of the injury comparison, and the other leagues of Table S27. The AFL's record can count soreness as injury or leave it out. The AFL counts once here, with soreness left out. The analyses were planned in three steps, each written down before its results were read; the plans are in the archive. Each script refuses to run if its plan's fixed part has changed. Two later changes, football on outfield starts and the G League without likely moves to the NBA, were also planned before they ran (below). The analyses are exploratory.

**The rows.** In football, the rows are the starts of the injury comparison (Section S3). The plan, written before any result was read, used every start, goalkeepers included: these are the registered rows. The main text prints outfield starts. The event is an injury-coded substitution, the exposure is the starter's hours, and recent match load is counted in full matches. In the other leagues, the rows are those of Table S27: starters where the starters are named, and all appearances otherwise. The event is the panel's injury, the exposure is the time played, and recent match load is counted in full games.

**Who the player was.** A Poisson model gave each player a level of their own. The rate this level sets is the main text's baseline rate. The level holds everything that stays the same about a player across their appearances. Position, build, history and usual load are among it, and so is how the player's club records injuries. This level is a random intercept, drawn from a normal distribution with standard deviation $\sigma$. The model also holds the season terms, the logarithm of the exposure as an offset, and recent match load centred on each player's own mean. It was fitted by maximum likelihood. Each player's level was integrated out by adaptive Gauss–Hermite quadrature with 40 points. A season without an injury carries no information, so the B.LEAGUE's three such seasons were left out of this fit.

**How far two players differ.** For two players drawn at random from the same season, the median rate ratio is the median of the higher injury rate over the lower. It equals $\exp(\sqrt{2} \times 0.6745\,\sigma)$ (Austin and colleagues, Statistics in Medicine 2018). This is the count-data form of the median odds ratio (Larsen and Merlo, American Journal of Epidemiology 2005). It is on the same scale as the rate ratio of a factor such as load. So the differences between people and the factor's effect can be set side by side (Merlo and colleagues, Journal of Epidemiology and Community Health 2006). The median rate ratio's 95% CI comes from the profile likelihood of $\sigma$. Players are known to differ in injury risk. Frailty models of recurrent sports injury rest on this (Ullah and colleagues, British Journal of Sports Medicine 2014). In them, a frailty is a player's own level of risk. Shared frailty corrects for the selection of less injury-prone athletes over time (Nielsen and colleagues, British Journal of Sports Medicine 2019). Differences between players were found in elite Australian football (Esmaeili and colleagues, cited in the main text) and in public NBA data (Lewis, Journal of Athletic Training 2018). A recent study in this journal found large differences between players in training concussions, and no association with match minutes in the previous 28 days (Hudson and colleagues, cited in the main text). What is new here is their size beside recent load, measured one way in six sports.

**The load.** The load's rate ratio is per extra full game of recent match load, within the same player. It comes from the conditional Poisson model of Section S9, with the player-clustered 95% CI. In each league outside football it reproduces Table S27. In football it was fitted the same way, on the starts of the injury comparison. The model gives every player the same load effect. But players may respond to load differently, as the NBA study found (Lewis 2018). The ratio is then an average over them. The load's window is the previous seven days. So the ratio misses effects that lag longer, such as fast bowlers' injury risk three to four weeks after heavy match loads (Orchard and colleagues, American Journal of Sports Medicine 2009).

**Which is larger.** The load's largest factor is the biggest change, up or down, that the load's 95% CI allows for one extra full game. A fall counts by how many times lower the rate becomes. In a group, who the player was outweighs the load when the lower end of the median rate ratio's 95% CI exceeds that factor. That held in 10 of the 11 groups, in all six sports (Table S29). Japan's B.LEAGUE recorded only 27 injuries. There the lower end was 1.00, against a largest load factor of 1.81. With so few injuries, the two were too imprecise to tell apart.

**The share of each.** In the model, the log injury rate of an appearance is a season term, plus the player's level, plus the load's term. Leaving chance and the season aside, the differences in injury rate across appearances therefore come in two parts. The player's part is the variance of the players' levels, $\sigma^2$. The load's part is the variance of the load's term over the appearances, $\beta^2 V$. Here $\beta$ is the logarithm of the load's rate ratio, and $V$ is

the mean square of recent match load centred on each player's own mean. The player's share is $\sigma^2/(\sigma^2+\beta^2 V)$, and the load's share is the rest. A share describes the load as it varied in its group. Where loads varied more, the load's share would be larger. Football's own inputs show how much. With football's spread of load, recent load would reach a tenth of the share only if a full match changed a player's rate by 60%. It would reach half only if a full match changed the rate 4.1-fold. So the share cannot tell a small effect of load from none. Chance is in neither part, and in a single start it dominates. With football's outfield fit, the player's level accounts for about 2% of the variation in whether one start ends in an injury-coded substitution. Over thirty starts it accounts for 34% (Austin and colleagues 2018). Differences between people are usually much larger than the differences between measurements on the same person (Bland and Altman, BMJ 1994). Injuries, too, gather in a minority: in one birth cohort, a fifth of its members made over a third of its injury insurance claims (Caspi and colleagues, Nature Human Behaviour 2016).

**The intervals.** Each group's $\sigma$ and $\beta$ were redrawn 100 000 times (seed 319). Half the draws of $\sigma$ fall below the estimate and half above. Each half follows a half-normal curve, one side of a bell curve, scaled to its own side of the CI. So their median is the estimate, and their 2.5th and 97.5th percentiles are the ends of its profile CI. A draw below zero counts as zero. The draws of $\beta$ are normal, with the standard error recovered from its 95% CI. In each draw, every group's share was recomputed. A pool's share is the mean of its groups' shares, each group counting once. Each 95% CI runs from the 2.5th to the 97.5th percentile of the draws.

**Results.** In the 10 groups where who the player was outweighs the load, the player accounted for 98.5% of the differences in injury rate (95% CI 97.6% to 98.9%). Recent load accounted for 1.5% (1.1% to 2.4%). Two groups, the G League and MLB's batters, give 90% of the ten groups' load shares added together. In each of the 10 groups, the lower end of the player's own share was at least 87.3%. With the B.LEAGUE, all 11 groups gave 98.6% (89.8% to 98.9%) for the player and 1.4% (1.1% to 10.2%) for the load. The B.LEAGUE's own interval reaches zero, and that alone widens the pool's. Across the 10 groups, two players drawn at random from the same season typically differed in injury rate by a factor of 1.86. That is the median of the groups' median rate ratios. After one extra full game, the same player's rate differed by a median of 5% either way. For football, the load's ratio in Table S29 is the within-starter ratio pooled over seasons, on every start, goalkeepers included, as the share was registered. With the same-season ratio of Section S6 (1.11) in its place, recent load's share in football would be 0.5%. With football's outfield same-season ratio (1.10) in its place, the groups' median change per full game would be 5.3%. The player's share held without the seasons and team-seasons whose injury records stand out (99.0%), and with a term for each team (98.4%; Section S12).

**Football on outfield starts, and the G League without likely moves.** Two changes were planned before they ran, each on the rows of its registered fit. All the main text's other football injury results leave goalkeepers out, so football's model was refitted on outfield starts. Its median rate ratio was then 1.83 (95% CI 1.76 to 1.90; 4591 players and 6710 injury-coded substitutions), and the load's ratio 0.99 (0.94 to 1.04). A later refit, also planned before it ran, used the narrower outcome on the same starts: injury-coded substitutions followed by a recorded absence. Its median rate ratio was 1.94 (1.81 to 2.07; 2537 events) and the load's ratio 1.11 (1.02 to 1.20). So the player still clearly outweighed the load (Table S29). In the G League, 60 of the 646 onsets of 2015 to 2024 (9%) were followed by an NBA game of the same player within 14 days. These likely moves to the NBA were counted as appearances without an injury. The other seasons' 112 onsets stay as recorded, because the NBA's record here cannot check them. It does not cover the earlier seasons, and its 2025 season is held back for a registered test. Without the likely moves, the G League's load ratio was 1.33 (1.18 to 1.49) and its median rate ratio 2.04. With the football change alone, the ten groups' median rate ratio was 1.82. With the G League change alone, it was 1.86. With both, it was 1.82 (1.59 to 4.16 across groups), the value the main text prints, against 1.86 as registered. With both changes, the median change per full game stayed 5%, up or down. Across the groups it ran from 18% lower to 33% higher, and four of the ten load ratios lie below one. With both changes, the player's share was 98.6% (97.7% to 99.0%), and 98.7% with the B.LEAGUE (89.8% to 99.0%). Over all 11 groups, the B.LEAGUE included, the median rate ratio was 1.83-fold and the median change 5%, against 1.82-fold and 5% over the ten. So the medians hold with the B.LEAGUE in. As registered, the median rate ratio was 1.90-fold over all 11 and 1.86-fold over the ten.

**The load on the players' own scale.** A full game is a different share of a week in each sport. An MLB batter plays most days, so one game is a small part of the batter's week. For an NFL player, one game is the whole week. So a median over sports of per-game effects mixes unlike comparisons. Each group's own spread of load was taken as roughly normal. On the scale of the median rate ratio, two weeks of the same player then typically differ through load alone by a factor of 1.02. Across the groups, this factor runs from 1.00 to 1.19. That is far below the 1.59 to 4.16 between players. So setting a full match beside the spread between players, as the main text does, if anything makes load look larger than it usually is.

**Table S29.** Who the player was against recent load, group by group and pooled. The median rate ratio is the median, over pairs of players from the same season, of the higher injury rate over the lower (95% CI from the profile likelihood). The load's rate ratio is per extra full game of recent match load, within the same player (conditional Poisson regression, player-clustered 95% CI). Leaving chance and the season aside, the two shares split the differences in injury rate between the player and the load. Their 95% CIs come from 100 000 redraws of each group's two estimates. A pool's shares are the means of its groups' shares, and its players and injuries are their totals. (a) All appearances, where the starters are not named; starters otherwise. (b) The lower end of the median rate ratio's CI does not exceed the load's largest factor, so the group enters the second pool only. (c) The rows the main text prints: football on outfield starts, on injury-coded substitutions and on the narrower outcome, and the G League without its likely moves to the NBA (Section S11). The pools beneath them use the first football row and the G League's. Football's registered row uses every start, goalkeepers included, and its load ratio is the within-starter ratio pooled over seasons. The AFL counts injuries without soreness. Counting soreness changed every share by less than a tenth of a percentage point.

| League | Players | Injuries | Median rate ratio (95% CI) | Load's rate ratio per extra full game (95% CI) | Player's share, % (95% CI) | Load's share, % (95% CI) |
|---|---|---|---|---|---|---|
| *Football* | | | | | | |
| European men's football | 5003 | 6859 | 1.97 (1.90 to 2.05) | 1.00 (0.95 to 1.05) | >99.9 (99.8 to >99.9) | <0.1 (<0.1 to 0.2) |
| *Basketball* | | | | | | |
| B.LEAGUE, Japan[b] | 691 | 27 | 2.84 (1.00 to 6.90) | 0.89 (0.55 to 1.44) | 99.6 (0.0 to >99.9) | 0.4 (<0.1 to 100.0) |
| G League | 2353 | 758 | 2.00 (1.77 to 2.27) | 1.34 (1.20 to 1.50) | 93.6 (87.3 to 97.5) | 6.4 (2.5 to 12.7) |
| NBA | 1027 | 1856 | 1.97 (1.83 to 2.14) | 1.03 (0.97 to 1.11) | 99.9 (99.1 to >99.9) | 0.1 (<0.1 to 0.9) |
| WNBA (women) | 748 | 599 | 1.82 (1.61 to 2.08) | 0.95 (0.86 to 1.06) | 99.7 (96.9 to >99.9) | 0.3 (<0.1 to 3.1) |
| *Baseball* | | | | | | |
| MLB, batters | 2354 | 3109 | 1.71 (1.62 to 1.81) | 0.91 (0.90 to 0.93) | 93.3 (89.9 to 95.9) | 6.7 (4.1 to 10.1) |
| MLB, pitchers | 1616 | 1634 | 1.59 (1.47 to 1.72) | 1.01 (0.84 to 1.21) | >99.9 (98.9 to >99.9) | <0.1 (<0.1 to 1.1) |
| *Australian rules* | | | | | | |
| AFL, Australia (men) | 1253 | 1824 | 1.66 (1.54 to 1.79) | 1.05 (0.94 to 1.17) | 99.9 (98.5 to >99.9) | 0.1 (<0.1 to 1.5) |
| *Ice hockey* | | | | | | |
| Liiga, Finland[a] | 916 | 139 | 4.16 (3.11 to 5.84) | 1.03 (0.52 to 2.02) | >99.9 (98.0 to >99.9) | <0.1 (<0.1 to 2.0) |
| National League, Switzerland[a] | 1016 | 1807 | 1.79 (1.66 to 1.93) | 0.82 (0.70 to 0.97) | 99.2 (97.1 to >99.9) | 0.8 (<0.1 to 2.9) |
| *American football* | | | | | | |
| NFL[a] | 6580 | 4613 | 1.90 (1.81 to 2.00) | 1.06 (0.98 to 1.14) | 99.9 (99.5 to >99.9) | 0.1 (<0.1 to 0.5) |
| *Pooled* | | | | | | |
| The 10 groups where the player outweighs the load | 22 866 | 23 198 | – | – | 98.5 (97.6 to 98.9) | 1.5 (1.1 to 2.4) |
| All 11 groups | 23 557 | 23 225 | – | – | 98.6 (89.8 to 98.9) | 1.4 (1.1 to 10.2) |
| *Football on outfield starts and the G League without likely moves, as in the main text* | | | | | | |
| European men's football, outfield starts[c] | 4591 | 6710 | 1.83 (1.76 to 1.90) | 0.99 (0.94 to 1.04) | >99.9 | <0.1 |
| European men's football, outfield starts, followed by a recorded absence[c] | 4591 | 2537 | 1.94 (1.81 to 2.07) | 1.11 (1.02 to 1.20) | 99.5 | 0.5 |
| G League, without likely moves[c] | 2353 | 698 | 2.04 (1.78 to 2.32) | 1.33 (1.18 to 1.49) | 94.2 | 5.8 |
| The 10 groups where the player outweighs the load | 22 454 | 22 989 | – | – | 98.6 (97.7 to 99.0) | 1.4 (1.0 to 2.3) |
| All 11 groups | 23 145 | 23 016 | – | – | 98.7 (89.8 to 99.0) | 1.3 (1.0 to 10.2) |

**Lower rates at higher loads.** Studies of training load in the British Journal of Sports Medicine have read lower injury rates at higher loads, across athletes, as protection (Gabbett, British Journal of Sports Medicine 2016). The same selection of fit players can produce that pattern: the players who play the most are those fit enough to keep their place.

# S12 Injury history

We compared players by their injury history, on the paper's own injury counts. Five sports have records that give the length of each absence: football, Australian rules, ice hockey's Swiss National League, American football and baseball. In each, players with two or more injuries of 28 days or more in the previous three years were compared with the other players. We say these players have a history, and we call their mark the flag. A previous injury is a known risk factor for the next. Collegiate American footballers with three or more earlier concussions were three times as likely to have another (Guskiewicz and colleagues, JAMA 2003). Runners injured in the previous year had about one and a half times the risk of a new injury (Walter and colleagues, Archives of Internal Medicine 1989). Premier-league clubs rank previous injury first among the risk factors they perceive (McCall and colleagues, British Journal of Sports Medicine 2015). A history of the same injury carries larger ratios than a history of any injury. In a meta-analysis, any history of hamstring strain injury went with nearly three times the risk of the next, and a recent one with nearly five times (Green and colleagues, British Journal of Sports Medicine 2020). An earlier injury also goes with a higher risk of a later lower-limb injury of a different kind or site (Toohey and colleagues, British Journal of Sports Medicine 2017). So the history here counts every injury of 28 days or more, whatever its kind. It does not class each later injury as a recurrence, as the subsequent-injury categorisation does (Finch and Cook, British Journal of Sports Medicine 2014). The analyses were planned in four steps, each written down before its results were read; the plans are in the archive. Two later checks, of age and of the weeks after a return, were also planned before they ran. The analyses are exploratory.

**What each record counts.** Outside football, an injury is each league's own record, read as the panel reads it (Section S9). In seven of the ten groups it is a game missed with injury as the recorded reason. It counts at the game the player played before it, within seven days and the same season. Each absence counts once, at its start. In the NBA, the WNBA and the G League, the record is the league's box score. In the Swiss National League it is the federation's game sheet. In the NFL it is the club's injury report. In Liiga it is the club's roster, which marks injured players. In the AFL it is the team list, with soreness left out (Section S11). In MLB, for batters and pitchers, and in the B.LEAGUE, it is a placement on the injured list, which counts at its own game. Where a recorder gives injury as the reason for rest or a move, the record counts it. In the G League, onsets followed by an NBA game within 14 days were recoded as likely moves (Section S11). Public records also capture prominent players' injuries better. Public NFL data held about two thirds of the anterior cruciate ligament injuries the league's own surveillance recorded. They held more of skill players' injuries than of linemen's (Inclan and colleagues, American Journal of Sports Medicine 2022). NHL teams withheld the details more often for higher-profile players (Pinkoski and colleagues, Orthopaedic Journal of Sports Medicine 2025). Where capture follows prominence, it becomes part of each player's level. It can then widen the spread between players that Section S11 measures.

**The records, checked first.** Before any comparison, each league's injury record was checked season by season and team by team. A season or a team-season stands out when its injuries lie far from the league's own rate. Its rate, with two standard errors either side, must lie wholly above twice the league's rate or wholly below half of it. Table S32 lists what stood out. Four findings bear on results reported in Sections S9 and S11.

- MLB's 2008 season recorded one injury among batters, where 208 were expected.
- In the G League, up to 21% of a season's injuries were followed by an NBA game within 14 days. These are likely moves to the NBA, not injuries. Section S11 refits the G League without them.
- In the NBA, 16 team-seasons stood out below the league's rate.
- In Liiga, several teams record injuries differently. With a term for each team, its median rate ratio falls from 4.16 to 2.91.

The share of who the player was in Section S11 does not depend on them. Without the seasons and team-seasons that stand out it was 99.0%, and with a term for each team it was 98.4% (Table S33). With a term for each team, the ten groups' median rate ratio was 1.85, against 1.86 without.

**Who has a history.** A long injury kept the player out for 28 days or more. Each player's season is cut into periods at each long injury. A period's history counts the long injuries that began in the 1095 days before it. A period counts only if its whole history lies inside the league's record, so each comparison starts three years after the record begins. Players with two or more long injuries in their history were compared with the others, who had none or one. In MLB, rows from 1 September on are left out, because rosters then grow.

**Where no history can be read.** No basketball league had a usable history. The NBA's and the G League's box scores drop long absences. The WNBA's record changed over the seasons and holds few long injuries. The B.LEAGUE's record holds too few injuries. Liiga's record starts in 2022-23, too late for a three-year history.

**The comparison.** In each group, a Poisson model with season terms compared the players with a history and the others. It compared their injuries per unit of time played, with standard errors clustered by player. The rows and the injuries are the paper's own (Sections S9 and S11): starters where the starters are named, and all appearances otherwise. In football, the rows are outfield starters of the five leagues, in the 27 five-substitution league-seasons whose reasons are near-complete. The outcome there is the injury-coded substitution per start. Three sensitivity analyses repeat each comparison: with team-season terms, with three or more long injuries, and per appearance.

**The history's part of who the player was.** The model of Section S11 was refitted on this section's rows, without and with a term for the players with a history. Here $\sigma^2$ is the variance of the players' levels. The history's part of who the player was is the share of $\sigma^2$ that the term removes. Its 95% interval comes from 200 resamples of players.

**Results.** In every sport with injury histories, players with two or more long injuries in the previous three years were injured more often. They were injured 1.3 to 1.7 times as often as the others in the six groups (Table S30). So the flag confirms a known factor, now in five sports' records. In football the ratio was 1.39 (95% CI 1.31 to 1.49). It was 1.35 to 1.49 in each of the five leagues alone. So uneven recording between the leagues does not explain it. The sensitivity analyses agreed wherever they could be measured, with one exception. Among MLB's pitchers, team-season terms left the ratio at 1.18 (95% CI 0.99 to 1.41), an interval that reaches one. The history accounted for 0.5% to 11% of the variance between players (Table S31). So it is one measured part of who the player was, but not most of it. With each player's own level in the model, the history's term lay wholly above one in three of the six groups (Table S31). Football was one of them (1.32, 95% CI 1.24 to 1.42). In those groups the flag also marks a time of higher risk, as a concussion does within the same rugby union player (Moore and colleagues, British Journal of Sports Medicine 2023). In the others it mostly marks players at higher risk all along. When people's own risk differs, comparisons across them must overstate the effect of a past injury (Hamilton and colleagues, American Journal of Epidemiology 2011).

**One long injury against two.** A single previous injury is already a known risk factor. So the threshold of two needed a check against a threshold of one. In each group, players with one or more long injuries in the previous three years were set against players with none, by the same test. This test was planned before it was computed, and the plan is in the archive. Players with one or more were injured more often in all six groups, 1.3 to 1.4 times as often (football 1.29, 95% CI 1.21 to 1.38). So the flag does not depend on the choice of two. The threshold of one marks far more starts: 57% of football's outfield starts, against 26% at two. At two, the flagged players made 26% of football's outfield starts and 33% of its injury-coded substitutions. In the other groups they made 2% to 23% of the rows and 4% to 28% of the injuries.

**The flag as a screen.** A factor linked to injury need not pick out the players who will be injured (Bahr, cited in the main text). Table S34 reads the flag as a screening test, row by row. In football, at two long injuries, the flag marked 26% of outfield starts and 33% of injury-coded substitutions. It left unmarked 74% of the starts that ended without an injury-coded substitution. Flagged players had 4.4 injury-coded substitutions per 100 starts, against 3.1 for the others: one more for every 82 starts. The area under the curve was 0.54, where a half is a coin toss and one is perfect. In the other groups the flag marked 4% to 28% of the injuries. So the flag marks a group at higher risk, not the players who will be injured. Models built on injury history, often with age, also separated injured from uninjured players only modestly in team-sport tests of prediction (Hughes and colleagues, Sports Medicine Open 2020; Lee Dow and colleagues, American Journal of Sports Medicine 2021; Opar and colleagues, Medicine and Science in Sports and Exercise 2022; Shanley and colleagues, Journal of Orthopaedic and Sports Physical Therapy 2022).

**Age, and the weeks after a return.** Two checks were planned before they ran. Players with two or more long injuries are on average further into their careers, and age is a known risk factor. So football's model was refitted with age bands beside the league-season terms: under 23, 23 to 26, 27 to 30, and 31 and over. Each period's age is the player's age at its first start. The three periods with no date of birth were left out. Football's ratio was then 1.39 (95% CI 1.31 to 1.49). These periods gave the same ratio without the age terms. The flag could also mostly mark a recent return from a long injury, which medical teams already manage. Without the rows dated in the 28 days after the end of each such injury, football's ratio was 1.39 (1.30 to 1.48). Without the 84 days, it was 1.38 (1.29 to 1.48), and 13 945 starts fell in those weeks. The flag held at both windows in all six groups. Without the 84 days, the other groups' ratios ran from 1.21 to 1.60. So the flag is more than the weeks after a return. But caution can also follow the flag. Staff who know a player's history may withdraw or rest that player sooner, and the record then counts an injury. That caution should be greatest in the weeks after a return, which these checks leave out.

**Table S30.** Injury history, group by group. Players with two or more injuries of 28 days or more in the previous three years (history) against the others, on the paper's own injury counts. The rates are injuries per 100 appearances, or per 100 starts in football. The rate ratio is per unit of time played, and per start in football. It comes from a Poisson model with season terms, league-season terms in football, and standard errors clustered by player. (a) Outfield starters of the five leagues, in the 27 five-substitution league-seasons whose reasons are near-complete; the outcome is the injury-coded substitution.

| Group | Players, history / others | Injuries, history / others | Per 100 appearances, history / others | Rate ratio (95% CI) |
|---|---|---|---|---|
| *Football* | | | | |
| European men's football[a] | 1509 / 3239 | 1954 / 3964 | 4.36 / 3.14 | 1.39 (1.31 to 1.49) |
| *Australian rules* | | | | |
| AFL, Australia (men) | 230 / 1043 | 212 / 1470 | 6.53 / 4.41 | 1.52 (1.28 to 1.80) |
| *Ice hockey* | | | | |
| National League, Switzerland | 99 / 466 | 150 / 728 | 2.24 / 1.57 | 1.44 (1.14 to 1.82) |
| *American football* | | | | |
| NFL | 158 / 3087 | 87 / 2039 | 4.41 / 2.58 | 1.69 (1.32 to 2.16) |
| *Baseball* | | | | |
| MLB, batters | 372 / 1457 | 383 / 1748 | 0.77 / 0.58 | 1.34 (1.18 to 1.52) |
| MLB, pitchers | 311 / 854 | 303 / 762 | 3.40 / 2.54 | 1.31 (1.12 to 1.52) |

**Table S31.** The history's part of who the player was. $\sigma$ is the standard deviation of the players' levels in the model of Section S11. It is fitted without and with a term for the players with a history, on the same rows in both fits. The history's part is the share of $\sigma^2$ the term removes, with a 95% interval from 200 resamples of players. A negative lower end means $\sigma$ was larger with the term than without it in some resamples. The term's rate ratio is the history's effect beside each player's own level. This table's $\sigma$ differs from Table S29's for the same group, because the rows differ. A row counts here only when its three-year history lies inside the record, and in MLB only before 1 September.

| Group | $\sigma$ without the history | $\sigma$ with the history | History's part, % (95% interval) | History term's rate ratio (95% CI) |
|---|---|---|---|---|
| European men's football | 0.71 | 0.68 | 9.0 (6.6 to 11.9) | 1.32 (1.24 to 1.42) |
| AFL, Australia (men) | 0.52 | 0.49 | 11.1 (1.1 to 23.0) | 1.25 (1.07 to 1.47) |
| National League, Switzerland | 0.63 | 0.62 | 2.6 (−7.5 to 13.2) | 1.06 (0.85 to 1.32) |
| NFL | 0.69 | 0.67 | 4.8 (0.4 to 9.1) | 1.37 (1.07 to 1.75) |
| MLB, batters | 0.60 | 0.60 | 0.5 (−6.2 to 9.0) | 1.01 (0.89 to 1.14) |
| MLB, pitchers | 0.56 | 0.55 | 4.6 (−8.6 to 19.3) | 1.05 (0.90 to 1.23) |

**Table S32.** The check of the injury records, league by league. A season or a team-season stands out when its injuries lie far from the league's own rate. Its rate, with two standard errors either side, must lie wholly above twice the league's rate or wholly below half of it. A team is judged over all its seasons together.

| League | Seasons | Seasons that stand out | Team-seasons, below / above | Teams, below / above |
|---|---|---|---|---|
| B.LEAGUE, Japan | 10 | none | 0 / 0 of 212 | 0 / 0 of 30 |
| G League | 13 | 2014 (below), 2015 (below), 2016 (below), 2020 (above) | 0 / 2 of 328 | 1 / 0 of 33 |
| NBA | 10 | none | 16 / 3 of 300 | 0 / 0 of 30 |
| WNBA (women) | 29 | 1997 (below), 2008 (below) | 0 / 1 of 365 | 0 / 0 of 19 |
| MLB, batters | 16 | 2008 (below) | 1 / 0 of 480 | 0 / 0 of 30 |
| MLB, pitchers | 16 | 2008 (below) | 0 / 0 of 480 | 0 / 0 of 30 |
| AFL, Australia (men) | 9 | none | 1 / 0 of 162 | 0 / 0 of 18 |
| Liiga, Finland | 4 | none | 0 / 1 of 62 | 5 / 1 of 16 |
| National League, Switzerland | 9 | 2017-18 (below) | 2 / 0 of 117 | 0 / 0 of 14 |
| NFL | 13 | none | 7 / 0 of 416 | 0 / 0 of 35 |

**Table S33.** The player's share of the differences in injury rate (Section S11), in %. It is given as fitted, without the seasons and team-seasons that stand out in Table S32, and with a term for each team.

| Pool | As fitted | Without the seasons and team-seasons that stand out | With a term for each team |
|---|---|---|---|
| The 10 groups where the player outweighs the load | 98.5 | 99.0 | 98.4 |
| All 11 groups | 98.6 | 99.0 | 98.2 |

**Table S34.** The flag as a screening test, group by group, at two long injuries and at one. A row is a start in football and an appearance elsewhere, each counted once, on the rows of Table S30. Rows marked are the flagged players' rows. Injuries marked is the share of the group's injuries in those rows, the sensitivity. Rows without injury left unmarked is the share of the rows without an injury that the flag leaves unmarked, the specificity. Injuries per 100 rows give the rate in the flagged rows and in the others. The first is close to the positive predictive value per row: the chance that a marked row holds an injury. For a single cut, the area under the curve is the mean of the sensitivity and the specificity. A half is a coin toss, and one is perfect.

| Group | Threshold | Rows marked, % | Injuries marked, % | Rows without injury left unmarked, % | Injuries per 100 rows, marked v others | Area under the curve |
|---|---|---|---|---|---|---|
| AFL, Australia (men) | 2 or more | 9 | 13 | 91 | 6.53 v 4.41 | 0.52 |
| European men's football | 2 or more | 26 | 33 | 74 | 4.36 v 3.14 | 0.54 |
| MLB, batters | 2 or more | 14 | 18 | 86 | 0.77 v 0.58 | 0.52 |
| MLB, pitchers | 2 or more | 23 | 28 | 77 | 3.40 v 2.54 | 0.53 |
| NFL | 2 or more | 2 | 4 | 98 | 4.41 v 2.58 | 0.51 |
| National League, Switzerland | 2 or more | 13 | 17 | 87 | 2.24 v 1.57 | 0.52 |
| AFL, Australia (men) | 1 or more | 43 | 50 | 58 | 5.39 v 4.00 | 0.54 |
| European men's football | 1 or more | 57 | 63 | 43 | 3.83 v 2.98 | 0.53 |
| MLB, batters | 1 or more | 44 | 51 | 56 | 0.70 v 0.53 | 0.54 |
| MLB, pitchers | 1 or more | 53 | 60 | 47 | 3.10 v 2.34 | 0.54 |
| NFL | 1 or more | 20 | 25 | 80 | 3.33 v 2.44 | 0.53 |
| National League, Switzerland | 1 or more | 40 | 48 | 60 | 1.95 v 1.45 | 0.54 |

# S13 Reproducibility

Every number in the paper is computed by the archived scripts. The archive holds the aggregated results those scripts produce and a file that lists each number with its source. None of the results describes a single player. The match records the scripts read identify players, so they are not in the archive. The archive says how to rebuild them from the public providers. The analyses used Python 3.12.10 with pandas 2.3.3, NumPy 2.3.5, SciPy 1.16.3, statsmodels 0.14.5 and Matplotlib 3.11.1. The archive's README maps each result of the main text to the supplementary tables and figures that hold it, the scripts that compute it and the plan it was registered under.